







Development of a Browser Extension for Phishing Website and Email Detection Using Supervised

By

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ABSTRACT

Phishing is one of the most intractable internet crimes threatening regular web users, as fraudulent websites and counterfeit emails keep stealing credentials and other vital information and money. This investigation report evaluates the possibility of creating a browser extension, which uses supervised learning to identify phishing websites and phishing emails by structural, textual, and selected behavioral features. The report conducts a review of the recent works published in 2023, compares other similar systems, supports the choice of technologies, assesses appropriate developmental methodologies, and examines the needs of users based on survey data obtained among 44 participants. The results indicate that users are highly aware of phishing threats, often receive suspicious emails and links, and are well-convinced that a lightweight extension might be able to deliver quick, conspicuous, and intuitive warnings. The project is in line with Sustainable Development Goal 9 because it helps to make digital infrastructure and innovation safer and achieve SDG 16 because it makes it easier to trust and mitigate cyber-enabled fraud. In general, the research proves that the use of a supervised-learning-based browser extension as the solution to cybersecurity in the context of cybersecurity on a daily web and email use is technically and practically applicable.

Keywords — phishing detection, browser extension, supervised learning, phishing websites, phishing emails, cybersecurity

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CHAPTER 1: INTRODUCTION

1.1 Introduction

The growing reliance on online platforms has turned the manner in which individuals interact, labor, study and conduct daily transactions. Banking portals, learning management systems, cloud storage, e-commerce websites, social media accounts, and organizational resources have come to be mediated by web browsers and email services. This reliance has brought about a digital world where speed and convenience have become the order of the day. Simultaneously, though, it has also opened up more avenues to cybercriminals to take advantage of users by interfaces and messages that look normal, familiar, and credible.

Phishing is one of the most problematic and harmful types of cyber threats that affect ordinary users. Phishing attack tries to fool the user into taking an unsafe measure like clicking an infected link, downloading an infected file or typing credentials into a fake form. The deceit is normally based on impersonation. The attackers replicate authoritative brands, organizations, college systems, delivery services, or financial systems so that they can make the victim think that the communication is real. Technical success of phishing can therefore be less about hacking into a system and more about convincing the user to open the door willingly.

Phishing has become more serious due to the improvement of the quality of attacks. The previous phishing websites were usually easier to identify as they had bad grammar, lack of brand consistency, or even suspicious links. Conversely, current phishing attacks are often conducted with high-quality imitated webpages, persuasive headings, temporal rogue web domains, credible-sounding sender names, and well-timed social-engineering wording. Recent publications by the Anti-Phishing Working Group and the 2024 Internet Crime Report by the FBI indicate that phishing and spoofing are still among the most commonly reported types of cyber-enabled frauds, which implies that the issue is both very relevant to individuals and organizations (APWG, 2025; FBI IC3, 2025).

The old-fashioned anti-phishing tools like blacklists, reputation services, static rules, and general spam filters are also valuable, but not enough by themselves. Blacklists are responsive and rely on the detection of a bad site or spammer and only after detection can blocking commence.

Patterns that are known can be detected by rule-based systems, yet they cannot cope with attackers who modify the phrasing of messages, new domains, or modify technical characteristics enough to be detected. Practically, this implies that users can still be exposed to zero-day phishing attacks before a defensive system has sufficient evidence to identify them via older, more static approaches.

This is a more adaptive problem with supervised learning. A supervised-learning model can be trained on labelled data and deployed to label new examples by examining the underlying patterns that distinguish phishing and legitimate content, as opposed to relying solely on manually written rules. These patterns can consist of structural information provided by the URLs, textual information provided by the email messages and behavioral or contextual information accumulated by the webpage content. Recent work still demonstrates that supervised-learning schemes are competitive and workable in phishing detection, especially when the aim is to construct precise classification pipelines, not with the computational expense of vast end-to-end deep-learning systems (Asiri et al., 2023; Fares et al., 2024).

Technical accuracy cannot suffice in itself. A detection system should also be able to reach the user at the appropriate time and in a form that is easily understood. That is why this project concentrates not only on the detection model, but also on the delivery mechanism. The solution here is a browser extension which uses supervised learning on phishing websites and emails. This is a viable method since users will see suspicious web pages, content on webmail and login pages in real time on the browser. The project will deliver timely protection and meaningful user support to users through a combination of a usable extension interface and a detection model based on supervised learning.

1.2 Problem Background

Phishing remains effective as the attack is focused on human trust, habitual behavior, and time constraints instead of relying solely on technical shortcomings within software. People are used to clicking on an email message, clicking on a link on a message that looks like they know, and they log into a service without a thorough examination of the page they have entered. Hackers take advantage of these habits to create interactions that appear natural to avoid suspicion. A phishing email that pretends to be a campus alert, a failed delivery notice or a bank security message can lure a user to rush into quick action before further verification occurs.

The second aspect of the issue is the rising complexity of phishing. Phishing websites can now be constructed in a visual form almost identical to that of legitimate websites, and phishing emails can be sent with no apparent errors that formerly revealed them. Attackers take advantage of branding resources, temporary domains, redirection paths and well-constructed urgency messages to enhance believability. Other campaigns combine several tactics as one e.g. an email might be used to generate urgency and a spoofed website to steal the credentials of the victim. Recent sources indicate that such mixed strategies have rendered the distinction between suspicious and legitimate content a more challenging task to assess by human judgment, alone (Naqvi et al., 2023; Kavya et al., 2024).

Conventional defense strategies have a number of drawbacks within this setting. The blacklist-based methods are still effective against well-known bad destinations but fail to stop newly established phishing infrastructure. Rule-based systems have the con of the converse, but similar set of flaws: rule-based systems may be comprehensible and can be executed quickly, but they become fragile when the lexical, structural, or content-level characteristics of attackers change. Even the general email filters cannot always recognize spam and real phishing intent accurately. This leads to the user being exposed to bad pages or messages even with the common defensive layers in place.

Technical detection research and practical delivery to users, too, have a gap between them. A large body of research indicates that classification does well in controlled settings, although the

applicability of a detection model is determined by the manner in which the output is delivered to the user in real browsing. An alarm that comes too late disrupts the workflow too often or does not define risk in a sensible manner may not have the desired protective result. That is, phishing detection is not only a classification problem, but also a human-computer interaction problem. The browser extension format is appealing because the model can work in a similar environment where the user makes a trust decision.

That is why an adaptive, quick and practical solution is obviously needed, which can be used every day in the online environment. This requirement can be satisfied with a supervised-learning-based browser extension which provides the combination of data-driven classification and direct functionality protection to the user. The project aims to explore that hybrid: how phishing sites and email-related content can be identified with high precision with the current supervised-learning algorithms, and how the detection output can be presented in a way that fits well into the user's web experience, like a browser extension.

1.3 Project Aim

Design and develop a supervised learning-based system capable of accurately and efficiently detecting phishing websites and phishing emails by analyzing their structural, textual, and behavioral features, in order to enhance cybersecurity and protect users from online threats.

1.4 Objectives

- To analyze the common characteristics and techniques used in phishing websites and phishing emails.
- To collect and preprocess relevant datasets for phishing websites and email detection.
- To develop supervised machine learning models for classifying websites and emails as either legitimate or phishing.
- To evaluate the performance of the developed models using standard classification metrics.

1.5 Scope

This project is conducted on a scope of exploration, design and prototype development of a browser extension to identify phishing sites and phishing emails and identify them through supervised learning. Since this paper is in the form of an investigation report and not an implementation thesis, the scope has been defined in such a manner that differentiates what is being investigated and supported in the current phase and what is going to be fully implemented in the next phase. The following description is thus activity-based, like the model employed in the sample investigation report, but specific to phishing detection and browser-extension development.

1. Requirement Gathering and Problem Definition. The initial step in scope is the requirement gathering. This will involve the analysis of recent literature on phishing-detection, the study of current trends in browser-based risk and gathering of user expectations by way of a confidential online survey. The survey targets user awareness of phishing, experience of suspicious websites and emails, self-detection confidence, anticipations of the alert style, and perceptions of the usefulness of a browser extension. These activities are critical since the proposed system should address a well-defined user problem and not just be a technical experiment.

2. Collection and Preparation of Data. The project also involves the identification and pre-processing of public datasets that can be used in detecting phishing sites and phishing emails. This includes the choice of labelled sources, cleaning up of raw data, checking balance of classes, missing values, simple text or URL cleaning, and initial feature preparation. The report takes into account also the shortcomings of publicly available datasets, such as representativeness, imbalance, stale samples, and cross-dataset generalization. The extent is not, however, to obtain live phishing materials of actual victims nor carrying out any malign or deceptive campaign to create new data.

3. Feature Engineering and Model Selection. One of the fundamental aspects of the project scope is the examination of structural, textual, and chosen behavioral indicators which can be converted into features of supervised-learning models. In the case of websites, these can be URL

length, token patterns, suspicious lexical markers, domain properties and webpage cues. In email detection, subject-line patterns, suspicious sender indicators, risky wordings, and embedded-link indicators are all included within the scope, unless they are impractical and unethical to process. The algorithms explored in the study include Logistic Regression, Random Forest and Support Vector Machine due to their practicality in classification, interpretability in analysis, and their feasibility within the time frame of an undergraduate project.

4. Planning of Browser Extension Prototype. The project will involve designing a browser extension prototype and explaining how it is going to serve the purpose of providing the user with phishing warnings. This involves research of the extension workflow, the user interface of the warnings, potential explanation messages, and the point of integration between the model output and the browser environment. It also encompasses taking into account constraints like browser permissions, processing overhead, privacy implications, and having non-obstructive but visible alerts. Publication and enterprise deployment of full-scale markets are not currently in scope.

5. Evaluation Planning. The investigation stage involves specification of evaluation criteria of the proposed system. The technical aspects of the project will be based on the usual classification measures, including accuracy, precision, recall, and F1-score. The report on the user side assesses the perceived usefulness, desired alert forms, expectations of speed and usability requirements based on survey results. The present report does not assert mass field implementation and longitudinal user-testing results. Rather, it lays down the assessment plan that will inform subsequent implementation and testing.

6. Exclusions and Constraints. There are a number of areas that are clearly out of scope of the project. At this stage, the system will not be implemented as a commercial security product. It will not eavesdrop or harvest personal credentials, serve as a network-level gateway, or substitute enterprise email security infrastructure. It will also not be involved in unethical practices like sending phishing mail to actual users. The primary limitations are time, quality and accessibility of recent public data, limited computing resources, browser-extension security limitations and balancing model accuracy with lightweight run-time behavior.

7. Reporting on the boundary of the investigation. In practice, this report is about what needs to be done to rationalize and inform you about the subsequent implementation phase. It defines the context of the problem, reviews the literature, makes comparisons between related systems,

chooses an appropriate methodology, studies the needs of users, and outlines a realistic technical path. The complete implementation, empirical model training, coding of the browser-extension, iterative interface testing, and benchmarking the final performance are going to be developed in the forthcoming project stage.

1.6 Potential Benefits and Target Users

The suggested project has its tangible and intangible advantages. Tangible benefits are those that can be easily noticed or quantified in the process of developing and evaluating a system. Here, the key concrete deliverables include the prototype of a browser extension, a trained supervised-learning model or models, quantifiable evaluation outcomes based on classification measures, and a specified user-facing warning workflow. These results are testable, compare and write. When effectively used the system can also mitigate the chances that a user clicks on a malicious site or acts recklessly on a site that is potentially suspicious by giving a warning when the risk occurs.

Intangible benefits are as well crucial. By demonstrating what kinds of signals users should be suspicious of, the project can raise their awareness of phishing tricks. It may also help to instill user confidence by providing support during the decision-making process, as opposed to compelling the user to wholly depend on his or her intuitions. Academically, the research will add value to the applied body of knowledge in cybersecurity by demonstrating how to combine supervised learning, user feedback and delivery via a browser into a single, practical anti-phishing concept. The report also enhances the developer with the knowledge of secure design, phishing and user centered detection systems.

The general internet users, university students, working adults, and staff members who use browsers and email to fulfill their daily activities are the target users of the proposed system. University students are particularly of interest since they often deal with login pages, institutional notices, cloud applications, and webmail applications, which makes them an appealing group of victims of phishing. Adults at work and general users also matter since phishing is not restricted to the formal organizational environment; it is often used to target personal finance, delivery services, social media, and productivity tools. Hence, the purpose of the proposed browser

extension is not to be as technical as it can be understood and useful to both academic and non-academic users.

1.7 Overview of IR

Chapter 1 outlines the introduction to the study and specifies the direction of the project. It describes the development of phishing as a user-facing cyber threat, the shortcomings of the traditional detection methods, the purpose and the goals of the project, the scope and the potential benefits of the suggested browser extension. The chapter serves as the logical point of entry of the report and sets the context of why the project is pertinent both in cybersecurity and a user-protection context.

Chapter 2 introduces the literature review. It initially explores the wider phishing area, shifting beyond the global threat patterns to focused coverage of phishing sites, phishing email, supervised-learning methods, browser-based security, feature engineering and dataset issues. It then reviews recent similar systems and related publications published after 2022, paying special attention to browser-extension solutions, client-side detection and email-centric classification studies. The chapter ends with the identification of the hardware and software environment, system requirement and technical considerations that help in the viability of the proposed solution.

Chapter 3 describes the approach of the investigation step. It makes comparisons on the relevant development methodologies, justifying the final selection and how the selected methodology suits the nature of the project. The chapter also outlines the data gathering plan, reasons why survey questionnaire was chosen as the main research tool, and a thorough discussion of all the responses obtained. The analysis is then converted to user requirements which will lead to the next phase of development.

Chapter 4 marks the end of part 1 of the project. It gauges whether the investigation and research undertaken is adequate to facilitate the realization of the proposed system and it also points to areas that still need to be finalized in the implementation process. They are dataset selection, feature engineering, model tuning, extension performance and future usability validation. By doing so, the last chapter serves as a wrap-up and as a transition to the next phase of the project.

1.8 Project Plan

This project plan is the Investigation Report (IR) part of the Final Year Project, which will start on 26 January 2026 and finish on 22 April 2026. This schedule indicates the real academic investigation tasks that will be performed at this stage, such as the refinement of the topic, confirmation of the title, the preparation of Chapter 1 to Chapter 4, literature review, designing and distributing surveys, analyzing surveys, consultations with the supervisor, preparation of appendices, formatting, and submission.

The timeline has been arranged according to the actual progression of the investigation work. On 11 February 2026, the title proposal was officially confirmed. Supervisor Meeting 1 was on 31 March 2026, Meeting 2 on 14 April 2026, Meeting 3 on 20 April 2026 and finally Meeting 4 on 22 April 2026 before submission. These dates are also directly included in the project plan and Gantt chart to ensure that the timeline is similar to the actual project record.

The updated detailed project plan is shown in Table 1 and the Gantt chart with the same sequence of tasks is shown in Figure 1 in the IR period. Appendix D also reproduces the same Gantt chart to be submitted.

WBS	Detailed Task / Activity	Start Date	End Date	Deliverable / Output
T1	Review the Investigation Report guideline, sample structure, and appendix requirements to understand the deliverables expected for Part 1.	26-Jan-2026	28-Jan-2026	Clear understanding of IR scope and required sections
T2	Explore the project domain and refine the study focus on phishing website and email detection using supervised learning.	26-Jan-2026	04-Feb-2026	Refined research focus and problem direction
T3	Prepare the title proposal, including the provisional title, initial rationale, and alignment with the project domain.	05-Feb-2026	10-Feb-2026	Submission-ready title proposal
T4	Submit the title proposal and obtain confirmation of the project title.	11-Feb-2026	11-Feb-2026	Confirmed project title
T5	Draft Chapter 1 sections, including the introduction, problem background, project aim, objectives, and scope.	12-Feb-2026	20-Feb-2026	Initial draft of Chapter 1
T6	Complete the remaining Chapter 1 sections, including potential benefits, target users, overview of the IR, and project plan narrative.	18-Feb-2026	24-Feb-2026	Completed Chapter 1 draft
T7	Prepare the ethics-related content and supporting wording needed for the ethics form submission.	20-Feb-2026	24-Feb-2026	Draft ethics-form content
T8	Design, review, and finalize the Google Form survey questionnaire for user awareness and extension expectations.	22-Feb-2026	02-Mar-2026	Final survey instrument

T9	Search for recent and credible sources published from 2022 onward to support the literature review and technical research sections.	24-Feb-2026	06-Mar-2026	Curated set of current academic references
T10	Write Chapter 2.1 Domain Research based on phishing trends, phishing websites, phishing emails, supervised learning, and browser-based detection.	03-Mar-2026	12-Mar-2026	Draft of Section 2.1
T11	Prepare Chapter 2.2 Similar Systems / Works, including the comparison table with strengths, weaknesses, citations, and paper links.	08-Mar-2026	17-Mar-2026	Completed Section 2.2 and comparison table
T12	Write Chapter 2.3 Technical Research, covering the selected tools, software, hardware, and technical feasibility.	13-Mar-2026	20-Mar-2026	Completed Section 2.3
T13	Distribute the survey to respondents and monitor response collection over the planned collection period.	10-Mar-2026	20-Mar-2026	Completed survey response collection
T14	Clean, review, and organize the survey responses to prepare them for descriptive analysis and reporting.	21-Mar-2026	24-Mar-2026	Usable survey-response dataset
T15	Conduct Supervisor Meeting 1 to review report progress and obtain comments on the investigation work completed so far.	31-Mar-2026	31-Mar-2026	Feedback from Meeting 1
T16	Write Chapter 3, including the selected methodology, methodology comparison, data gathering design, and detailed survey analysis.	25-Mar-2026	08-Apr-2026	Completed Chapter 3 draft

T17	Write Chapter 4 and prepare the initial conclusion, limitations, and transition to the implementation phase.	06-Apr-2026	10-Apr-2026	Completed Chapter 4 draft
T18	Conduct Supervisor Meeting 2 to review the updated full draft and obtain corrective feedback before final revision.	14-Apr-2026	14-Apr-2026	Feedback from Meeting 2
T19	Revise the report based on feedback, update references, check citations, and refine formatting, headings, and spacing.	11-Apr-2026	19-Apr-2026	Revised near-final IR draft
T20	Conduct Supervisor Meeting 3 to discuss the revised document and confirm any final corrections needed before submission.	20-Apr-2026	20-Apr-2026	Feedback from Meeting 3
T21	Finalize the appendices, project plan, Gantt chart, demographic profile, and overall document formatting for submission.	20-Apr-2026	21-Apr-2026	Submission-ready Investigation Report
T22	Conduct Supervisor Meeting 4 for final confirmation of the completed report before submission.	22-Apr-2026	22-Apr-2026	Final confirmation from supervisor
T23	Submit the Investigation Report (IR).	22-Apr-2026	22-Apr-2026	Investigation Report submitted

Table 1: Detailed Project Plan

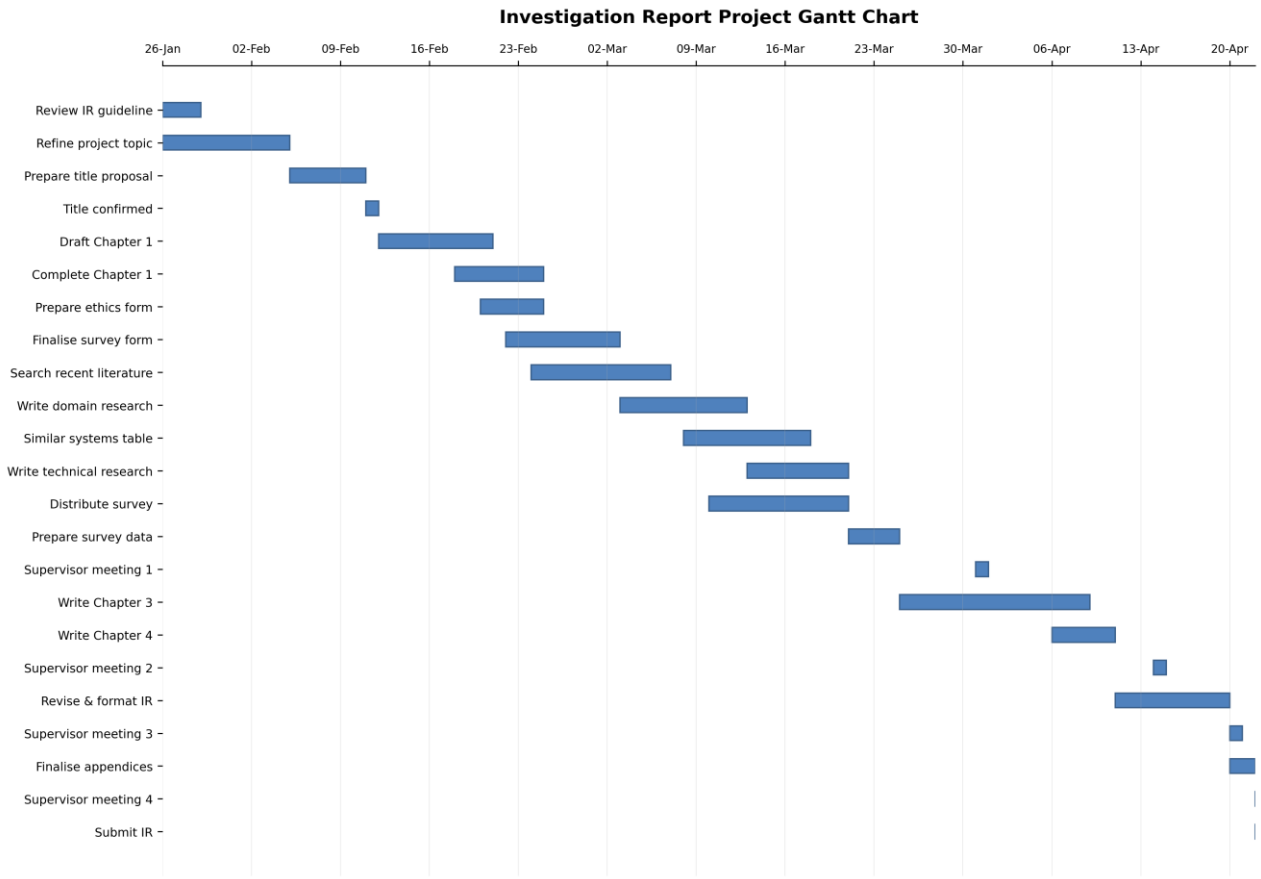


Figure 1: Project Gantt Chart

CHAPTER 2: LITERATURE REVIEW

2.1 Domain Research

This chapter oversees the domain of phishing-detection starting general to particular, starting with the existing phishing context, followed by phishing websites, phishing email, supervised-learning design decisions, browser-extension delivery, and the real-life concerns of privacy, performance, and generalization. The chapter is not merely to enumerate past work, but rather to establish a logical base to the system proposed. At the close of the chapter, the connection between the research problem, the technologies chosen, and the design of the proposed extension as presented to the user should be obvious.

The priority of the sources used in this chapter is only recent and credible materials that are released after 2022 to ensure that the investigation is up to date with the current research on phishing. This is crucial since the world of phishing threats is rapidly changing. Attackers modify the lexical format of URLs, switch channels of delivery, enhance the quality of email wording and integrate technical tricks with social engineering more and more. Thus, literature on the topic created in the recent past is more helpful than older literature in comprehending the current limitations of browser-based phishing protection.

2.1.1 Present-day Phishing Scenario and Threat Development.



Figure 2:Visual Representation of Phishing Threats

Recent findings have always indicated that phishing is no longer an email specific issue. It has become a large ecosystem of fake interactions that encompasses cloned sites, brand impersonation, counterfeit document-sharing alerts, account-checking, and fraudulent delivery notices, and credential-gathering campaigns spread via various channels. According to Naqvi et al. (2023), phishing is a dynamic and adaptive type of attack that has remained uneliminated due to its use of people, interfaces, and trust relationships as opposed to just focusing on technical vulnerabilities. They also conclude in their review that effective mitigation needs a mix of prevention, detection, user education and responsive reporting, which also suggests that a single fixed defensive layer might not suffice.

Although awareness of phishing has grown, the Anti-Phishing Working Group reported almost one million phishing attacks in the fourth quarter of 2024 alone, which shows that the phishing activity, on a high scale, does not decrease (APWG, 2025). Similarly, the 2024 Internet

Crime Report by FBI reported phishing and spoofing among the most commonly reported internet crimes, which confirms that the threat is still impacting significant user numbers in practice (FBI IC3, 2025). Such official reports are helpful since they supplement academic results with working evidence: phishing is not merely a lab issue, but a constantly occurring user-facing threat.

Attackers are becoming more adept at using technical evasion in conjunction with social engineering finesse as phishing campaigns continue to become increasingly sophisticated. Contemporary-day campaigns deploy temporary infrastructure, redirection chains, Unicode or lexical manipulation, realistic forms, and credible-looking sender identities. Sometimes the email itself is not the goal, it is a bait that directs the user to a page that harvests credentials or performs some other action that is malicious. This cross-channel behavior is the key in the current project since a browser extension that checks websites and not email-linked behavior would only cover a portion of the decision chain, which results in victimization.

Recent reviews also suggest that the development of phishing is driving researchers towards more adaptive, explainable and deployable solutions well near the user. Kavya et al. (2024) suggest that future-oriented phishing detection should be concerned not only with the classification accuracy, but also with the robustness, interpretability, and the response to the novel variants of attack. This is important in the context of an undergraduate project as it implies that a viable solution must be able to compromise between model complexity and maintainability as well as human usability. An example of a lightweight supervised-learning pipeline that can be delivered as a browser extension is thus quite consistent with the pragmatic path indicated by the existing literature.

2.1.2 Phishing Websites

Phishing sites are a type of fraudulent webpage created to mimic the appearance of a valid online site to gather information, spread malware, or make dangerous user choices. The fundamental deceit is normally pictorial and circumstantial. A web page can replicate the color scheme, layout, and logos of a bank, cloud service provider, university portal, or online shopping site, and make the user think they have arrived at a secure location. The misrepresentation in most instances is

supported by lexical similarity in the URL, counterfeit security language or a well-timed email spoil that causes the page to seem anticipated.

Safi and Singh (2023) indicate that the research on phishing websites detection has extensively been done on machine-learning methods due to its flexibility in modeling lexical, host-based and content-based features than the fixed rules. They also review that datasets have different sizes, ages, source, and feature design that have direct implications on generalization. The performance of a classifier in a benchmark might not be the same when applied on another dataset, which is gathered on a different threat setting. This note applies to the present project since the suggested browser extension is meant to be used on normal web interactions and not on a single curated benchmark environment.

Contemporary phishing sites tend to manipulate URLs at the level of trust or to mask danger. Patterns that are common are long URLs, suspicious subdomains, deceptive tokens like login, verify or secure, use of IP addresses or shortened links, and visually similar domains. But risk of websites does not conclude with the URL. Signals can also be provided by page structure, form behavior, embedded links, favicon sources, script patterns and external-resource dependencies. The complication is that not every one of these features is as lightweight or as secure as it would be to gather in a browser extension. This implies that feature requirements of the proposed system should be well selected to ensure that the features are useful but not invasive or heavy to compute.

Ongoing studies continue to add to feature engineering of phishing sites. Omari (2023) investigated gradient boosting to detect phishing and found that structured sets of features can lead to a great classification strength being relatively interpretable. Rashid et al. (2024) further stressed on the significance of generalization in datasets and the collection environment. Their work particularly comes in handy with this project since they warn against overconfidence on models that work well in just one set of data. In order to be useful in practice, a browser extension should not be too specific to a particular training distribution.

A second key aspect of recent literature is that user-facing site detection should also take into account how the signal is conveyed as well as whether a site is malicious or not. A browser extension cannot presuppose that the user is going to examine raw features or comprehend

technical output. Rather, the outcome of the classification should be diminished to meaningful actionable feedback like an explicit warning, an intrinsic threat condition, or a brief description. The suggested system thus considers phishing web site detection to be both machine-learning problem and a decision-support problem.

2.1.3 Phishing Emails

Phishing emails are also one of the most frequent and most effective delivery mechanisms of phishing attacks as email is a daily tool that is used in academic, personal, and business life. The phishing email can demand a verification of the password, purport that an account will be closed, impersonate a document-sharing service, or generate a sense of urgency about paying, tax, delivery, or security. The message is meant to break habitual behavior, and it is meant to compel the user to take immediate action before thorough scrutiny is carried out.

Email phishing is not the same as generic spam since the goal is typically more targeted and more dangerous. The message can include a link to a bogus login form or prompt the user to open a weaponized attachment or gather data by use of a reply-based scam. Fares et al. (2024) demonstrate that machine-learning algorithms like the Random Forest, the Support Vector machine and the XGBoost can still be used to classify email-phishing when trained on meaningful features and assisted by an appropriate choice of features. Their work is particularly applicable to this project since it shows that only classical supervised-learning models can be used to detect email-based phishing and not large and resource-intensive ones.

The feature-design questions are also presented by the email domain of phishing detection. Other characteristics are lexical and textual e.g., suspicious words, urgency language or strange grammar. Others include the structural ones, e.g. inconsistency in senders, mismatch in domain, linkage patterns or indication of attachment. Meléndez, Ptaszynski, and Masui (2024) compared models built with traditional machine-learning with transformers-based models in detecting phishing emails and concluded that more recent language-centric models have the potential to offer high-quality results, albeit with the caveat of needing to carefully design datasets, trade-offs in computation, and resilience to the changing content of emails. In the present project, this justifies

a practical way: classical supervised-learning practices may be used in case the objective is to develop a viable and fairly lightweight extension-oriented solution.

A deployment challenge also exists. A browser addon is not necessarily allowed complete access to native email clients and even within webmail systems the information available might be limited by permissions or privacy expectations or page layout. Thus, the email-based aspect of the suggested system is deliberately designed in the context of email-based phishing cues accessible via the browser environment, including warning signs of a suspicious sender in webmail messages, malicious links, and content patterns, which can be approached both ethically and feasibly. This more limited, yet more plausible scope is better than promising the ability to inspect an inbox at full detail without thinking about privacy and implementation issues.

The phishing emails are particularly significant in terms of the user-centered approach since they are often the initial step in a multi-stage assault. A suspicious email might not explicitly steal credentials, but it can provide conditions of trust that cause the user to be more susceptible to accept a harmful webpage as legitimate. The suggested system thus considers phishing emails and phishing websites as related, as opposed to independent phenomena. This combined approach is among the primary factors why a web-based delivery model is suitable.

2.1.4 Structural, Textual, and Behavioral Indicators.

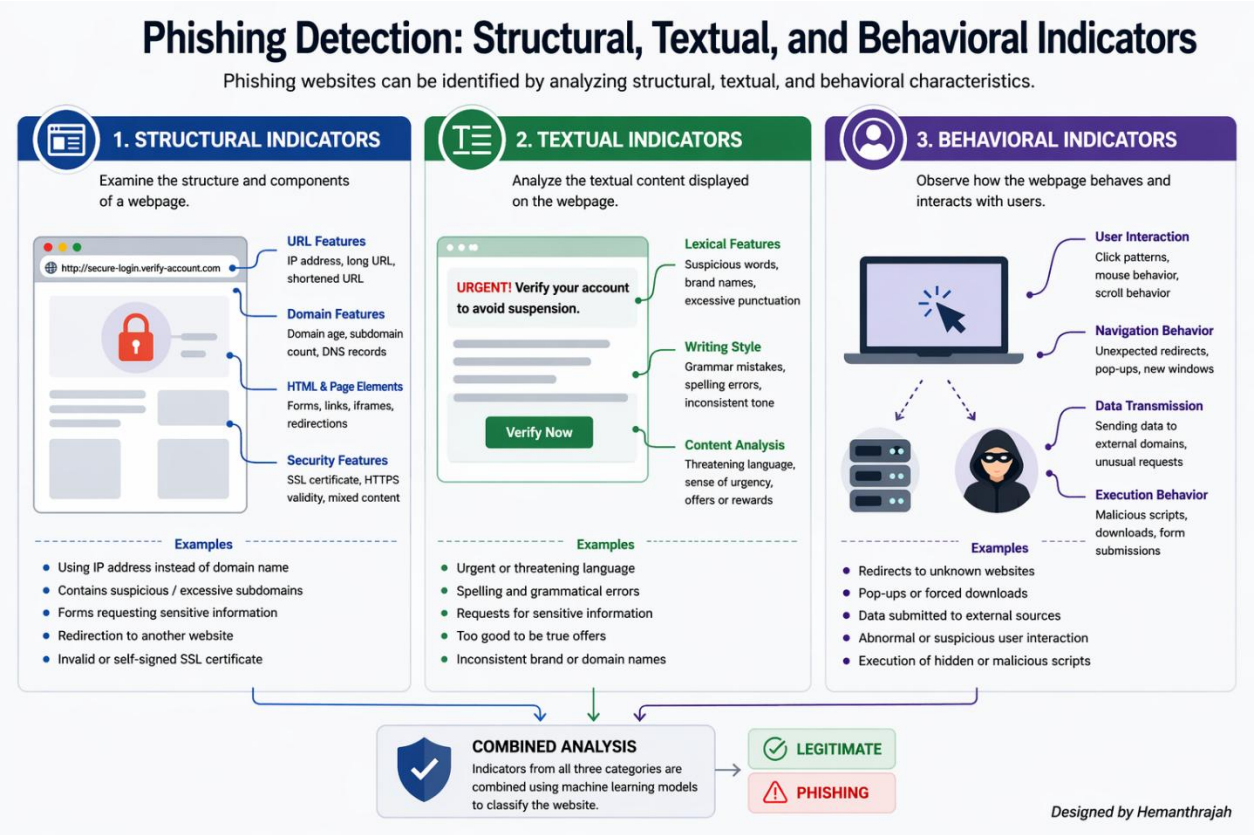


Figure 3: Graphical Representation of Structural, Textual, and Behavioral Indicators in Phishing Detection

The present project suggests that not a single type of indicator should be used to detect phishing. Rather, the combination of structural, textual, and chosen behavioral cues can be used to establish a stronger system. Structural indicators are any measurable features of URLs, domains and page layout. Some of them are the number of subdomains, pattern of tokens in URLs, the presence of suspicious characters, complexity of the path, relationship among form actions and page identity. These indicators are particularly appealable in browser-based detection since most of them can be extracted in a short time without necessarily heavy visual processing.

Textual indicators are linguistic indicators in an email or webpage. This can encompass urgency words, words that request action, or words that attempt to evoke fear or compliance, lexical patterns related to credential gathering, and wording that does not match the purported identity of the sender or service. Text messages come in handy since phishing is usually based on

persuasion rather than technical deception. Meanwhile, textual analysis should be approached with precaution, since valid messages may also include an element of urgency or security jargon. This is the reason why textual features should be applied in conjunction with other cues and not a decision rule by itself.

Behavioral indicators are more contextual. Behavioral in the context of this project does not imply gathering invasive long-term browsing telemetry. Rather, it means observable context surrounding a suspicious interaction: such as whether a page immediately requests credentials, whether the destinations of links and links match that destination visible text, whether a sender is new, or whether the page tries to prompt a quick decision to trust or not via interface design. These indicators are used to fill the gap between raw features and risk to the user. They can be especially helpful in a browser extension since they are more intuitively aligned to the type of warning language a user can comprehend.

The recent research on explainable phishing detection makes the argument of cautious feature selection more robust. Shafin (2025) suggested an explainable feature-selection framework of identifying web phishing based on machine learning and established that not every available webpage feature is equally useful in a model. This is significant as large feature sets may also lead to complexity, which does not enhance feasible performance. In a project that is based on extensions, features and explainability are particularly desirable: the model must be correct enough to be useful, but simple enough to warrant and implement without unwarranted overhead.

Collectively, these studies imply that a practical phishing-detection system must be able to combine various types of signals but yet selective concerning what is actually processed. The proposed project is informed by this insight. The extension does not require all the features that are possible in theory. It requires a set of defensible features that trade off performance, interpretability, runtime cost, and user trust.

2.1.5 Dataset Quality, Drift, and Generalization

Among the most significant problems of phishing detection research, the fact that datasets may become obsolete in the short term should be mentioned. Domains are dropped, campaigns are

changed and over time the linguistic nature of phishing messages changes. A model that has been trained on an older benchmark can thus have a good historical performance but a poor generalization in the real world. This problem has been brought to focus of recent studies. Rashid et al. (2024) directly consider the aspect of generalization among phishing URLs datasets and demonstrate that cross-dataset transferability is not an easy thing. The result is of significant importance to the present project since a browser extension is useful only when it is able to support the data that is not in the identical format as the training set.

The composition of the dataset also matters since phishing detection tends to be based on the class imbalance and heterogeneity of the sources. Phishing datasets can be collected based on reporting feeds, open repositories, synthetic collections, or hand-curated benchmarks. Popular-domain lists, or ordinary email corpora, or web archives may serve as legitimate samples. Every decision influence the task in a different way. As an example, just training on blatantly malicious URLs and very popular legitimate domains can provide a false sense of an easy line between the two. However, in a real-world browser context the model can come across ambiguous content that is between those extremes.

The other problematic area is the availability of features in datasets. Other datasets include webpage content, host features, or email bodies whereas some offer URL-level attributes only. This has a direct repercussion on reproducibility and deployment. An extension that is based on capabilities not provided by most of the public sources could be hard to test reliably. In comparison, a system that is built based on a smaller yet more reproducible set of features could be simpler to train, compare, and justify. Due to this reason, the current project focuses on feasibility and consistency rather than on the maximum number of features.

It is also mentioned in the literature that interpretability is important in the case of drift and generalization. When a model starts to give up on new data, a set of interpretable features makes it easier to diagnose why. Recent explainable-ML research in phishing detection supports the perception that transparent feature contribution is not only a scholarly matter, but also practically applicable to maintain and refine via iteration (Shafin, 2025). This underpins the choice of supervised-learning models in the project as more easily comparable, tuned, and explained within the constraints of an undergraduate cybersecurity project.

2.1.6 Supervised Learning and Algorithms Choice.

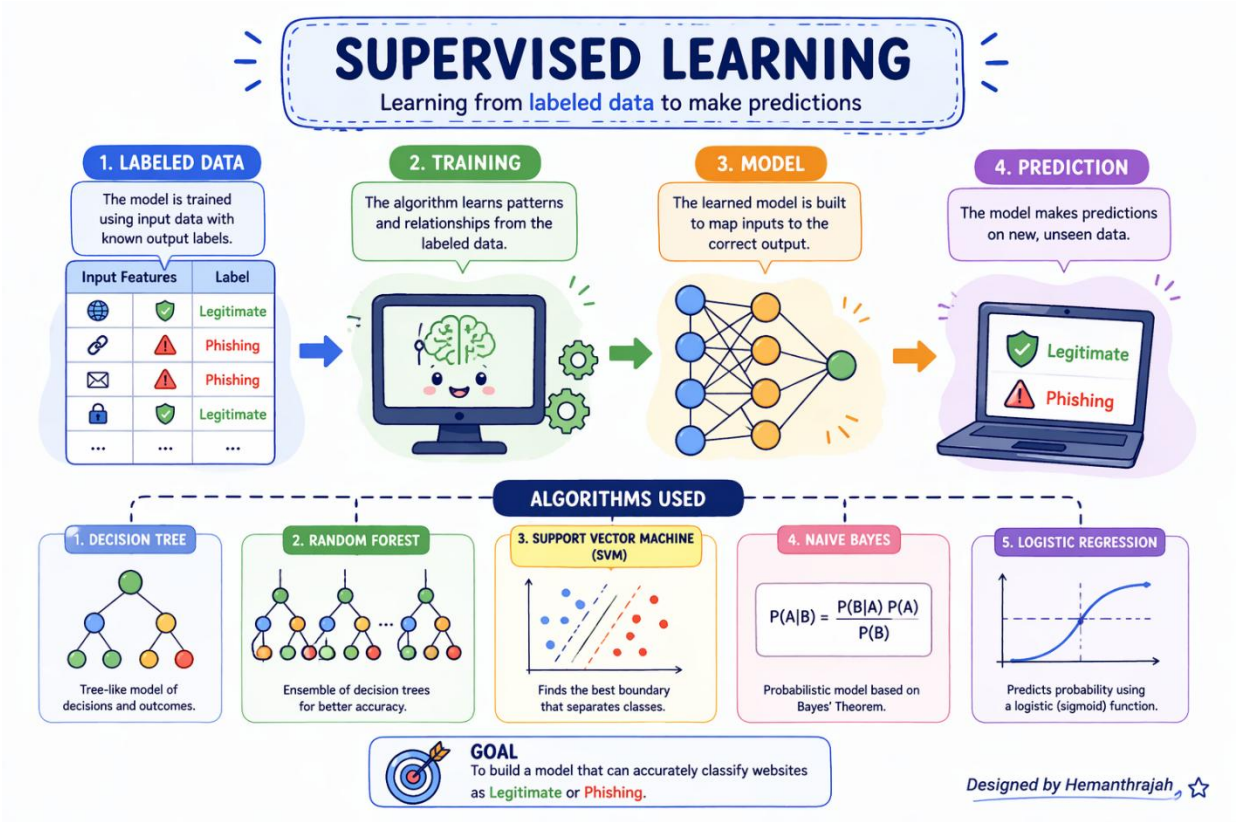


Figure 4: Graphical Representation of Supervised Learning and Classification Algorithms

Supervised learning is still appealing to phishing detection since it occupies a viable space between flexibility, interpretability, and resource utilization. The model in a supervised environment acquires the knowledge of labelled phishing and legitimate samples and uses the association between features and labels to categorize new examples. This simple paradigm is highly applicable to phishing since several existing public datasets already include binary labels and the overall system goal in this project is also binary: is a webpage or email-linked artefact to be considered as legitimate or suspicious.

Logistic Regression is an example of a classical model that can be used as a baseline since it is easy, quick, and somewhat interpretable. It tends to work well in situations where the feature space is highly linearly related to the class label, and it can be used to give a clear point of reference to more complex models. Random Forest is useful due to its ability to model non-linear interactions between features and its resistance to moderately noisy features. Support Vector Machine is

appealing in the sense that it can work well in classification, with well-crafted feature spaces, and can be efficient when the problem is well-constrained. These features render the three appropriate to the current project.

Current studies still confirm these models in phishing situations. In a study by Fares et al. (2024) on the use of both random forest, SVM, and XGBoost to detect email phishing, all three algorithms were useful and SVM had the greatest accuracy in their context. Omari (2023) proved that the use of gradient boosting techniques can also be quite efficient when it comes to phishing websites classification. Even though the results of transformer and deep-learning models can be very strong in certain circumstances, Meléndez et al. (2024) demonstrate that the decision between the traditional and transformer-based approaches should be made based on the context, data characteristics, and constraints of the deployment and not just on the trend.

In this project, a fully deep-learning-heavy design is undesirable, and instead, supervised learning is more appropriate due to the following reasons. To begin with, the extension should be lightweight and responsive. Second, the goal of the investigation phase is not to come up with a maximum scale benchmark outcome, but to create an acceptable and executable anti-phishing concept. Third, classical supervised-learning models are more readily compared systematically with precision, recall, F1-score, and confusion analysis. This makes them more applicable to a final-year project setting where transparency, viability and critical analysis are of value.

2.1.7 Browser Extension as a Delivery Model.

An example of a viable delivery mechanism of anti-phishing is a browser extension since the browser is usually where a trust decision is made by the user. The user clicks a link and arrives on a page, or reads a message in webmail, or a login prompt. When protection is provided retrospectively most of the value is lost. An extension in the browser removes that distance by enabling the detection logic or risk-evaluation workflow to execute within the identical space of interaction where the suspicious content is shown.

New studies confirm this trend. Wu, Kuo, and Yang (2023) have introduced a machine-learning-based browser extension (to detect phishing websites) and demonstrated that even real-

time browser-integrated warning can be of practical benefit in the context of anti-phishing. Linh et al. (2024) applied the existing concept of browsers to a deep-learning framework implemented as extensions, showing that the desire to deploy at the client-side remains. These systems can be applied to the current project since they demonstrate that phishing detection is no longer restricted to either offline classifiers or server-side filters, it can also be provided as a user-centered tool that can be used during browsing.

But browser extensions also bring about limitations in design. They should be light, secure, permission-conscious and understandable to the user. Any extension that introduces perceptible latency, demands excessive access, or generates noisy alerts can be technical but practically ineffective. Thus, the extension model does not eliminate the complexity in design, it merely shifts the focus. The project should not only concentrate on the performance in classification but should also concentrate on the interface simplicity, warning timing, risk explanation, and trust in the extension itself.

2.1.8 Extension-Based Detection: Security, Privacy, and User Trust.

A web browser extension that will investigate web pages or email-based content has to be attentive to security and privacy. Users tend to be paranoid about installing extensions as extensions have access to browsing context, page content, or interaction events. This warning is warranted. The system proposed will have its security value compromised when the user thinks that the extension is invasive or too powerful. Therefore, the current project considers minimal necessary access as a design value.

Privacy wise, the extension must not gather too much personal information and must keep browsing data that has no connection to phishing detection. In case of possibilities, feature extraction can be restricted to the amount required to do classification and generate warnings. There should be no retention of sensitive data unless it is necessary to conduct tests and the system must be transparent on what it is analyzing. This is especially critical in any functionality involving email, since users might consider inbox-linked data more sensitive than regular webpage information.

Explanation is also important to user trust. When the extension merely states that a page is dangerous with no explanation, a portion of the users might not pay attention to the warning, particularly in cases where the page itself seems valid. The responses to the surveys used in this project show that the vast majority of users like to receive direct warnings in the form of pop-ups, and a considerable number of users also like to understand why the content is suspicious. This implies that the extension must record layered communication: a brief initial warning on speed, and optionally, on confidence and learning.

Another practical consideration is security of the extension lifecycle. The extension must be developed in such a way that the extension can be updated later, new features added or the model can be replaced without posing a significant level of risk or complexity. The importance of maintaining the runtime path to be manageable and the deployment model to be coherent has been stressed in the literature on newest browser-integrated phishing systems. The same principle is embraced by the proposed project. A simpler and more defendable system is better than a feature-laden design which is not justifiable or supportable.

2.1.9 Summary of Domain Research

The sources examined in this section demonstrate that phishing is a large-scale and constantly developing cyber threat, particularly via websites and email-operated exchanges. It also demonstrates that supervised learning remains a viable and plausible route to the classification of phishing, as long as the model is backed by the right features and the latest data. Lastly, the literature shows that browser-integrated protection is a good direction due to its proximity of detection to the actual decision point of the user. The combination of these findings proves the main assumption of the given project: a browser extension built on supervised learning has a chance to be both academically and practically justified.

2.2 Similar Systems / Works

In this section, the review of the recent systems and works that are highly related to the proposed browser extension are reviewed. Following the command of the user, the accent is made on recent and credible work that has been published after 2022 and not on old landmark systems. The idea is to know what the latest phishing-detection solutions can and cannot do, what their strengths and weaknesses are, and what the proposed system might differ with them in terms of scope and design focus.

Since the current project integrates phishing protection of websites and emails within a browser extension, the comparison is not based on the study of a single narrow type of previous work. Rather, it compares browser-extension systems, real-time web-detection systems, email-phishing classifiers and hybrid frameworks. Such a broad comparison can be helpful since it shows the place in which the proposed system falls in the existing research field: it is neither a new web site classifier, nor an email filter, but a user-friendly browser-based detection idea that cuts across various channels of phishing.

2.2.1 Typical features of Comparable Systems / Works.

One of the best recent examples of phishing detection offered in the form of a browser extension is provided by Wu et al. (2023). Their work is highly applicable as it shows that machine-learning-based scoring of risks can be directly incorporated into the operation of a browser and be applied to give a user real-time feedback. The main characteristic of this system is its practical client-facing nature. Rather than making phishing detection a phishing offline classification activity, it puts the model as a part of the browsing process.

The study by Linh et al. (2024) is also dedicated to the extension-based phishing detection, though it is more deep-learning-oriented. A different route of design is emphasized in their work: they do not use lightweight classical models as the primary source of power, instead deep-learning-based detection is investigated in the context of extensions. This makes it handy as a benchmark of comparison in terms of the trade-off between sophistication and runtime overhead.

Fares et al. (2024) concentrate on phishing email detection and compare the models of supervised learning to classify email content. Even though their work is not packaged as a browser extension, it holds significance as it focuses on the email aspect of the phishing issue with approaches that are not too complicated to be implemented by undergraduates. The property that renders the work relevant is that it focuses on feature selection and classical supervised learning and not on opaque large-scale architectures.

Melendez et al. (2024) contrast the use of traditional machine-learning models with transformer models to detect phishing emails. Their contribution is valuable as it demonstrates that the classical approaches are not necessarily nullified by the newer language models. Rather, a model is effective based on context, data, and deployment objectives. This analogy is very applicable to the current project in the sense that it supports the notion that feasibility, interpretability and user-facing deployment constraints should be used in determining the model choice.

Asiri et al. (2024) suggest a PhishingRTDS, a deep-learning based real-time phishing detection system. This system is applicable as it is a more sophisticated and holistic detection framework with the objective of providing protection in real time. It assists in putting the ongoing project in comparison to a more complicated state-of-the-art orientation. Equally, Rashid et al. (2024) and Shafin (2025) are valuable, as they discuss two concerns that are of great importance in real-world implementation: generalization across datasets and explainable feature selection. These are not the same extension products as Wu et al. (2023) but are what a sound browser-side solution must and must not rely on.

2.2.2 Strengths and Weaknesses of Similar Systems

Immediacy is the best strength of new browser-extension systems. When detection is implemented in the browser, the user can be cautioned during the presence of risk, as opposed to once a suspicious interaction has taken place. Wu et al. (2023) demonstrate this strength by demonstrating the possibility of a machine-learning-based extension to present direct phishing warnings as a person browses. The shortcoming, however, is that extension-based systems have to

be lightweight and permissioned. The richness of a model does not necessarily make it more effective when it creates delay, complexity, or privacy issues.

The systems developed based on deep learning, like those mentioned by Linh et al. (2024) and Asiri et al. (2024), tend to be more powerful in modelling and detecting in real-time. Their great advantage lies in the fact that they can learn more intricate patterns than simple rules which have to be engineered manually. The weakness of their practicality in the context of this project is that heavier models may be more difficult to justify, explain, and integrate into a lightweight extension prototype. They might also need additional calculation and additional fine-tuning than would be optimal in a final-year project based on feasibility and usability.

Another strength of email-centric classifiers like Fares et al. (2024) and Meléndez et al. (2024) is that they demonstrate that it is possible to do phishing detection effectively on textual content through supervised learning. These researches facilitate half of the existing project that is email oriented. They are limited by the fact that they cannot resolve the delivery problem individually. Even a high-performing email classifier requires an ethical and practical means to bring risk to the user in a live browsing or webmail context. It is precisely that delivery challenge that the proposed concept of browser-extension can work on.

The studies with a focus on generalization and explainability bring another type of strength. Rashid et al. (2024) outline the significance of robustness across datasets, whereas Shafin (2025) lays emphasis on the transparency of features and the effective selection of features. The value of these contributions lies in the fact that they identify weaknesses that are frequently concealed by high benchmark scores: lack of transferability and complexity of features that are not needed. These lessons can be useful to the proposed project because they explicitly prefer supervised-learning techniques which can be compared, explained and maintained with less difficulty.

System / Work	Recent paper / citation / link	System type	Main strengths	Main limitations	Relevance to proposed system
Browser-extension	Wu, Kuo, & Yang (2023)	Machine-learning	Real-time browser-side	Focused mainly on phishing	Very high

phishing detection	Link: https://doi.org/10.1109/AsiaJCIS60284.2023.00023	browser extension	warning, practical user delivery	websites, extension constraints remain	
Deep-learning extension-based real-time detection	Linh et al. (2024) Link: https://doi.org/10.11591/ijece.v14i3.pp3021-3035	Browser extension with deep-learning pipeline	Immediate client-side protection with stronger modelling capacity	Heavier architecture and potentially higher runtime cost	High
PhishingRTDS	Asiri et al. (2024) Link: https://doi.org/10.1016/j.cose.2024.103843	Real-time phishing detection system	Strong modern architecture and emphasis on real-time detection	More complex than a lightweight undergraduate prototype	High
Email phishing detection with classical ML	Fares et al. (2024) Link: https://doi.org/10.1016/j.procs.2024.11.179	Supervised-learning email classifier	Good performance with SVM, RF, and XGBoost; feasible baseline approach	Does not directly address browser delivery or interface design	High
Traditional vs transformer email detection	Meléndez et al. (2024) Link: https://doi.org/10.3390/electronics13244877	Comparative phishing-email study	Clarifies trade-offs between classical ML and transformer models	Not itself a deployable user-facing system	Medium-High
Cross-dataset	Rashid et al. (2024)	URL-based phishing	Highlights robustness and	Not packaged as	Medium

URL generalization	Link: https://doi.org/10.1016/j.comnet.2024.110398	detection generalization study	transferability issues	a client-facing product	
Explainable feature selection for web phishing	Shafin (2025) Link: https://doi.org/10.1016/j.dsm.2024.08.004	Explainable phishing-feature engineering framework	Improves interpretability and feature efficiency	Focus is feature selection rather than full delivery workflow	Medium
Proposed system	Current project Link: N/A	Supervised-learning browser extension for websites and email-linked phishing cues	Combines user-facing browser delivery with practical supervised learning	Prototype scope, dataset quality, and permission limits still apply	N/A

Table 2: Comparison of Similar Systems and Proposed System

2.2.3 Conclusion

As the comparison reveals, the proposed project is midway between two significant lexicons of recent work. There are browser-integrated systems that emphasize instantaneous warning and client-side protection on one side. Classification studies, which explore which algorithms and feature sets are effective on phishing websites or phishing emails, are found on the other side. The suggested system is a synthesis of the two strands. It does not strive to match the most sophisticated deep-learning structures in literature. Rather, it seeks to develop a viable supervised-learning solution, which can be incorporated into a browser extension and tested according to the user-centered criteria, including speed, clarity of warnings, and usability.

This is a suitable position for a final-year project. It is not too ambitious to be meaningful, as it deals with websites and email-linked phishing indicators, but is limited enough to be implementable within resource and time constraints. The recent literature thus augurs in favor of the project direction: the lightweight, supervised-learning, browser-based anti-phishing solution is no trade-off in relevance, but a logical and justifiable design choice.

2.3 Technical Research

In this section, the technical environment that the proposed system requires is identified and the reasons why the chosen technologies are applicable. The technical research in the project does not include listing tools. It also covers system requirements analysis, runtime constraints analysis, and model-selection implications analysis, extension design analysis, and practical relationship between machine-learning layer and the browser interface. The technical environment shall be capable of supporting both data-science experimentation and client-side integration, as the proposed system is a browser extension that is powered by supervised learning.

Systems wise, the project is a combination of two areas that may be addressed differently. The first is the offline machine-learning workflow, including the dataset preparation, feature extraction, training the model, validation, and comparison of the performance. The second one is the browser-extension workflow that involves content access in the browser, warning presentation, no or minimum delay in running, and easy user interaction. An effective technical design should ensure that these domains are linked together without producing very complicated architecture.

2.3.1. The Non-Functional Requirement Analysis.

The suggested system will have to meet a number of functional requirements. First, it should be capable of analyzing the indicators of a web site and determining a browsing target as phishing or legitimate. Second, it must be capable of analyzing email-linked or email-visible indicators in a browser environment where such an analysis is both ethically and technically

feasible. Third, it should notify the user in a clear manner using a warning or safe-status message or just a plain explanation of its result. Fourth, it must enable offline assessment using standard metrics of classification to enable rigorous comparisons of model performance during development.

Non-functional requirements are also critical in that the system should be meant to be used by actual users and not just to be used offline to benchmark it. The extension must not be heavy such that it noticeably slows down the browsing experience. It should be detected quickly, as it was revealed in surveys that users place great importance on speed of response. The interface must be user-friendly, since even good performance with models cannot work in practice when the warning message is unintelligible or overly technical. In the design, confidentiality must also be observed by not necessarily collecting or retaining personal browsing information during tests.

2.3.2 Software Environment



Figure 5: Python Logo

Python is chosen as the main language that the machine-learning workflow is written in since it provides an opportunity to process data effectively, experiment with it, analyze the model, and compare its results. Pandas and NumPy are libraries that can be used to work with datasets, clean inputs, and make features. The fact that scikit-learn offers established implementations of Logistic Regression, Random Forest, Support Vector Machine, model-selection tools, and measures of evaluation in a single package makes it particularly appropriate. These features render it suitable to a project that appreciates the comparative supervised-learning experimentation.



Figure 6: Jupyter Logo

During the exploration phase Jupyter Notebook comes in handy as it enables the developer to explore and investigate data, experiment with sets of features, and interactively compare model results. Visual Studio code will be the primary development tool since it can be used to code Python-based data science, as well as web-extension in HTML, CSS, and JavaScript. This will ease tension between the model development and the browser-extension side of the project.

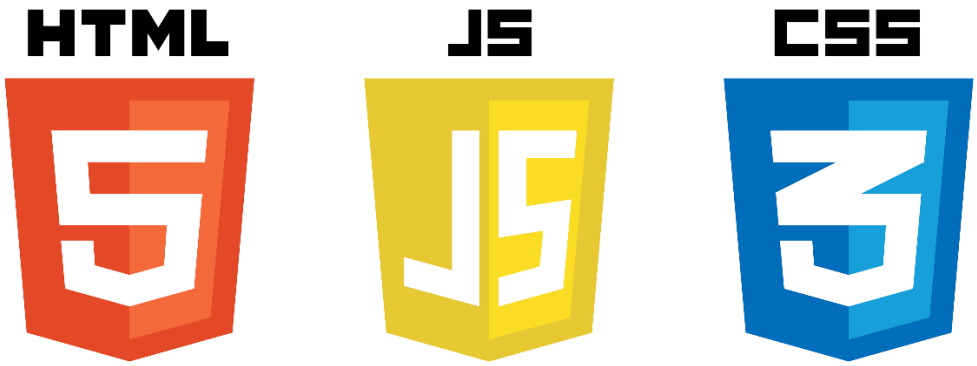


Figure 7: HTML, CSS & JavaScript Logo

In the case of the browser component, normal web technologies suffice. The extension interface is structured in HTML, visual hierarchy is provided with the help of CSS, and browser events, content extraction, and interaction with the prediction workflow are managed with the help of JavaScript. A browser based on Chromium like Google Chrome or Microsoft Edge will be suitable since both support extensions, developer tooling and a well-documented testing path to local prototypes. The technical environment is also very affordable and reachable, hence suitable for a final year project.

2.3.3 Hardware Environment

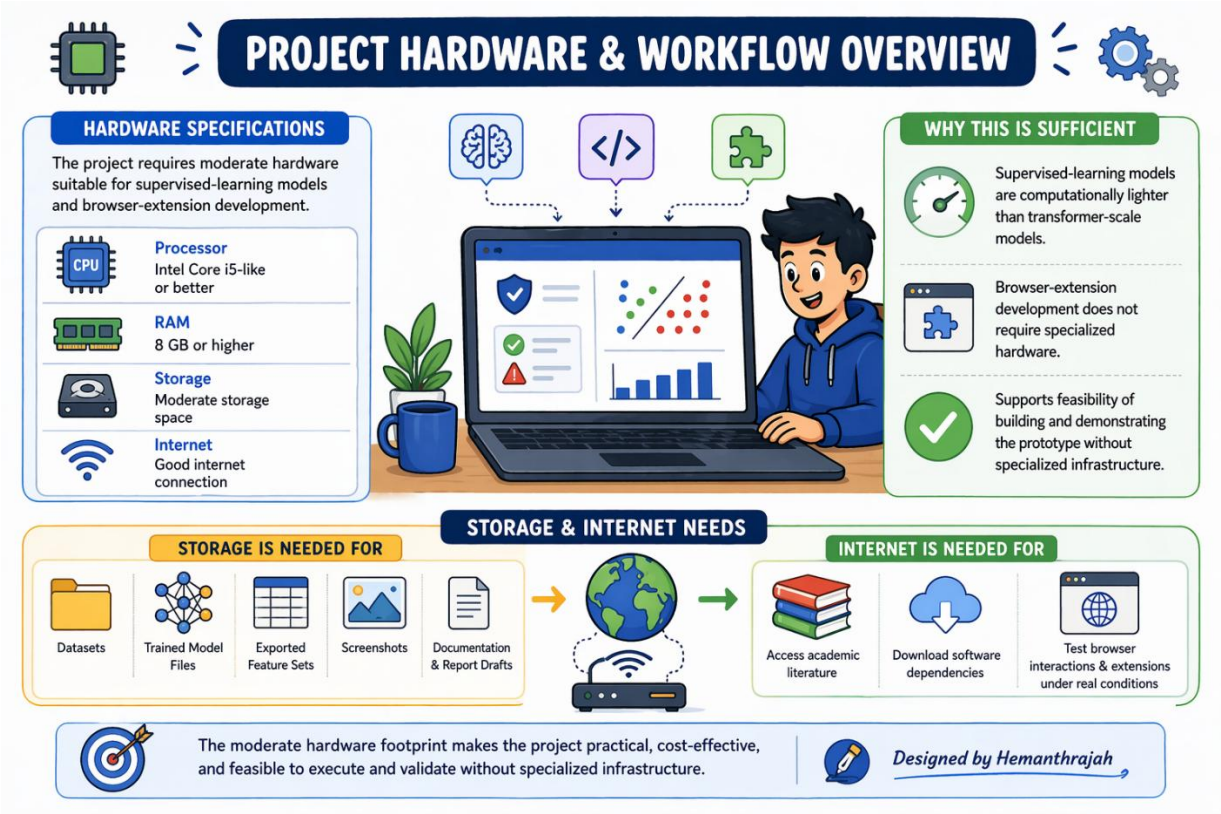


Figure 8: Hardware Specifications Required for Browser Extension and Supervised Learning Development

The hardware specifications needed by the project are medium in comparison to large-scale deep-learning systems. The planned workflow requires a typical laptop or desktop with an Intel Core i5-like processor or better, 8 GB of RAM or higher, moderate storage space, and a good internet connection. These are sufficient since the project focuses on supervised-learning models that are computationally lighter than transformer-scale models and since the development of browser-extensive does not also necessitate specialized hardware.

Storage is needed for datasets, trained model files, exported feature sets, screenshots, documentation, and report drafts. The need to access academic literature and software dependencies via the Internet is not the only one: it is also required to test browser interactions and check the behavior of extensions under realistic browsing conditions. The pragmatic benefit of the

moderate hardware footprint of the project itself is that it supports the viability of executing and proving the prototype in the absence of specialized infrastructure.

2.3.4 Technical Architecture Considerations

One way is not the only possible way of designing a completely integrated browser-extension phishing detector. An alternative is a pure local architecture where feature extraction and prediction are done all within the extension. This offers good privacy advantages but can limit the size of models and complexity of features. The second possibility is a hybrid architecture where the extension gets a small number of features a locally available and then forwards such features to a lightweight local service or prediction endpoint. This will be able to lessen the client-side computing load but adds complexity to communication and deployment. In the current project, both models are conceptually taken into consideration in the investigation phase, although a lightweight and privacy-friendly workflow is the direction of the prototype.

Another consideration of architecture is featuring availability. The easiest to extract features are URL-based and may need more parsing to extract webpage content features. Detection using email presents even more limitations due to the differences in webmail interfaces and browser permissions should also be proportional. This architecture must be incremental. The initial prototype phases can focus on URL and visible-page indicators, and subsequent phases may add email-linked indicators where the practice is still within the limits of ethical and technical considerations.

Maintenance and updates of the model too need to be taken into account. The architecture of a browser extension aimed at detecting dynamic phishing strategies cannot afford a hard-to-upgrade system. Due to this, the current project will enjoy the benefit of supervised-learning models with features and thresholds that can be re-trained or modified without reconstructing a highly complex end-to-end pipeline. This practicality of the selected technical direction is its maintainability.

2.3.5 Design Browser Extension.

The browser extension must be visible to facilitate the decision making without being annoying. The survey shows that customers like pop-up notifications and instant feedback, yet they also appreciate rapid performance and simplicity. Such preferences mean that the interface should not be loaded with technical information as a default. It is more suitable to be layered: the main output may be a short warning or a safe-status message, and a secondary display may include more information for those users who might need to know why the classification was made.

There should be no pretentiousness of the extension being a certainty as well. All automated decisions related to cybersecurity, and phishing detection, in particular, are probabilistic to a certain degree. The interface must however be able to convey risk in a clear and responsible manner. This can involve the use of language like suspicious, high risk, or possibly deceptive as opposed to making the tool appear to be flawless. This type of design enhances trust since it brings the expectation of the user in line with the actuality of machine-learning-based judgement.

Lastly, extensions and permissions should be done sparingly. Too many permissions may ruin user confidence and introduce unwarranted risk. The design must thus conform to the principle of least privilege, which asks only that access which is required by the desired phishing-analysis workflow. This technical decision can also be considered as a defense of the ethical stand of the project, considering that the system is designed to enhance trust and not introduce any more privacy issues.

2.3.6 Evaluation Meters and Technical feasibility.

Evaluation design is closely related to technical feasibility. Accuracy is not sufficient in phishing detection since phishing datasets can be skewed, and false negatives can be particularly expensive in security terms. Precision, recall and F1-score consequently gives a more balanced perspective of performance. Precision refers to the frequency with which the system is accurate in marking something as phishing. Recall is used to show the frequency with which the system is

able to detect phishing of all phishing samples. F1-score is particularly effective when the aim is to trade off the missed threats versus unreasonable false alarms.

Runtime behavior should also be considered under the umbrella of technical feasibility in the context of a browser extension. An offline model that runs too slowly to produce an answer might not be suitable to run on the user side. Consequently, the speed of inference, the cost of feature-extraction and the responsiveness of the interface ought to be considered as pragmatic evaluation dimensions alongside the traditional classification measures. This once again justifies the choice to concentrate on supervised-learning techniques that will allow a reasonable trade-off between performance and efficiency.

2.3.7 Technical Research Summary.

The technical research demonstrates that the suggested system can be constructed with the tools that are readily available and that are mostly free and still be in accordance with the academic and practical requirements. Python-based data-science tools assist with dataset preparation and supervised-learning evaluation, and standard browser-extension technologies assist with user-facing delivery within the browser. The hardware specifications are average, architecture can be incrementally scoped and chosen model families can be compared transparently. All in all, the technical environment confirms viability of the project and helps the transition between investigation to implementation.

2.3.8 Operational considerations and Security Controls.

The artifact, which is the proposed extension, is security sensitive in itself. Since it will interoperate with webpages, and possibly webmail content in restricted ways, it will need to be designed in such a way that its own behavior will not cause undue risk. Minimal-permission design is one of the controls. Only the permissions that are needed to support the features being implemented in the current iteration should be requested by the extension. In case the prototype just requires access to the active tab, the current URL, and the content scripts that were selected

on the relevant pages, the prototype should not demand unrelated broad permissions. This design option enhances trust and minimizes attack surface.

The second control is local-first control where it is feasible. In case the chosen model and feature set are not heavy, it is better to do local classification within the extension since it is necessary to transmit the content to other locations less frequently. Although there may be later architecture using a companion local service or lightweight endpoint, the principle is the same: any data transfer must be justified, minimized and documented. This is an operational consideration that is consistent with the ethical framing of this project and privacy expectations as indicated in the survey.

Transparent update and version management is a third control. The implementation phase might need to be refined to the new phishing patterns that are observed during the implementation. This must occur in a managed manner such that model changes are not made to silently change behaviors in an undocumented manner. Good version control and change logging also come in handy in the academic setting, when writing reports and during evaluation, since it enables the developer to provide an explanation of why a specific model or feature set was chosen, and how it changed over time.

False positives and false negatives also must be taken into consideration operationally by the extension. False negative exposes the user to risk and false positive will harm trust and reduce subsequent use. These are not just metric issues; they are operational behaviors, which influence adoption. This extension must then be tested not just in terms of statistical performance, but also in terms of its behavior in realistic browsing conditions. This is another justification why the system needs to be developed using Agile iterations as opposed to a single fixed build.

2.3.9 Next Development Phase Priority Interpretation.

With the help of the survey results, it is also possible to prioritize the development effort in a more strategic manner. The core site-warning workflow should be the initial development priority, since warning users against suspicious sites is the most directly requested feature, and the most naturally aligned to the browser-extension delivery model. Creating this initially enables the

developer to create the bare minimum of the protective loops that the system needs: inspect, classify, warn.

Lightweight email-linked phishing support should be the second priority. As a lot of the respondents have been suspicious of emails and they use email services on a daily basis, email-related detection cannot be delayed indefinitely. Nonetheless, it is to be applied cautiously and in a manner that does not violate privacy and technical capability. In the case of the prototype phase, this could involve concentrating on email indicators and suspicious links that can be seen in the browser as opposed to trying to conduct an actual analysis of the mailbox.

The warning explanation should be given the third priority. Even though the most demanded feature is not explanation, it is a very important element in trust. A warning should be based on the understanding that the user will take the warning more seriously and learn. Explanation must thus follow the fundamental warning cycle but preceded lower-priority facilities like more detailed reporting, customization, or sophisticated visual analytics.

The fourth priority must be enhancement features like report buttons, auto-block actions, richer dashboards, or longer logging. These are obviously good based on open ended responses, yet secondary to the basic security and usability cycle. Their treatment as subsequent concerns make the project realistic and contributes to ensuring that the initial working version of the project addresses the most valuable problem long before it starts growing to optional complexity.

CHAPTER 3: METHODOLOGY

3.1 System Development Methodology

Methodology chapter gives an account of the systematic approach adopted in the investigation and subsequent development stage of the project. Methodology is particularly significant in this report since the proposed system is neither a software interface project, nor a machine-learning project. It brings together research, data preparation, model evaluation, user-facing extension design, and refinement. An appropriate methodology should underpin experimentation and controlled development. When it becomes too rigid, the developer might not be able to refine features and warnings following evidence. When it becomes too loose, the project will lose focus, get bigger than it should, or make unjustified design decisions.

Methodology plays a two-fold role in this investigation report. First, it gives a rational explanation of how the developer will proceed with requirements and literature findings to a working browser-extension prototype in the later stage of the project. Second, it helps explain why a certain approach to development is more adequate than others based on the peculiarities of the project. Such attributes are changing demands, iterative model optimization, the centrality of user feedback, and the real-life difficulty of integrating a supervised-learning workflow with a Web-based interface.

To comply with the need for the investigation report guideline, this section will provide a comparison of the relevant software-development methodologies after which the most appropriate one will be chosen. It is not just a theoretical discussion. It also relates strengths and weaknesses of each of the methodologies to the real needs of the proposed phishing-detection extension. This analysis is then shifted to the activities and process of the chosen methodology in a way that the path of the research to the implementation is comprehensible and justified.

3.1.1 Importance of System Development Methodology

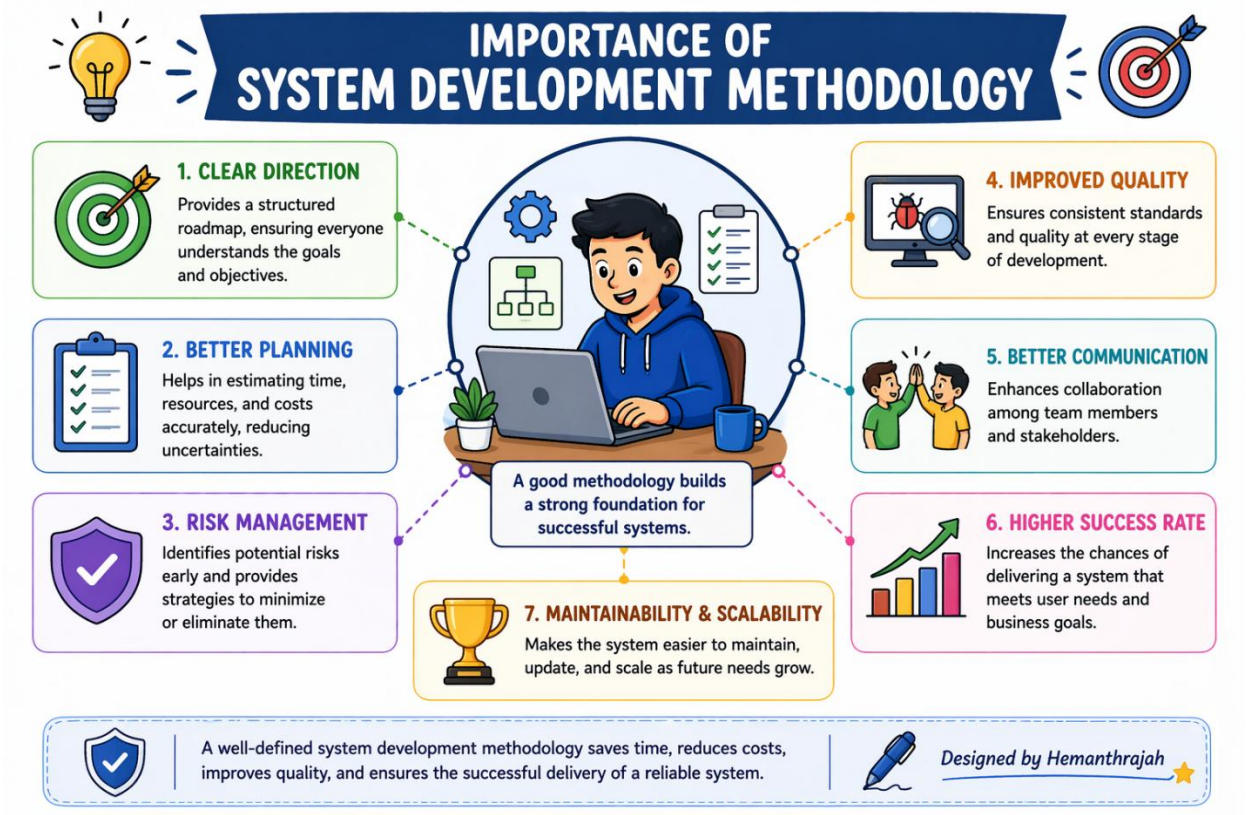


Figure 9: Key Reasons for the Importance of System Development Methodology

There is a set methodology that brings order. Structure is not a small administrative issue in a final-year project; it is the means of transforming a general concept into a series of deliverable tasks. The present project will involve literature review, survey design, requirement analysis, dataset work, feature engineering, model evaluation, browser-extension design, integration, and user-oriented refinement. In the absence of a methodology, such activities may end up being done in a haphazard sequence which may either result in overlapping of work or poor justification. A systematic approach minimizes that risk by defining clear stages, deliverables and check points.

Risk management is also supported by methodology. There are several types of risk in this project: dataset risk (when publicly available phishing datasets might be out of date or skewed); model risk (when a seemingly high-quality classifier does not generalize); usability risk (when warnings are obtrusive or too weak); and schedule risk (when time might be wasted unless experimental usage is constrained). The development approach must be appropriate in order to

expose these risks at an early stage as opposed to letting them occur until the implementation process is in its latter stages.

Another reason why methodology is important is quality assurance. Quality in a project such as this is multi-dimensional. It comprises technical quality, e.g. the relevance of features and the classification functionality; interface quality, e.g. comprehensibility of warnings and responsiveness; and documentation quality, e.g. traceable decisions and requirements that are well-defined. A methodology which promotes frequent inspection, small steps and correction through iteration is more likely to encourage these various types of quality than a methodology which does not provide feedback until the end.

Lastly, methodology aids in communication with stakeholders. Despite being an academic project, significant stakeholders, such as the supervisor, the project manager, and the users, who will be represented by survey responses, will be involved. The approach taken must enable the feedback with the said stakeholders to be easily integrated without leading to scope and direction chaos. Due to this fact, the methodology adopted by the project should be flexible and at the same time disciplined.

3.1.2 Methodology Options Considered

Agile, Rapid Application Development (RAD) and Kanban are the three methodologies that were taken into consideration in this project. They were chosen on the basis of their being iterative to a certain extent and as such, each of them commonly relates to projects where refinement and user input can be improved. Waterfall was thought of conceptually as a conventional baseline, but since the current project involves changing feature selection, experimentation of the model, and interface polishing, the primary comparison is between approaches that are more adaptive in real practice.

Agile is commonly linked to the short iterations, the frequent re-evaluation, and the constant stakeholder engagement. It is especially applicable to the projects that have requirements that can only be clarified by development, not by being fully stable. Agile would be appealing in

the context of this project since the performance of phishing detection, warning presentation, and extension behavior will tend to be enhanced by repeating the process of testing and revising.

RAD focuses on the speed of prototyping and speedy user-driven change cycles. Visual prototyping and immediate feedback are key to the project, and this makes it often useful. In the case of a browser extension, RAD seems appealing, as the interface concepts can be prototyped in a short time. Nevertheless, there is an element in the current project, machine-learning, which introduces an element of technical experimentation which does not necessarily lend itself well to speed. An approach which excessively emphasizes the importance of fast prototyping may generate pressure to make commitments to interface decisions before the classification pipeline has been fully justified.

Kanban on the other hand is particularly robust in visibility of tasks and management of workflow. It assists teams to visualize work, manage work in progress, and prevent bottlenecks. This may come in handy handling the small tasks in a project that is in the final year. But Kanban alone lacks the boundary of iteration and formal review moments as Agile has. In a project that requires a frequent check of the model outputs and warning messages to users, more powerful iteration cycles are useful.

3.1.3 Comparative Analysis of Agile, RAD, and Kanban

The primary advantage of Agile is that it does not consider change to be disruptive but expected. This is precisely the type of project which is helped by such an attitude. The developer can start with a set of planned URLS features and then find out in the course of testing that some of them are weak, redundant, or impractical in the extension workflow. Likewise, a style of alert that appears reasonable in theory might turn out to be obnoxious when a prototype is tried. The structure of iteration in Agile enables the absorption of such findings as productive rather than failures. The project is able to proceed through the planning cycles, building cycles, checking and refining cycles without losing the overall direction.

RAD does have its share of that flexibility, but the focus is altered. Its major advantage is speed. In the event that the primary issue of the project was to create and perfect an interface in

the shortest period of time, RAD could be the most appealing alternative. However, interface iteration is not the only thing needed in the current project. It involves evidence-based comparison of supervised-learning models, meticulous feature refinement, classification performance evaluation, and a rational tradeoff between performance and extension efficiency. These are iterative activities, but do not necessarily happen fast as required by RAD. A methodology that gives too much preference to visible prototypes at an early stage may downplay the quality of data and evaluation.

Kanban can be very useful in work visibility and management of the flow of tasks. It could be handy to use it to track literature-review activities, dataset-cleaning activities, documentation activities, and implementation milestones. However, a phishing-detection extension can be not only the problem of task management. It is an issue of staged learning and confirmation. The developer must have clear review points where the model family selected, the model interaction flow, and the warnings shown to the user are re-evaluated. Kanban can aid with that, but it lacks the iterative rhythm that Agile offers.

Recent comparison studies concerning software-development methods support this argument. Jain, Jotwani, and Agrawal (2024) highlight that Agile models are particularly effective when dealing with dynamic environments that need continual feedback and flexibility. Tetteh (2024) also emphasizes that various agile models are different in style yet have in common the suitability to projects where the requirements become clear as time goes by. Sitorus et al. (2025) compare Agile to more conservative methods and highlight the flexibility and speed of Agile. These are the observations that are highly consistent with the requirements of the current project.

According to the comparison, Agile is chosen as the main methodology of the project. RAD and Kanban still prove to be handy as points of reference: RAD can be used to understand why quick prototyping is important, and Kanban can be used to understand why visible task control is important. Nonetheless, Agile provides the most optimal combination of formal iteration, stakeholder, and maintainable refinement. This is why it is the most appropriate single methodology of the proposed phishing-detection browser extension.

Aspect	Agile	RAD	Kanban
Primary emphasis	Iterative development with structured review cycles	Rapid prototyping and quick user feedback	Workflow visibility and limiting work in progress
Strength for this project	Supports model tuning, interface refinement, and requirement evolution	Useful for fast interface mock-ups	Useful for tracking many concurrent tasks
Main weakness for this project	Requires discipline in iteration planning	May encourage interface speed over careful ML evaluation	Provides weaker iteration structure for validation-heavy work
Fit with phishing detection model development	High	Medium	Medium
Fit with browser extension refinement	High	High	Medium
Overall suitability	Selected methodology	Useful comparison method	Useful supporting management perspective

Table 3: Comparison of Agile, RAD, and Kanban

3.1.4 Justification for Selecting Agile

Agile is chosen since the suggested system is almost definitely going to change as it is developed. The combination of specific features to be used in detecting websites and emails might be modified after the inspection of the data sets. What might seem like a strong model initially might not be the strongest when one considers runtime speed, browser integration, and false-positive behavior. Similarly, extension interface might require multiple iterations to achieve the balance between visibility and simplicity that was observed in the results of the survey. Agile enables such revisions to take place within a rigorous process instead of being an unplanned shock.

The other reason why Agile was selected is that it is suitable for the structure of the project outputs. The extension does not have to appear in a comprehensive step. Rather, the project will

be able to proceed in small steps. A prototype can be early with a view to dataset preparation and a baseline model. An extension in a later form can result in a plain warning-only extension. Further iterations can enhance message clarity, add explanation text, or optimize threshold selections. This is a more realistic route than trying to design and write all the components in a sequence with no iterative review.

Agile is also compatible with the existing evidence base. According to the survey, speed, clarity and ease of use are highly valued by the users. Such factors cannot be optimized in a single run. Agile will allow the developer to ensure that the styles of warnings are always what the users want, and the developer can make corrections to the extension. Agile in this sense not only helps with technical development, but also user-centered design.

3.1.5 Agile Process and activities in the Selected Agile approach.

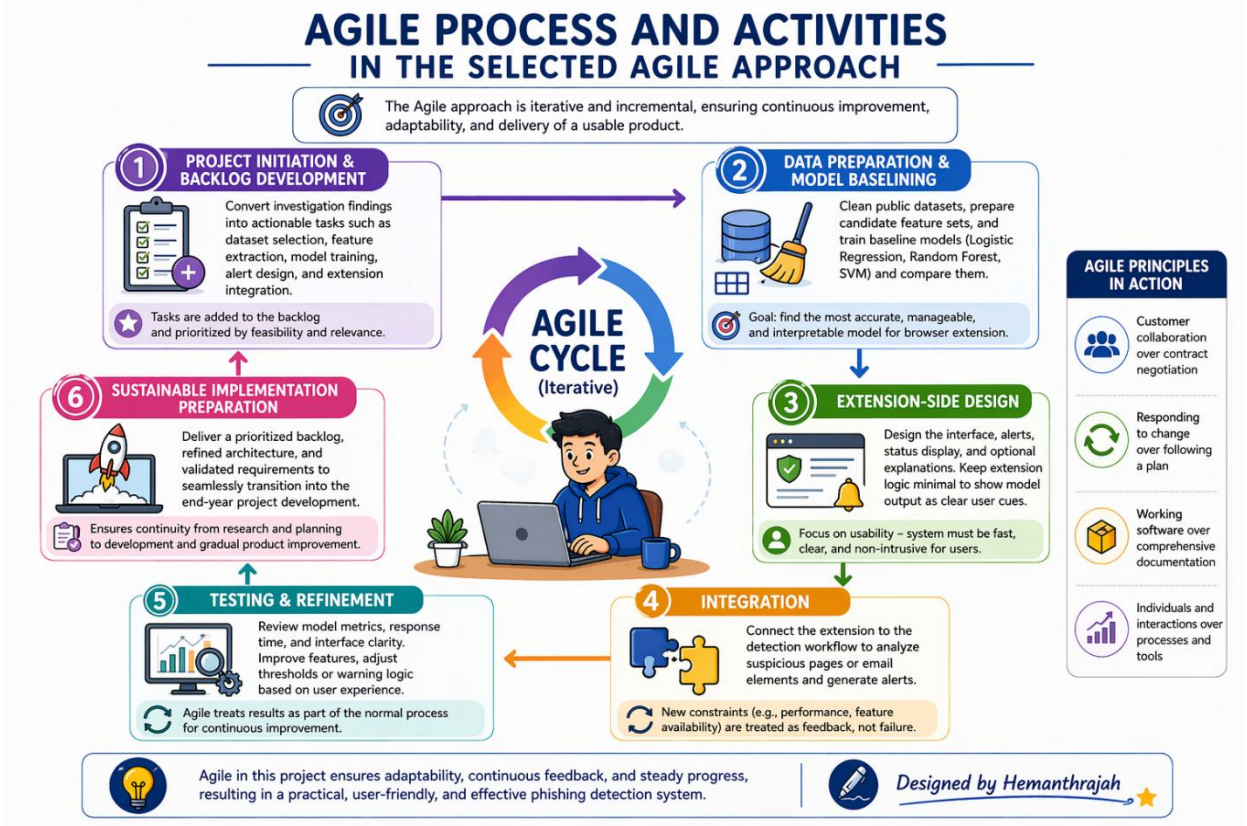


Figure 10: Graphical Representation of the Agile Process and Activities in the Selected Agile Approach

The Agile process to be used in this project is structured into iterative cycles, each having a definite purpose and a manageable output. The initial cycle is the project initiation and backlog development. At this phase, the developer will convert the results of the investigation report to actionable tasks, including selecting datasets, extracting features, training a baseline model, designing the alert, and integrating extensions tasks. Such items are the project backlog and are ranked based on feasibility and relevance.

The second cycle is on data preparation and model baselining. In this cycle, the public datasets are cleaned, candidate feature sets are made ready and basic models like Logistic Regression, Random Forest, and Support Vector Machine are trained and compared. The objective is to not only discover the most accurate one, but also to discover the models that are manageable as well as interpretable enough to allow the deployment of browser-extensions.

The third cycle works on extension-side design. In this case the interface design is written, the alert message, the status display, and any optional explanation display. The logic behind the interaction of the extension is maintained to a bare minimum to allow the developer to test the interaction of model output into a user-facing cue. The importance of this cycle is particularly great due to the results of the surveys that indicated that the system may be technically correct but still unacceptable in case it is slow, confusing, or visually intrusive.

Integration is in the fourth cycle. Here, the extension is connected to the chosen detection workflow, in which case suspicious pages or the elements connected to the email can be processed and identified. Integration is likely to create new constraints, such as about runtime performance or feature availability. These limitations are perceived as feedback and not failure in Agile. They update the subsequent refinement cycle.

Testing and refinement is the fifth cycle. Review of model metrics, response time, and interface clarity. In case of poor precision, feature engineering can be reconsidered. Thresholds or warning logic can be modified in case the users are likely to be annoyed by a pattern of pop-ups. Since Agile considers each of these results as the normal process, the developer is able to continuously improve the system without losing the project coherence.

The last Agile cycle within the context of this investigation report is sustainable implementation preparation. Practically, this involves bringing on a prioritized backlog, a more

sensible architecture and a proven requirement list into the subsequent phase of the end-year project. Agile is useful in this case as it is not based on the idea that the investigation report is the final stage; it can help to maintain the flow of research and planning into development and the gradual improvement of the product.

3.1.6 Recommended Agile Iteration Roadmap in the Next Phase.

The second stage of the project can be divided into a series of small Agile iterations to make the chosen methodology operational. Iteration 1 is aimed at data check and feature-baseline preparation. The objective is to make sure that the web site and email data sets are clean, labeled and can be used with the supervised-learning workflow envisaged in the project. By the end of this iteration, the developer ought to have a working data-preparation notebook or script, a documented feature inventory, and a collection of baseline assumptions that can be taken into model training.

Iteration 2: This iteration should be based on comparing the baseline models. The Logistic Regression, Random Forest, and Support Vector Machine may be trained on the chosen sets of features and compared in terms of precision, recall, F1-score, confusion matrices, and the cost of running. This iteration is not only aimed at finding the most numerically robust model but the ones that would be viable to be deployed as eventual browser extensions. Instead of a more complex model, a slightly less accurate one might be better, provided it is much easier to interpret.

Iteration 3 needs to be devoted to prototyping a warning interface. The survey indicates that users would like simple color coding, direct pop-up warnings and minimal friction, although they would appreciate explanation as well. Various interface designs may be tested in a low-fidelity or mid-fidelity manner during this cycle. The goal is to determine which information is to be included in the initial alert, what must be kept in the background of optional information, and how the extension can be visually incorporated into standard browsing, without being a distraction or aesthetically incongruous.

Iteration 4 will be on extension integration. The selected model workflow is linked with the browser environment in this cycle to enable a page or email-related context to be processed and the classification outcome presented. It is at this stage of integration that performance

constraints are observed to be very visible. Some features will probably have to be cut, postponed or simplified during this step. Agile is particularly useful in this context as it enables the developer to make use of these arising constraints in the regular process of improvement.

Iteration 5 will deal with refinement by evaluation. The developer is able to revisit the results of the survey as a usability standard: is the extension fast enough, simple enough, and visible enough? Were the warning messages appropriate to the alert style that the users liked? Do we have any explanation layers to give more context to the users? With a complete Agile cycle, every iteration is not only completed with a code or analysis, but with learning. The outcome is an evidence-based development path as opposed to a feature-driven development path.

3.2 Data Gathering Design

Information collection is required since the proposed system will be used by actual users as opposed to a purely theoretical demonstration. The developer thus requires evidence concerning the frequency with which the target audience uses browsers and email, what indicators they already have of phishing, how confident they are to identify suspicious material, and what sort of extension interface they would like actually. The aim of the data collection process is not to demonstrate that phishing is in the abstract. Rather, it is to determine palpable design anticipations that can mold the planned extension in a user-friendly manner.

The investigation-report guideline permits a number of procedures including surveys, interviews, reading of documents, and observation. In the case of the current project, the main data-collection tool is a secret online survey, which is complemented by an analysis of documents and current literature and official reports. Such combination is suitable since the survey will provide the direct user opinions, and the analysis of the documents will enhance the academic and technical foundation of the system design. Data collection via interviews was also a possibility but the survey approach was chosen as it provided a wider scope of coverage of the target population given the limited time and ethics involved.

3.2.1 Target Population and Sampling Method.

This study has a target audience of individuals who regularly use browsers and email services. This covers university students, working adults and ordinary users who are likely to come across phishing attempts within their normal academic, personal or work settings. Students are particularly applicable since they come into contact with the login pages, university portals, cloud solutions, and email-based messages regularly. Adults who work are also applicable since phishing is often targeted at business email, financial dealings, and daily web services.

The survey employed a convenient sampling method; a practical convenience-based sampling and expansive sharing on the internet. The link of the survey was given out via direct contacts and free online. Even though it is not a probability sample, it can be used in an undergraduate investigation report in which the aim is to collect requirements as opposed to making inferences about a large population. The 44 responses gathered are more than the minimum number of participants required according to the guideline and offers sufficient range in terms of age, occupation, and daily use of technology to facilitate requirement analysis.

The survey was not aimed at gathering sensitive identifying information. No names or student IDs, no personal email addresses, no passwords were requested. This design was done intentionally since the subject matter itself is related to phishing and trust. It would have been unethical to request unneeded personal information, and this would be contrary to the spirit of good ethics and the targeted privacy-focused nature of the project.

3.2.2 Survey Instrument Design

The questionnaire was structured into parts to enable the results to be viewed progressively. The initial part entails demographic data, including age bracket, gender, and occupation. The second one looks at the frequency of technology use, i.e. email and site use, as they influence phishing exposure. The third examines the awareness and experience, such as whether participants have heard of phishing websites and phishing emails, whether they have come across suspicious content and how confident they are about it. The fourth one is devoted to the expectations of the

offered browser extension, such as the preferred types of warnings, desired capabilities, the significance of response speed, and readiness to use the system. The last part is an invitation to feedback and improvement suggestions which are open-ended.

This design was selected as it reflects the reasoning behind requirement identification. The first step of determining how a browser extension should perform is to determine who the probable users are, how vulnerable they are to the issue, what indicators they already depend on, and what form of intervention they would be willing to receive. The combination of closed-ended and open-ended questions is in favor of the quantitative and qualitative interpretation. Through multiple-choice and Likert-style questions, clear tendencies can be measured, whereas the open-ended feedback questions can pick up ideas which are not reflected by the fixed alternatives.

The phrasing of questions was maintained to be simple and not elaborate to ensure that it does not bar other respondents that are not experts in the field of cybersecurity. This was necessary since the proposed extension is not targeting the experts only. The sample could have been skewed towards individuals possessing above-average security knowledge by using a very technical questionnaire and thus diminished the applicability of the results to real-life users.

3.2.3 Data Collection Procedure

The survey was shared online and remained open on 15 April 2026-17 April 2026. Participants were free to fill out the form during this time, read the ethics notice, give consent, and take their time to fill out the questionnaire. Since it was an online form, respondents were able to answer at any point of location thereby eliminating participation barriers and enhancing convenience. This strategy was particularly appropriate when it came to dealing with a project that dealt with the use of the web and email since it aligned with the digital lifestyles of the target market.

The importance of confidentiality was highlighted during the introduction of the form. The respondents were told that the continued participation was at their own will and that no personal identity was going to be reported in the investigation report and that the responses would not be utilized in any personal capacity. The shape had a contact email to provide feedback or enquiries.

This design facilitated the development of respondent trust and fit the data-collection approach with the privacy sensitive theme of the project itself.

After the response period was over the dataset was exported to spreadsheet to be analyzed. The cleaning of responses was then performed by verifying incomplete responses, duplicated consent responses and empty-value patterns. The last dataset consisted of 44 valid responses. This data is used in the analysis in Section 3.3.

3.2.4 Reason behind the selection of Survey Method.

There are a number of reasons why the survey method is justified. To begin with, it is efficient. It gives the opportunity to gather data on a greater number of people than would otherwise be possible with interviews within the same time frame. Second, it yields structured data which can be summarized easily to be analyzed during requirement analysis. Third, it will be appropriate for a project where the target users will be familiar with working with online forms and browser-based systems. This minimizes grit in engagement and enhances practical suitability.

Another benefit of the survey is that it facilitates breadth and comparability. By having all the respondents respond to the same set of core questions, the developer is able to compare the degrees of awareness, confidence and preference across the sample in a uniform manner. This helps in identifying those system requirements that are widely shared and those that are more optional or secondary. As an illustration, when the majority of the respondents consider speed and ease of use to be very important, they become obvious non-functional priorities.

Observation is also inappropriate as compared to the survey method in the given project. This would ethically be difficult and possibly unrealistic without putting participants in contrived circumstances in order to observe phishing behavior directly. Similarly, interviews would be richly detailed but not as broad. The survey thus presents the most viable balance of practicality, ethics and usefulness to the current investigation stage.

3.2.5 Supporting Document Analysis

Besides the survey, document analysis was employed to enhance the study. It consisted in the analysis of recent journal articles, conference papers, and official reports about the trends in phishing, phishing website detection, phishing email detection, browser extension design, and supervised learning. The purpose of document analysis is different compared to the survey. Whereas the survey will determine what is demanded and expected by the user, document analysis will determine what is supported by the current state of research both technically and conceptually.

The approach is especially significant since phishing detection is a dynamic sphere. Limiting the primary literature base to the most recent literature published after 2022 ensures that the report is not overly dependent on old assumptions and sets the technical direction of the project in the same direction as the current challenges: dataset drift, explainable feature selection, generalization, and browser-side delivery. Document analysis can thus make sure that the suggested system is based on the feedback of the users and on the existing scholarly evidence.

3.2.6 Ethical Considerations

Ethics were significant during the design of the survey and the framing of the project in general. Respondents had been assured that their answers would be utilized in an academic way and that they could withdraw their answers at any time by merely failing to fill in the form. There were no false prompts, fake phishing, or gathering of sensitive security credentials. These limitations were required because the project was not supposed to add more risk to the study of a topic of security sensitivity.

The technical orientation of the project is also ethics conscious. Since the system is designed to monitor potentially sensitive browsing or email-related content, the proposed design focuses on the minimum access that is required, the absence of any unwarranted retention of personal data, and disclosure of what is being analyzed. These principles are not yet a complete implementation policy but already form part of the requirement and architecture rationale developed by the investigation report.

3.3 Analysis

The questionnaire resulted in 44 valid answers and offers the empirical foundation of the requirement analysis of the suggested browser extension. The discussion herein is deliberately elaborated since the purpose of the investigation report is not to provide a description of statistics, but to provide an explanation of the meaning of the findings on the system design. All questions are thus addressed both in terms of the number they will give and their applicability to the offered extension.

The answers are deciphered in the identical general sequence as the questionnaire: demographics, awareness and experience, and the expectations of the proposed solution. This order is helpful since it introduces respondents, then demonstrates their level of exposure to phishing and their confidence in it and finally discloses to the reader the features and modes of interaction they expect a browser extension to have. The development of background design implications also assists in developing the survey results to a final list of requirements at the end of the chapter.

Demographic item	Most frequent response	Interpretive note
Age group	18-24 (63.6%)	Majority are digitally active young adults and likely frequent browsers and email users.
Gender	Male 56.8%, Female 43.2%	Relatively balanced samples reduce single-group bias.
Occupation	Student (54.5%)	Strong relevance to academic browsing and email contexts.
Email use	Daily (68.2%)	Supports strong need for email-related phishing awareness.
Website use	Daily (81.8%)	Strong justification for a browser-based delivery model.

Table 4: Summary of Survey Demographic Results

Indicator	Key result	Implication
Heard of phishing websites	86.4% yes	Awareness is high but not universal.
Heard of phishing emails	86.4% yes	Supports combined website and email focus.
Received suspicious email	75.0% yes	Email is a major practical attack channel.
Visited suspicious website	72.7% yes	Browser-side warnings are highly relevant.
Clicked suspicious link by mistake	52.3% yes	Awareness does not fully prevent risky behavior.
Average seriousness rating	4.64 / 5	Respondents see phishing as a serious current threat.

Table 5: Phishing Awareness and Exposure Summary

Indicator	Selections	Interpretation
Suspicious URL or website address	38 (86.4%)	Strongly recognized as a phishing cue.
Requests for passwords or banking details	37 (84.1%)	Strongly recognized as a phishing cue.
Unknown sender email address	34 (77.3%)	Strongly recognized as a phishing cue.
Unexpected attachments or links	33 (75.0%)	Strongly recognized as a phishing cue.
Offers that seem too good to be true	30 (68.2%)	Recognized by many users but not universally.
Urgent or threatening language	29 (65.9%)	Recognized by many users but not universally.
Poor grammar or spelling mistakes	26 (59.1%)	Useful cue, but more dependent on user judgement.

Table 6: Preferred Indicators of Phishing

Feature / Preference	Result	Design implication
Extension usefulness	95.5% Yes	Strong justification for the project concept.
Most useful feature	Warning message when a suspicious website is detected (52.3%)	Direct intervention is preferred.
Preferred alert style	Pop-up warning message (47.7%)	Warnings should be visible and immediate.
Importance of speed	Average 4.64 / 5	Inference must be fast and lightweight.
Importance of ease of use	Average 4.52 / 5	Interface simplicity is essential.
Willingness to use extension	77.3% Yes, 22.7% Maybe	Adoption potential is high if trust and usability are maintained.

Table 7: Desired Browser Extension Features

Question 1 – Participant Consent

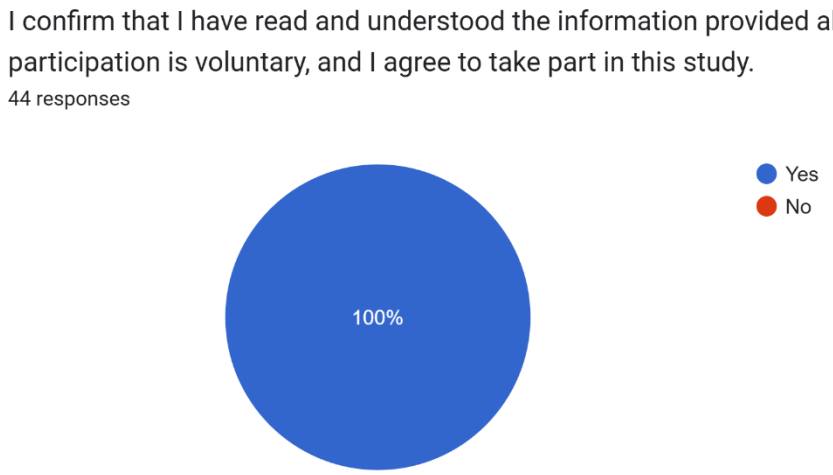


Figure 11: Participant Consent

Valid consent was present in all 44 responses that were given. It is a mere statistical finding, but a methodological point. It implies that all the analyzed entries were the result of a participant who read the introductory information and was clearly willing to participate in the study. There were no missing consent responses that would have to be disqualified, and this reinforces the ethical soundness of the data and eliminates uncertainty as to whether the results can be legitimately applied in the investigation report.

Regarding the research-design approach, the unanimous consent outcome also indicates that the front page of the form was not too complicated and comprehensible. This is important since a confusing consent statement may deter participation or may result in low consistency in the quality of response. The transparency of the consent process is critical in this project since the subject of the survey itself is trust, phishing, and suspicious online behavior. A properly designed consent step thus helps the credibility of the survey process and the project in general.

Question 2 – Age Group

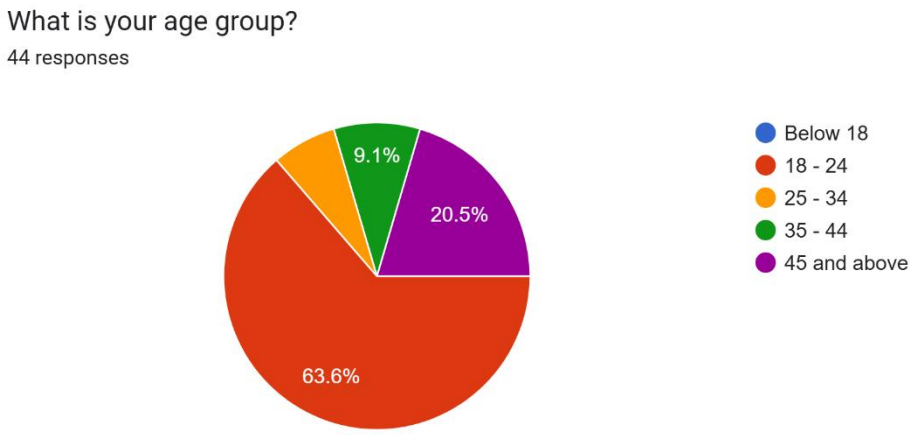


Figure 12: Age Group

The age group distribution reveals that 18 -24 is the most prevalent with 28 representing 63.6% of the sample. Smaller subsets consist of respondents that are between 25 and 34, between 35 and 44 and above. This composition can help the project as it would provide good representation to younger digitally active users but also include older adults whose browsing behavior and level of caution might vary.

The preponderance of younger respondents would indicate that the extension would be effective in high-frequency, high-paced browsing where convenience is important. Meanwhile, the fact that older respondents are there implies that the design cannot presuppose extremely technical instincts. An overly subtle warning interface can be overlooked by users in a hurry, whereas an overly technical warning can be misinterpreted by less security-confident users. The age distribution hence justifies the requirement of a design that is both immediate and comprehensible among mixed user groups.

Question 3 – Gender

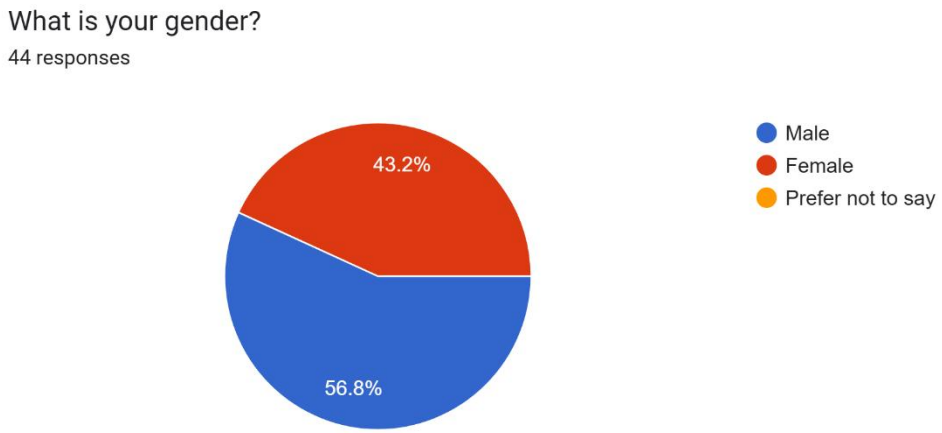


Figure 13: Gender

Gender distribution is fairly even, and the number of male respondents is 25 (56.8%), and the number of female respondents is 19 (43.2%). The sample is not exactly balanced, but it is not skewed to the extent that one gender viewpoint takes an overpowering position in the research. This enhances the usefulness of the survey since the proposed extension is to be used by the general population and not by a very specific demographic.

This balanced pattern, in terms of requirement-analysis, minimizes the probability of the project inheriting preferences which are highly skewed toward a single respondent segment. Rather, the usefulness, warning style, speed, and ease of use outcomes may be viewed as widely applicable design indicators. The gender finding is not then a design requirement, but it enhances the degree of confidence in the representativeness of the overall preference patterns that will be found later in the survey.

Question 4 – Occupation

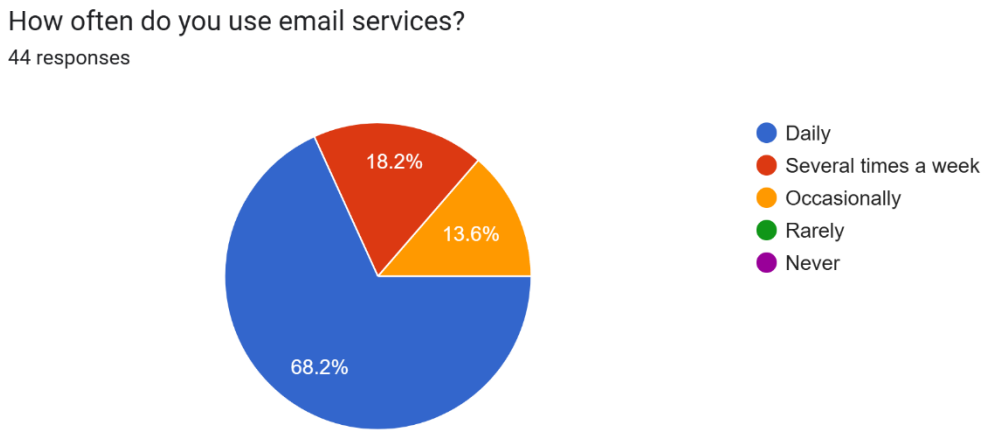


Figure 14: Occupation

Occupation results indicate that students represent the biggest group (24 respondents 54.5%). The number of employed respondents is also a significant population, with self-employed and unemployed respondents having smaller but significant percentages. Such a distribution is feasible since it is appropriate in the desired breadth of the project: it is very relevant to students, but not to them only.

The practical implications of the occupational diversity are practical. Phishing may affect students through business email, finance-related messages, and workflow tools as well as employed users through contact with university portals, cloud services, and links to shared documents. An extension that is too specific to context might not work in the other context. The survey thus recommends a solution that is generic enough to fit in the various real-world environments yet at the same time is particularly applicable to the student users.

Question 5- Email Usage Frequency.

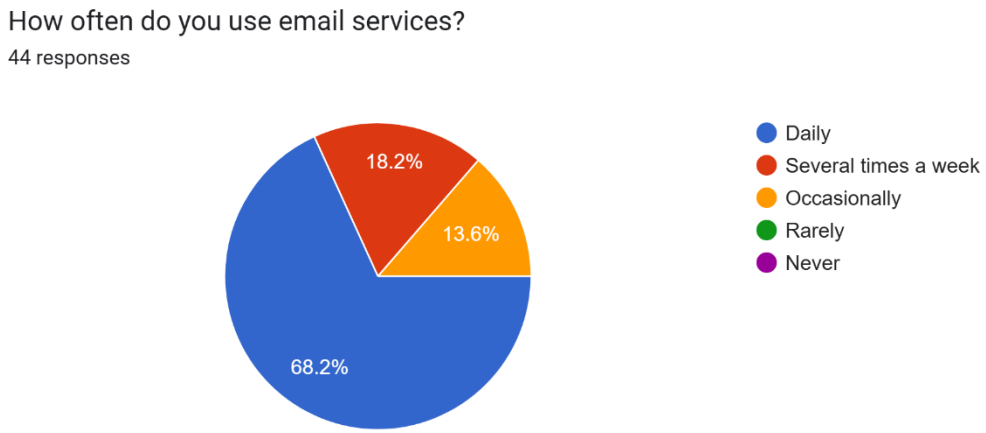


Figure 15: Email Usage Frequency

There is a high frequency of email-use in the sample. The number of respondents who use email daily was 30 (68.2%), and the rest used email several times a week or sometimes. None of the respondents said that he or she never used email. This is a good reason why email-related phishing cues should become a part of the project. In case the target audience uses email regularly, then email can be one of the most plausible locations of the phishing exposure.

The finding also indicates that email support cannot be considered as a side-by-side feature in the proposed extension. Although the webmail-visible content and link-based cues implemented on the browser-side may be more important than the actual inbox analysis, the email threat pathway is obviously an element of the daily experience of the user. Practically speaking, an extension that will examine regular sites but not be concerned with suspicious email-related transactions will create a significant vulnerability.

Question 6 - Frequency of Web Site Browsing.

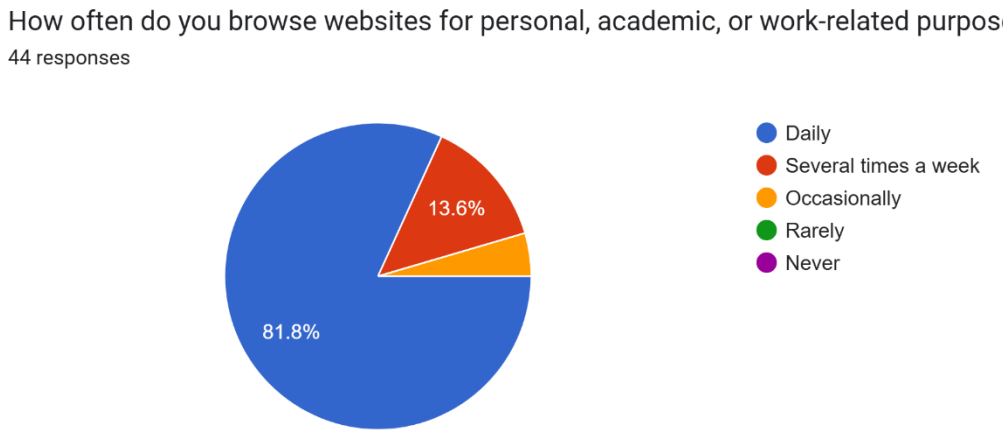


Figure 16: Frequency of Web Site Browsing

Browsing the web is even more common than emailing. Those who have been browsing on a daily basis were 36 respondents (81.8%), and almost all the others browse a few times a week or sometimes. This outcome presents one of the most illustrative reasons as to why the project should be delivered in this model. Since the browser is heavily used by the target population, it is a suitable location to place phishing protection.

The outcome also validates the significance of performance as a prerequisite. Slow, unstable, and disruptive extensions will soon become annoying when users browse on a daily basis and probably on a large number of pages. The fact that the browsing frequency is high thus lends more weight to the fact that the adopted supervised-learning method should be coupled with an efficient inference and a low-friction interface.

Question 7 -Phishing Website Awareness.

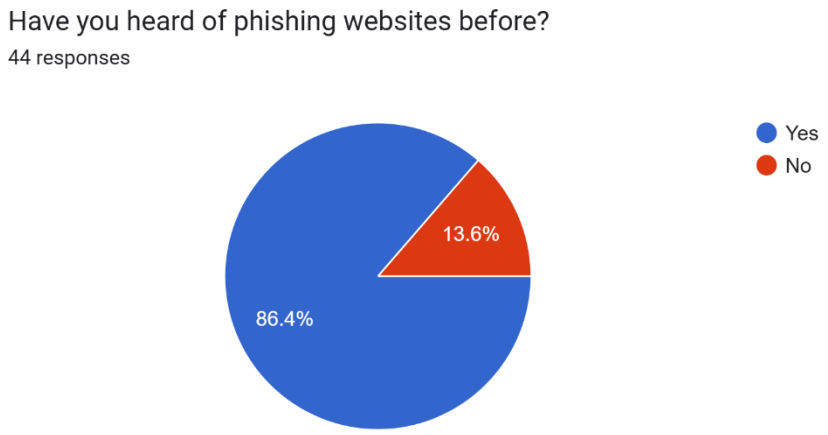


Figure 17: Phishing Website Awareness

Phishing websites awareness is quite good, with 38 respondents (86.4%) saying that they have heard of phishing websites. Nevertheless, there were a fewer number of people who responded with 'No'. This is a significant gap. It demonstrates that general awareness is not universal even in a digitally active sample, although it is very high. It should therefore not be assumed that all users have a common security vocabulary/experience on which the project is based.

Design wise this outcome favors the introduction of a text that is explicit and not too technical as a warning. In case a part of users still does not understand the terminology of phishing, then the extension should inform about danger in easily understandable language. As an illustration, a message that describes that a web page is purporting to be a legitimate site would probably be more efficient than a message that utilizes only the language of specialists or numeric model outputs.

Question 8 - Phishing Emails Awareness.

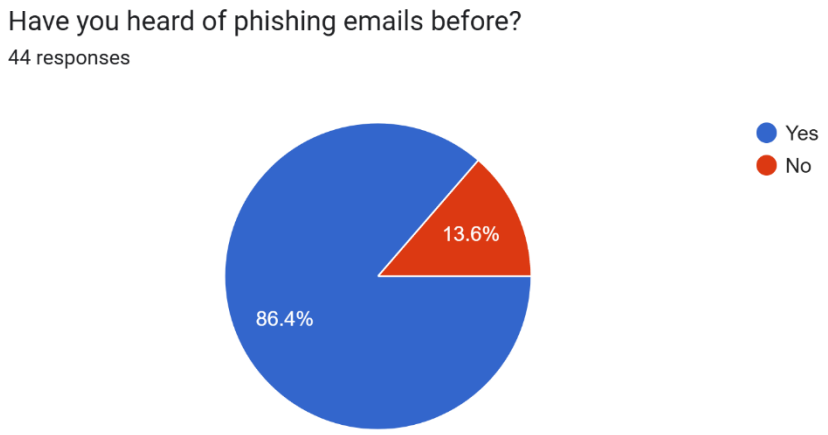


Figure 18: Phishing Emails Awareness

The level of awareness of phishing emails is almost similar to that of websites awareness, with 38 respondents (86.4%) reporting that they have heard about phishing emails. This similar finding indicates that respondents are more likely to acknowledge phishing as a general threat that is present on the internet instead of a channel-specific threat. Awareness, however, does not necessarily mean the ability to identify the less obvious or contemporary forms.

Project implication is that awareness should be supported by the system. Even when the user is aware of the existence of phishing, he or she may not evaluate a particular message or link appropriately in a realistic browsing situation. This is why the extension can be considered useful even to respondents, who already realize the overall idea of phishing.

Question 9 - Generalized knowledge of Phishing Attacks.

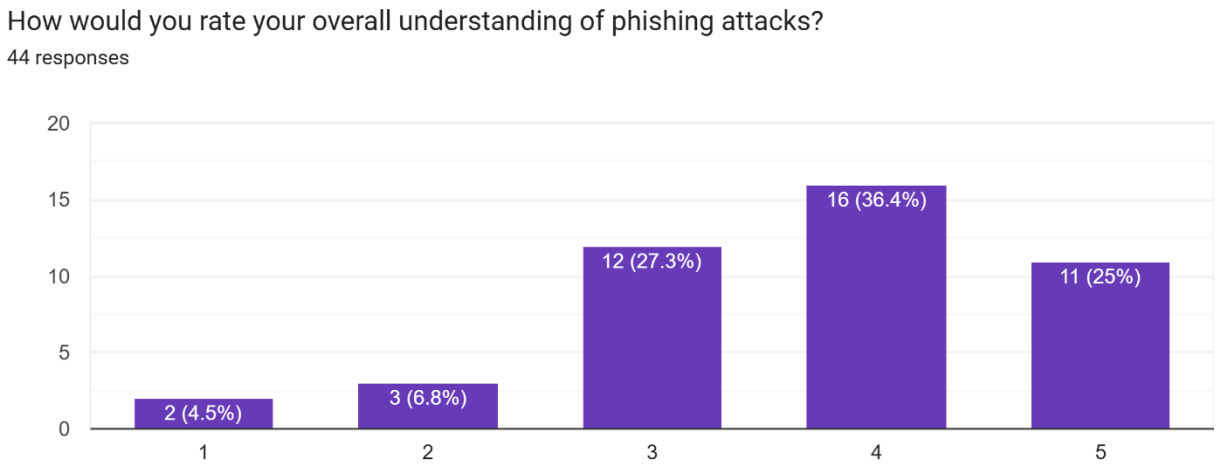


Figure 19: Generalized knowledge of Phishing Attacks

The overall self-rated level of knowledge of phishing attacks is 3.70 out of 5 with the majority of the answers being 3, 4 and 5. This shows that there is moderate to high perceived knowledge but not universal confidence. Proliferation is practical since it represents a realistic base of users: neither completely ignorant nor knowledgeable. This type of user base is just the type of audience that can use supportive detection tools.

Awareness and understanding are also a difference here. Some respondents might know about phishing and yet be confused about what to make of the suspicious links, cloned web sites, or persuasive wording. It is at this point that the explanatory role of the extension plays a significant role. The system need not simply tell the user that something is suspicious but should tell the user why they are suspicious, particularly in cases where the user is not a knowledge expert but is merely an intermediate knowledge user.

Question 10 – Suspicious emails.



Figure 20: Suspicious emails

Suspicious email experiences are common. 33 (75.0) respondents answered yes to the question that they have been asked to provide personal, login or financial information in a suspicious email. It is among the most formidable practical evidence in the survey since it indicates that phishing exposure is not hypothetical among the sample. It has already become a part of the online experience of the respondents.

This discovery hugely supports the rationale behind the project. The extension that is able to assist with phishing indicators in emails is a responding solution to a widespread, not a niche issue. It also implying that the value of the extension can be judged not only by the performance of the abstract model, but by whether the extension can be able to assist the respondents in case they encounter a suspicious email in practice.

Question 11 - Vexed Suspicion of Web.

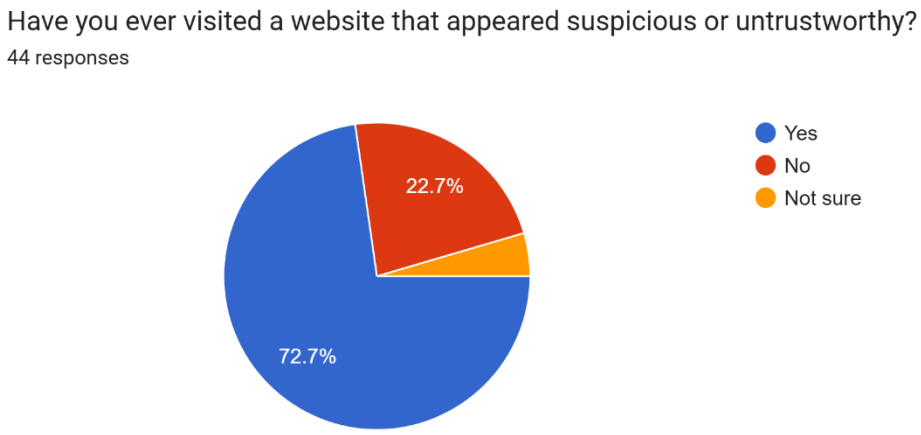


Figure 21: Vexed Suspicion of Web

The exposure to suspicious sites is also prevalent. 32 (72.7%) out of the 45 respondents reported having visited a site that looked suspicious or untrustworthy, the rest reported no or they were not certain. It is particularly telling that there is some ambiguity in certain answers. It indicates that users cannot necessarily know whether their suspicion is warranted, creating an opportunity to make decision-support tools.

This question hence greatly supports the browser-extension model. When users often visit dubious sites and are not necessarily sure how to read them, an in-context warning system can be useful. The extension is not only aimed at substituting judgement but empowering it in the most vulnerable moment the user could be deceived at.

Question 12 - Trust in Self-Detection.

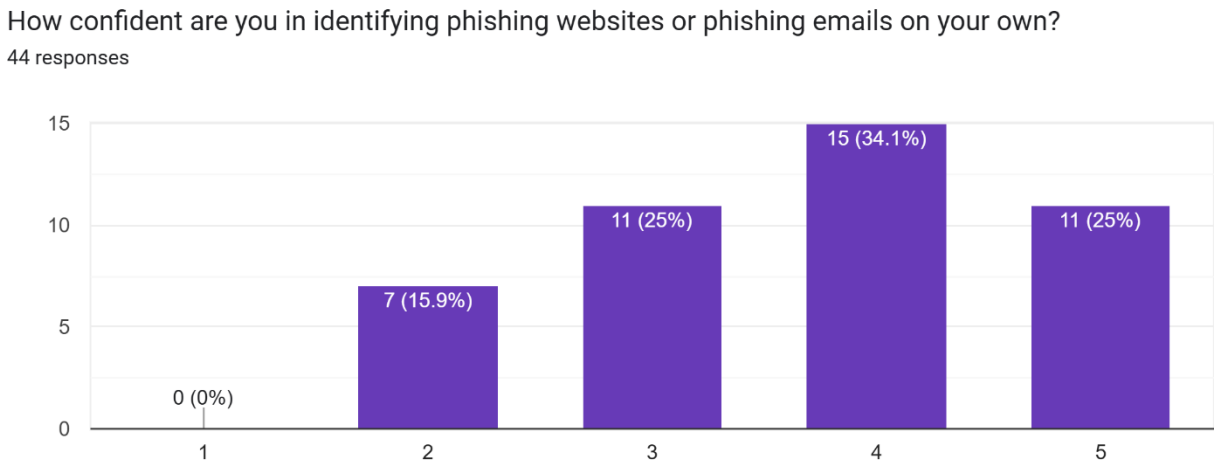


Figure 22: Trust in Self-Detection

Self-detection of phishing has an average score of 3.68 out of 5. The distribution indicates that a substantial percentage of the respondents feel moderately confident, and the scores are not all at the upper end of the scale. Such an outcome is beneficial due to the disclosure of a realistic tension: users tend to believe that they can be aware of phishing, yet this is not always the case. In the context of cybersecurity, biased trust can be harmful when it results in a decreased level of caution.

The question is further amplified when it is taken along with Question 10, 11 and 15. Users have moderate confidence, yet, they also have a high exposure to suspicious content, and a large percentage of inadvertent clicks. Collectively, these findings indicate that perceived competence does not eradicate real risk. The extension is thus warranted as a supplementary layer even with those users who feel that they are fairly competent individually.

Question 13 - Signs That Users Can be Suspicious of Phishing.

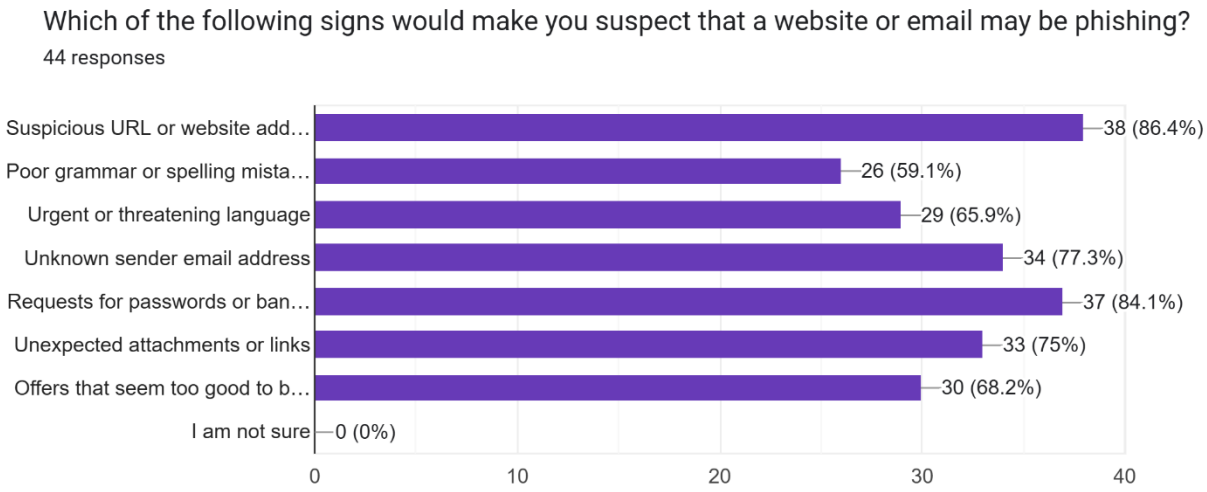


Figure 23: Signs That Users Can be Suspicious of Phishing

The cues that prompt them to suspect phishing were varied, though suspicious URLs, requests of passwords or banking information, unfamiliar sender addresses and unanticipated attachments or links were particularly highly chosen by the respondents. This trend indicates that users depend on a set of indicators other than a single indicator. That is a great design lesson. When it is already a multi-factor user judgement, then the extension should also be multi-factor in that it should also combine various types of signals instead of being based on a simplistic rule.

The high preference of URL-related cues, as well as credential-related cues is specifically pertinent to the proposed project. They are signals which could be frequently processed effectively and converted to warnings which users already know. Meanwhile, the moderate rate of grammar-related cues selection implies that even old stereotypes concerning the obviously poor phishing writing cannot be used independently anymore. The extension must thus be able to detect classic cues as well as subtle structural patterns.

Question 14- largest perceived risk of phishing.

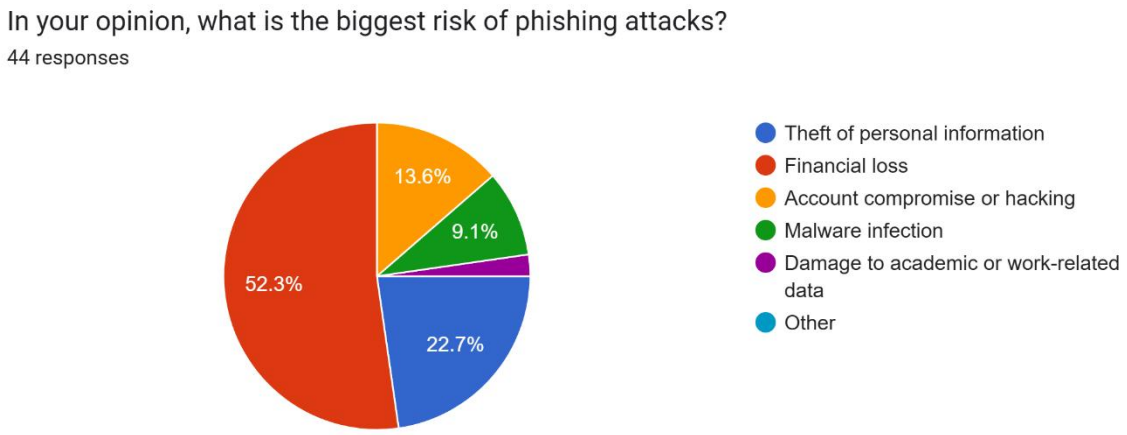


Figure 24: Largest perceived risk of phishing

In a question on the greatest risk of phishing, the greatest number of responses were on financial loss. Other reactions were stealing personal information, compromising the account, malware, and destroying academic or work data. This demonstrates that the users understand phishing as a risk of an immediate threat of harm to their fraud, but they are also aware of the impacts going into privacy, account management, and operational interference.

The significance of this finding is that it affects the motivation of users. When individuals perceive the possibility of a significant personal loss because of the underlying threat, they tend to appreciate a protection tool. With a properly designed extension, adoption barriers can be reduced, as the respondents already associate phishing with serious consequences. Protective value should consequently be communicated in the project, since the users are already aware of the severity of the possible consequences.

Question 15 -Mistakenly Clicking Suspicious Links.

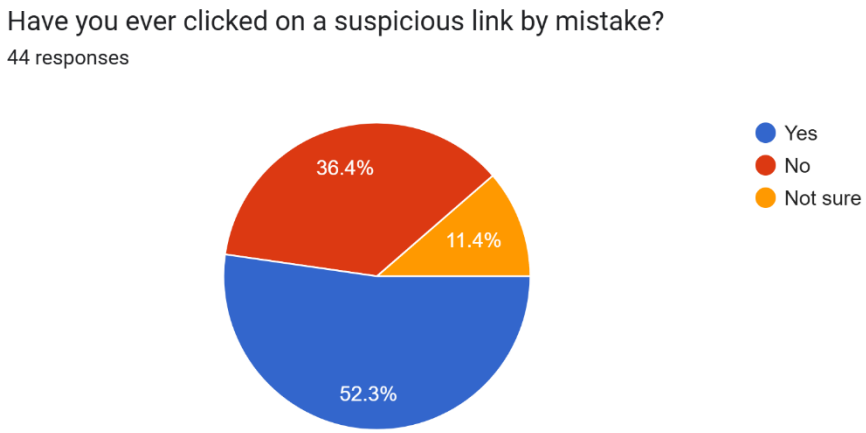


Figure 25: Mistakenly Clicking Suspicious Links

More than half of the respondents (23 respondents, 52.3%) admitted that they had clicked on a suspicious link by mistake. This has been found to be one of the most revealing in the whole survey. It demonstrates that the knowledge of phishing does not always protect against unsafe behavior. The real-world decisions are made fast, and in some cases, they may be distracted or even under extreme pressure but even the users of the theory are still likely to make a mistake in practice.

This result provides very strong support for the project. The extension is not there since the users have no idea that phishing exists; the extension is there since there is no use of awareness only. The system can interrupt risky behavior at the point in the browsing workflow when human judgement is most likely to be pressured, routine or overconfident by making warnings part of the browsing experience.

Question 16 - Soberness of Phishing in the present Digital Age.

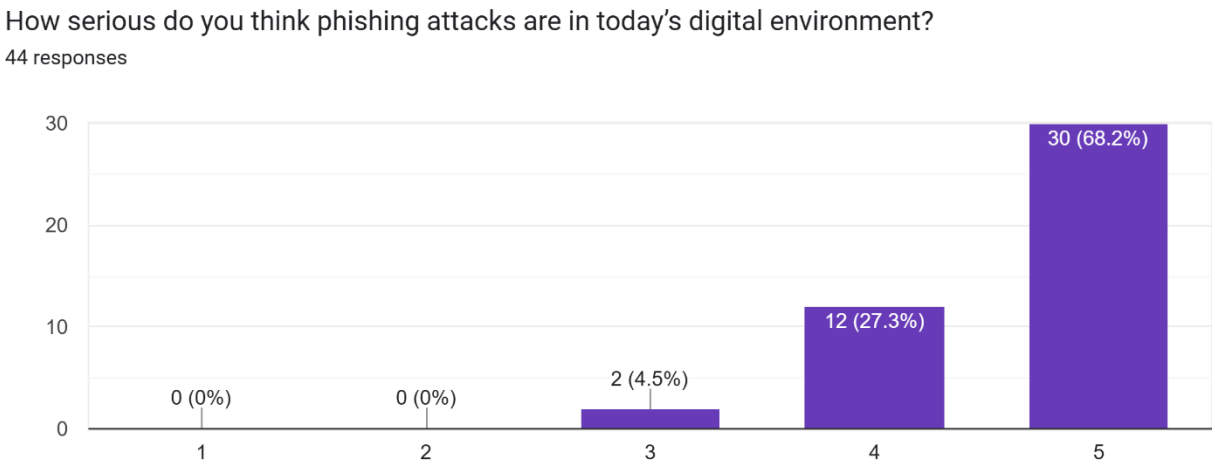


Figure 26: Soberness of Phishing in the present Digital Age

The gravity of phishing was also considered to be very high with the mean score of 4.64 out of 5. The majority of the respondents picked the greatest value on the scale and very few picked low rating. This demonstrates a very close correspondence between the perception of threat by the user and the essence of project motivation. The respondents do not consider phishing as a small or an occasional inconvenience; they consider it to be a big problem in the present digital environment.

The implications of this high seriousness rating are also design implications. Users who already perceive phishing as significant will probably accept more forceful warning and might be more open to a minor level of defensive friction when the extension is efficient. That is, the extension can be assertive but in the warning style without appearing irrelevant as long as the warnings are correct and timely.

Question 17 - Usefulness of the Proposed Browser Extension.

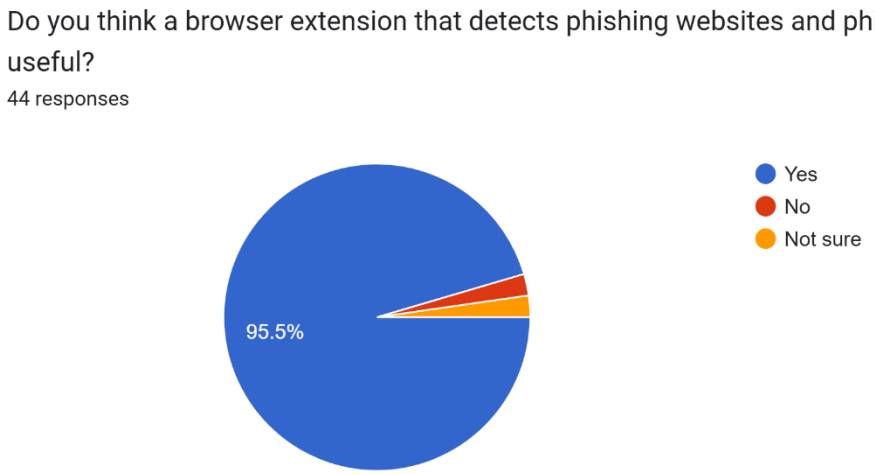


Figure 27: Usefulness of the Proposed Browser Extension

An overwhelming majority of the respondents (42 yes, 95.5 percent) perceive the proposed browser extension as useful. Only isolated responses choose no or not sure. This is first-hand confirmation of the project idea. The user base does not just realize that phishing is an issue; the user base also thinks that a browser extension is a reasonable kind of intervention.

The quality of this finding is particularly useful since viability of academic projects is not solely technical. A system must also have perceived usefulness. High usefulness perception implies that subsequent prototype testing can be more concerned with the way of implementing the idea effectively, but not whether the delivery model per se is sensible to users.

Question 18 -Most Helpful Extension Feature.

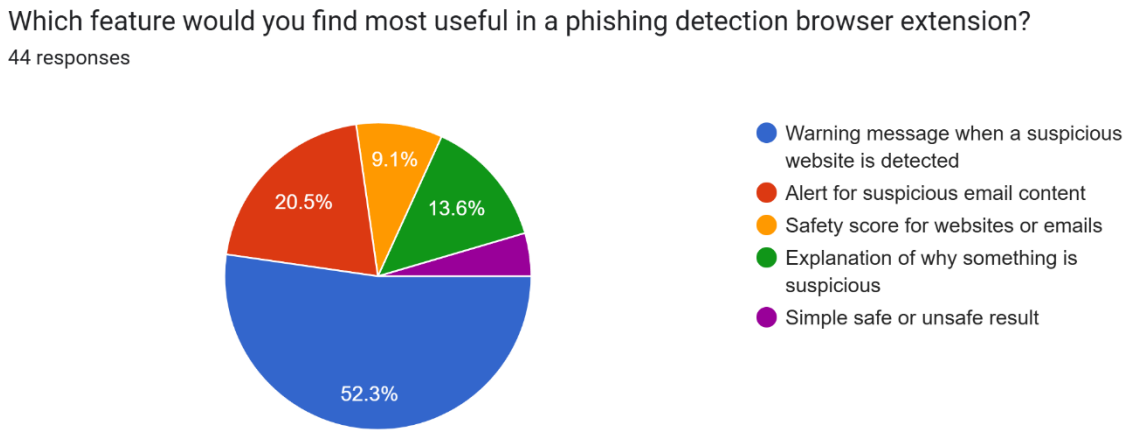


Figure 28: Most Helpful Extension Feature

The most frequently chosen one-useful feature was a warning message about suspicious websites, which was chosen by the respondents the most. The alerts about suspicious email contents, description of why something is suspicious, safety ratings, and easy safe or unsafe were also chosen, however, with lower frequencies. This ranking is significant as it indicates that direct intervention and supplementary explanation are preferred by the users first and second, respectively.

In terms of design, the message is clear: the extension must focus on giving immediate warnings rather than using passive indicators or concealed scores. But explanation must not be expelled altogether, since interpretation is still important to a significant minority. A layered design is thus the most suitable whereby an immediate warning is displayed initially, and more details can be obtained when necessary.

Question 19 - Type of Preferred Alerts.

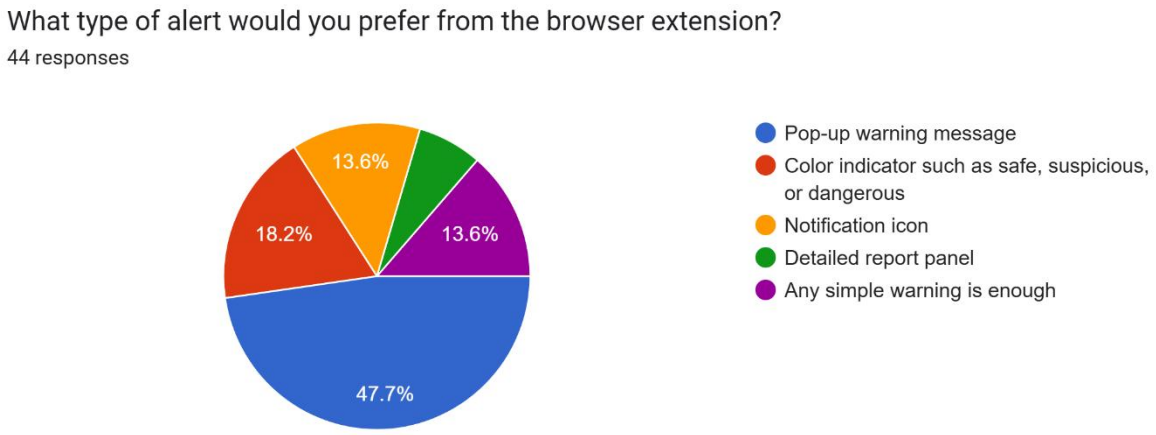


Figure 29: Type of Preferred Alerts

In terms of alert type, a pop-up warning message will be the most preferred type, color indicators, and simple notifications. The number of respondents who chose a detailed report panel as the main alert style is very low. This means that users desire the extension to display risk in a manner that is intuitive and instantaneous and not hidden from an elaborate technical perspective.

What it means is that the extension should not rely on the user to proactively open a dashboard to learn about risk. Rather, the main warning must manifest itself during the suspicion. The detailed views of the report may be additional features of the report that should be available to users that require more detailed description, yet the primary interface should be simplified and action oriented.

Question 20 - Significance of Detection Speed.

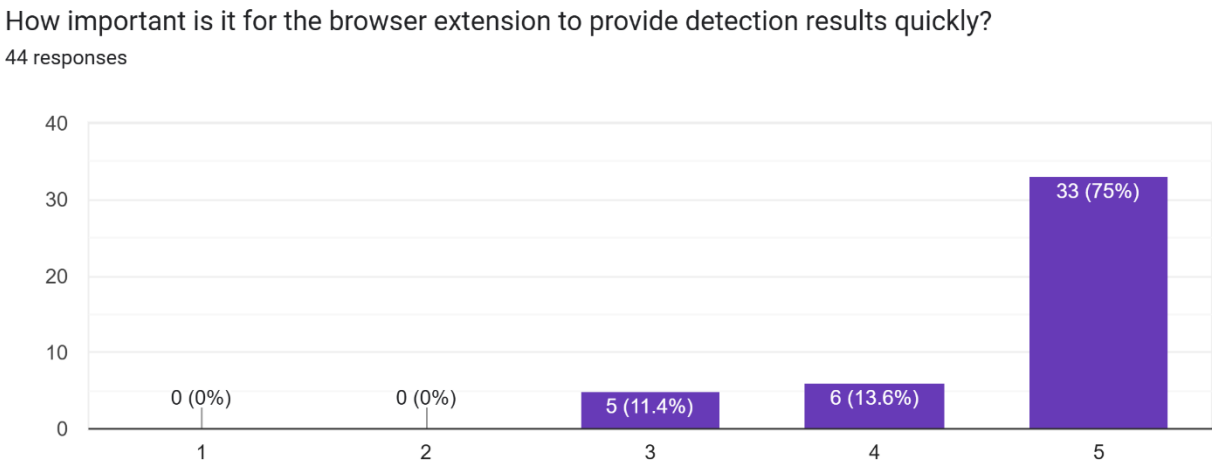


Figure 30: Significance of Detection Speed

Very high detection speed with an average score of importance of 4.64 out of 5. Most of the respondents chose the highest score. This renders speed as non-functional requirement and not an advantage. In that case, when the extension causes the page to interact at a slower pace and/or it feels heavier, the user can turn it off despite the classification model being correct.

This finding is especially significant to technology selection. It favors the selection of computationally manageable supervised-learning models and discourages prototypically heavy architecture. It also warrants the focus on efficient feature extraction and lean extension design rather than feature extravagance.

Question 21 - Ease of Use Importance.

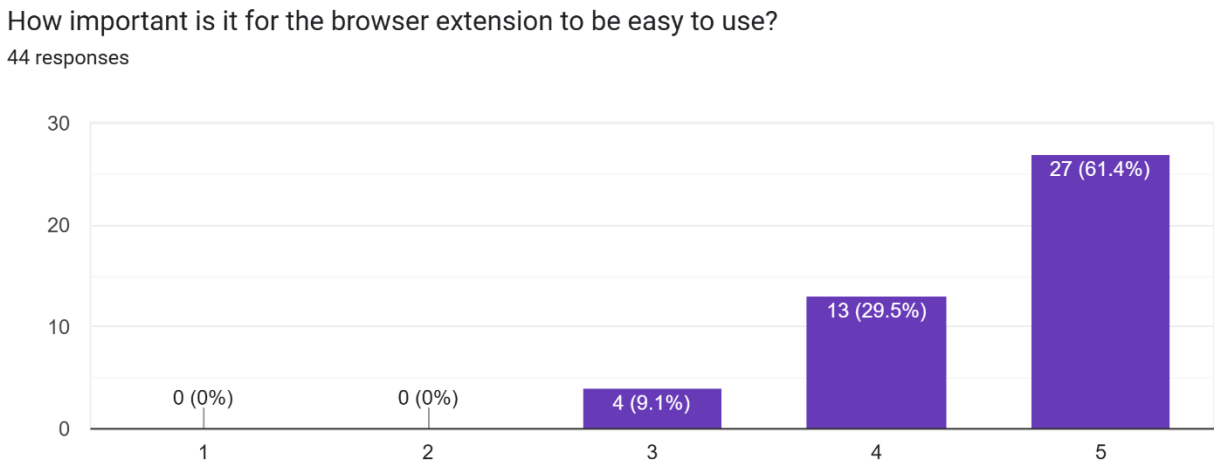


Figure 31: Ease of Use Importance

The ease of use was also very highly rated with the average score being 4.52/5. This proves that the extension should be easily comprehensible by users without the need to have cybersecurity knowledge. An instrument that is precise yet incomprehensible would not be satisfactory to the expectations highlighted in the survey.

The implication is that the extension should minimize interaction burden. Warnings messages must be concise, the textual meaning of statuses must be self-evident, and the user must not have to work out technical values to take safe actions. The ease of use thus becomes a fundamental success factor in oneness with model performance.

Question 22 - Readiness to Use the Extension.

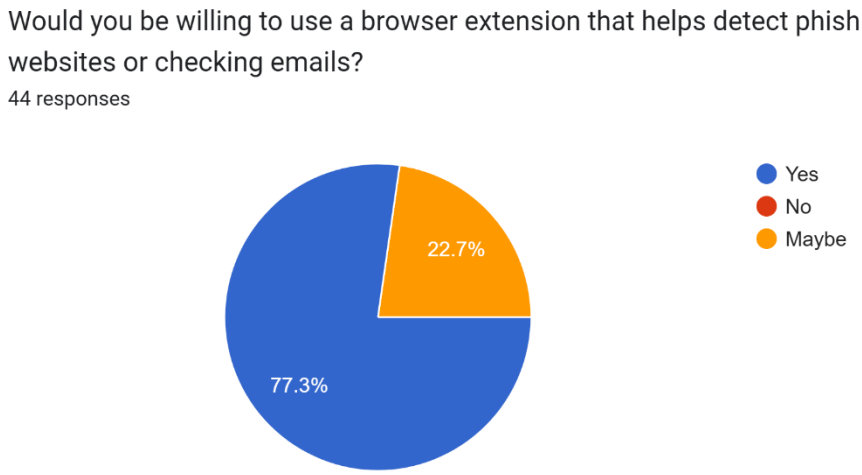


Figure 32: Readiness to Use the Extension

There is a high willingness to use the extension, 34 respondents (77.3) responded yes, and the other respondents answered maybe and not no. It is interesting that there is no explicit rejection. It implies that the idea of phishing-detection extension is tolerable to users provided the information of its implementation including privacy, speed, and quality of alerts are managed properly.

The fact that maybe responses were there is still important though. This means that there are conditionally interested and not entirely committed users. The said users will be convinced of such attributes as pragmatism does the extension feel trustworthy, is it lightweight, is it reasonable in explaining risk, is it considerate of privacy. This supports the significance of the quality of implementation in the following project stage.

Question 23- Other Features Wanted.

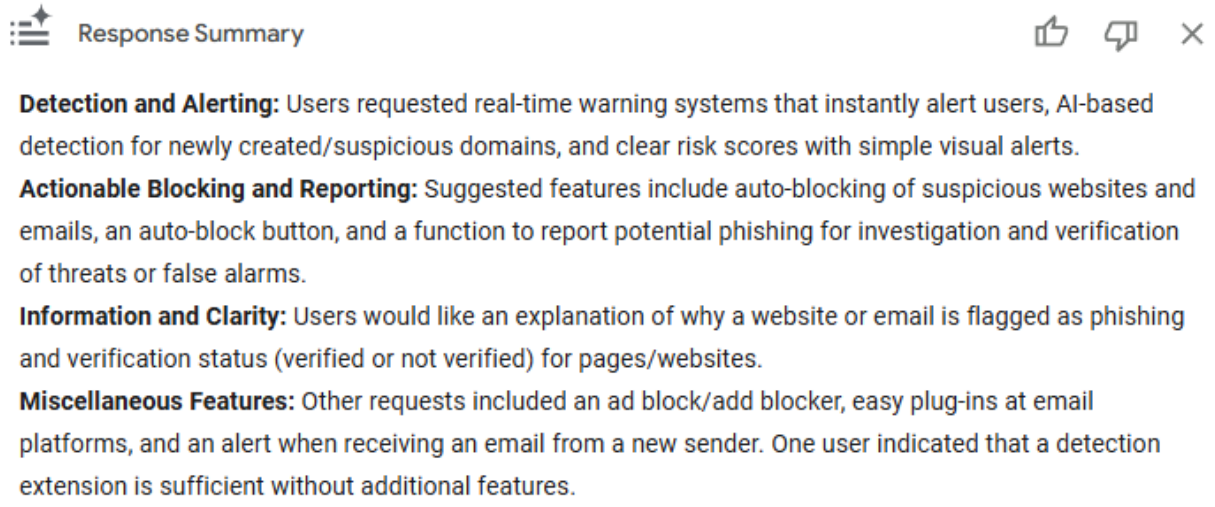


Figure 33: Other Features Wanted

The open-ended answers on other desired features show that there are the same priorities of users. The features requested by the respondents included more robust real-time alerts, optional automatic blocking, suspicious links or pages scan features, sender-awareness features, and how to report suspicious materials. Many of the respondents also showed interest in risk scores, page verification cues, and silent background scanning which does not involve a lot of manual work.

These recommendations show that the users desire the extension to not just label a page as suspicious. They anticipate having a friendly aid that can step in, clarify and potentially lead to safer user behavior in the long run. Concurrently, certain of the more sophisticated requests, like automatic blocking and scanning email more deeply, can add complexity to implementation or sensitivity to privacy. This implies that they must be handled as future-improvement candidates unless they can be incorporated in the prototype in a responsible manner.

Question 24 -Recommendations.

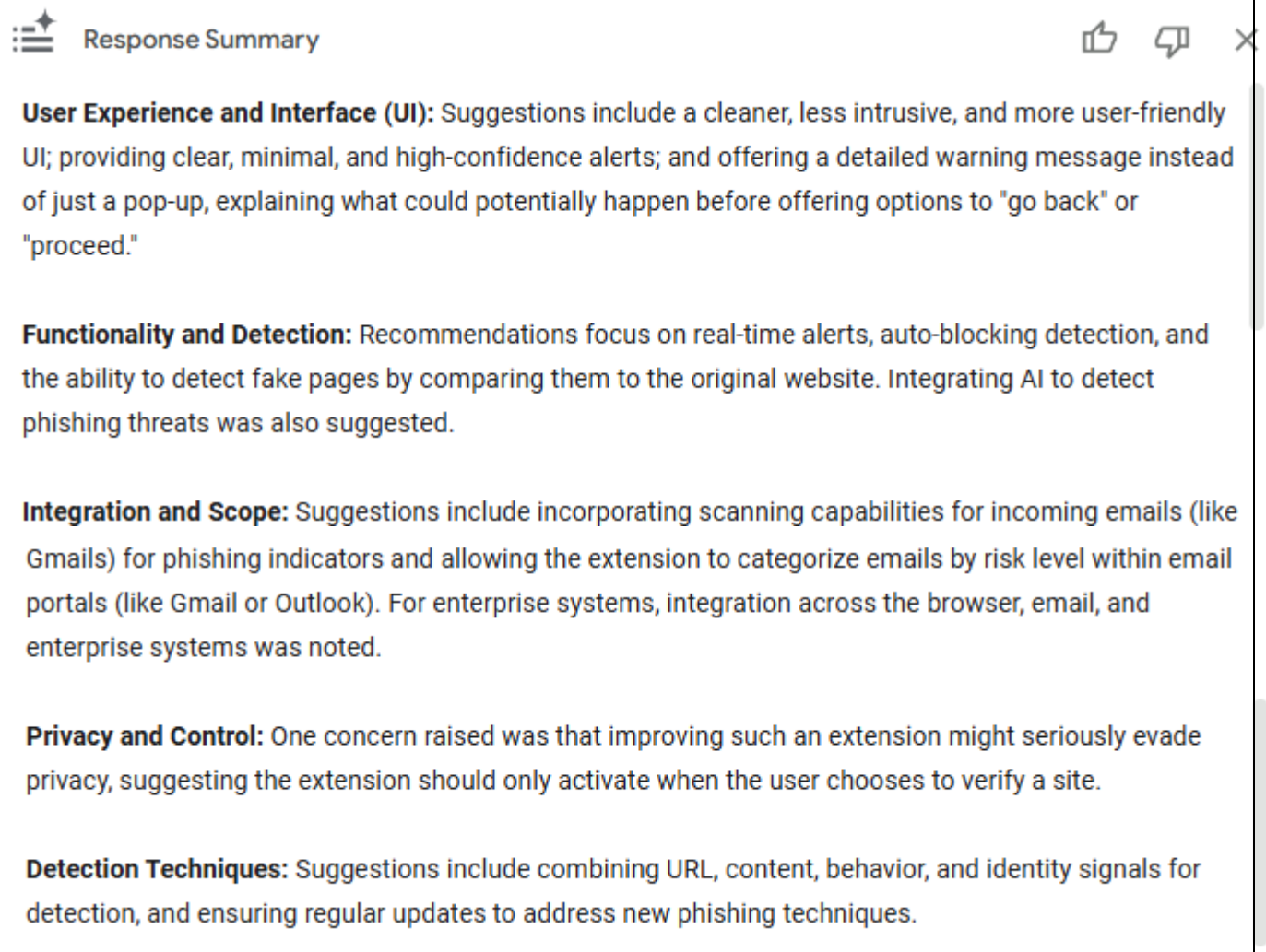


Figure 34: Recommendations

The last open-ended question on how it can be improved gives great insight into refinements. Some of the replies focused on less intrusive and clean user interfaces, more understandable warning messages, more concise explanations of the reasons behind a page being flagged, and periodic updates to match the current phishing methods. There were privacy concerns by others, particularly with regards to any functionality that reads email content or does more extensive background checks. These issues are very topical as they demonstrate that the users do not consider the extension by detection value only; they also take into consideration the way it works respectfully and transparently.

These recommendations, in practice, support the overall design philosophy of the proposed system. The extension must not be conspicuous and not excessive, it must not be informative and overloaded, it must be useful but not obtrusive. The real-time warnings, optional explanation, and conservative permission design combination is thus not only a technical convenience, but the direct reflection of the expectations stated by the target users in the survey.

3.3.1 Cross-Question Interpretation and New Themes.

When the survey is viewed as a unit and not as individual questions, there are a number of powerful themes. The first one is that phishing is not a new and unknown issue to the target audience. The level of awareness regarding phishing sites and phishing emails is high, yet the direct exposure level in regard to suspicious messages, suspicious websites, and clicking dangerous links remains high as well. This combination is significant as it demonstrates that awareness does not solve the problem on its own. Users require support when they interact.

The second theme is that the browser is a legitimate place of intervention. The frequency of website use is very high, with most respondents using websites on a daily or weekly basis, and they are also highly supportive of the notion of a browser extension as a helpful security tool. This offers firsthand empirical evidence to the delivery model adopted by the project. The extension is not being forced upon the users in an unnatural manner; it is in line with their real technology usage, and their own perception of what will be beneficial.

The third theme is that usability cannot be discussed without security effectiveness. Speed and ease of use were continually rated as very important by respondents. They also favored conspicuous yet simple notifications like pop-up warnings and color notifications in place of technical visages. That is, it will not be about the classification metrics only but about the ability of the extension to act in a manner acceptable by the user in the course of normal browsing.

Fourth theme is that users appreciate the explanations, but not at the expense of clarity. Although a direct warning message is the most desirable option, open-ended responses, and multiple-choice responses demonstrate a significant need to know the purpose of something suspicious. This implies that the extension would be communicating in layers: immediate risk first,

explanation second. An overly silent system cannot be relied upon, but an overly talkative system cannot be utilized.

Lastly, the open-ended answers show a progressive anticipation of active protection. Users cited automatic blocking, sender-awareness indicators, real time scanning, report features, and on-going updates. These are not all to be or should be implemented in the initial prototype. Nevertheless, they demonstrate that the target audience wants an anti-phishing extension to be a proactive tool of security but not a passive informational extension.

3.3.2 Final list of user requirements.

Judging by the analysis conducted above in detail, the final user requirements can be divided into functional and non-functional ones. Functional requirements outline what the system is supposed to do whereas non-functional requirements outline how the system is supposed to do it. The list below is a prioritized list of requirements based on the survey and not on the implementation testing since the project is still in the investigation stage. However, the requirements are detailed enough to be used at the next level of the project.

Requirement type	Requirement	Priority	Rationale from analysis
Functional	The extension shall analyze website-related indicators and classify the current browsing target as legitimate or phishing.	Must Have	High daily browsing frequency and strong concern about suspicious websites.
Functional	The extension shall analyze browser-visible email-linked phishing cues where doing so is technically and ethically practical.	Must Have	Email is used frequently and suspicious emails are common.
Functional	The extension shall display a clear warning message when high-risk phishing indicators are detected.	Must Have	Users strongly prefer direct warning messages and pop-up alerts.
Functional	The extension should provide a short explanation or supporting reason for the warning.	Should Have	Many users want to know why something is suspicious.
Functional	The extension may provide optional safe/suspicious/dangerous status indicators or simple risk scoring.	Could Have	Some users prefer quick visual interpretation.
Functional	The system may include a user-reporting or feedback mechanism for suspicious items in later iterations.	Could Have	Open-ended feedback requested report functions and improvement loops.
Non-functional	The extension should return results quickly and should not noticeably slow normal browsing.	Must Have	Speed was one of the highest-rated importance factors.

Non-functional	The extension should be easy to use and understandable without cybersecurity expertise.	Must Have	Ease of use was rated highly and awareness is not universal.
Non-functional	The warning design should be visible enough to interrupt risky action but not so intrusive that users disable the extension.	Must Have	Users want direct alerts, but several comments requested a cleaner interface.
Non-functional	The extension should request only necessary browser permissions and avoid unnecessary retention of personal data.	Must Have	Open feedback highlighted privacy sensitivity.
Non-functional	The selected supervised-learning models should be transparent enough for comparison and feasible for extension-oriented deployment.	Should Have	Project scope values feasibility, maintainability, and evaluation clarity.

Table 8: Consolidated User Requirements

The requirement list demonstrates that the priorities of the project are highly compatible with the survey. Direct phishing warning capability of websites and email-linked cues is the most crucial functional requirement. Speed, ease of use, visible yet acceptable warnings, and conservative privacy behaviors are the most vital non-functional requirements. Such a correspondence is appropriate since it indicates that it is not being developed as a matter of technical whim; it is guided by what has been written about the needs of the users.

It is also important what the requirements do not require. The survey does not warrant construction of an enormously complicated extension on the initial prototype. Rather, it endorses a narrow scope system that is capable of a limited number of high-value tasks in a positive way: inspect practical phishing indicators, caution the user in a way that is easily comprehensible and prompt, and be lightweight and understandable. This is one of the strengths as it provides the implementation stage with a realistic and defendable direction.

3.3.3 Surveying Survey Results to Decision Design.

When the survey is translated into tangible design decisions, this makes the survey most useful. The former significant mapping is delivery location. Since the respondents visit the sites on a regular basis, and believe that browser-based protection is helpful, the browser is the right place to place the proposed system. This substantiates the initial project direction and eliminates less integrated options like a standalone classifier disconnected to user-facing intervention when live browsing.

The second key mapping is that of the balance between the websites and emails. Suspicious emails are slightly more frequently reported than suspicious sites, although both are high. It implies that phishing cannot be viewed as only a problem related to websites in the context of the project. Rather, the browser extension must be marketed as an extension that safeguards against both direct phishing web pages and email-based phishing communications experienced in the browser. This wider framing also is more in line with the literature, which is moving towards treating phishing as an inter-channel issue and not as a single-format attack.

A third big mapping involves alert design. The high popularity of the pop-up warnings and explicit suspicious-website alerts is a sign that risk communication should be direct and swift. Hidden monitoring or technical dashboards are not the primary requests by the users. They desire intervention during the time of danger. This implies that the interface must favor interruption to passive presentation in case there are strong phishing signs. Meanwhile, the fact that there are respondents who desire explanation, verification status or risk scores implies that the design must accommodate layered information to the users who desire to scrutinize the logic behind the warning.

The fourth mapping is performance and model choice. Since speed and ease of use were rated to be very important, the project must not use model architectures that are too heavy at the prototype stage. This justifies the methodological choice of ranking supervised-learning models which could be compared and implemented more effectively. Algorithms are not explicitly stated in the survey, but the preferences of usability indirectly influence the admissible model space. The technical design should thus align with the expectation of the user of speedy and easy interaction.

The fifth mapping is on privacy and trust. There were also open-ended answers which indicated that an extension that is able to scan content would be too intrusive. Such issues are not to be considered in contrast to the project; these are design directions. They suggest that the proposed extension must not seek excessive permissions, needless data retention, and clarify what it inspects and why. Practically, it implies that privacy-conscious design is not an ethical supplement. It is one of the user requirements based on data gathering process.

Combined, these mappings reveal that the survey is of true design worth. It does not simply attest to the fact that phishing is a serious issue. It explains how the solution that has been proposed should be provided, the qualities that it should focus on and the kind of trade-offs that users are most likely to accept. This is what requirement-oriented data collection in an investigation report is all about.

3.3.4 Implications for Model Design and Extension Workflow

In terms of model-design, the survey recommends that the system must be designed in a manner that puts practical robustness ahead of the highest level of complexity. The large invisibility of AI was not what users requested; the rapid, precise, comprehensible defense. This implies that the model must be built around features that are applicable as well as accessible in the extension workflow. Features that are too expensive to maintain or pose challenging privacy implications can diminish the usefulness of the system in practice despite the improvement in classification scores in controlled experiments.

A lesson learned is that URL-based and visible-page features are good candidates that can be used in early iterations. Already, the respondents identify suspicious URLs, unfamiliar senders, unforeseen links, and credential requests as characteristics of phishing. These are aspects that can be easily transferred to supervised-learning input and subsequently to explanations accessible to the user. In comparison, features that are heavily hidden or infrastructure-level might be more difficult to communicate and might be inapplicable in the browser-extension environment without extra services or permissions.

One more implication is related to setting thresholds. A browser extension has to balance the false negative and false positive correctly. It is risky to miss a true phishing page, and too many alarms might easily diminish trust and create alert fatigue. The survey will not identify the mathematically optimal threshold but does suggest that users would like decisive warnings when the risk is high and a clean experience when the risk is low. The extension workflow must then assist in threshold tuning according to the performance measures as well as usability anticipations.

There should also be primary and secondary actions in the workflow. The most crucial activity of the extension is warning the user. Secondary actions involve presenting a rationale, disclosing a risk status, or providing a report feature. This difference assists in complexity management. It does not allow the extension to be loaded with features until the core detection and warning loop has been properly functioning. It is also suitable for the staged priorities proposed by the survey where visibility and ease were prioritized over more advanced features that are richer.

Lastly, the results of the survey suggest that extensions should be able to aid learning in addition to protection. Numerous respondents are already aware of phishing indicators, but they still report false clicks and doubt. An inhibitory signal can work in the short run, but an inhibitory signal that educates the user about what was suspicious can be more beneficial in the long run. The proposed system is thus capable of creating value not just by averting individual incidents but also by enhancing user judgement over time by improving the user judgement over a series of interactions.

3.3.5 Suggested System Architecture in the Future.

An effective design of the proposed system can be perceived in four layers: input capture, feature preparation, classification, and user feedback. The input-capture layer has the role of gathering the minimum number of signals visible to the browser required to make the extension work. Early versions of this layer can prioritize the target URL, visible cues of hyperlinks, suspicious lexical tokens, and non-invasive page-level cues. In the case of email-related scenarios on webmail pages, the extension may examine browser-visible sender signals, visible text patterns, and suspicious link targets where those are available and pertinent.

The feature-preparation layer converts raw browser-visible features into data that is model-ready. This step is important in classical supervised-learning systems since the quality of the classification result is very dependent on the meaning and stability of the features. In URL-based inputs, feature preparation can involve tokenization, length computation, flagging, domain-character pattern verification and encoding of particular structural cues. In the case of email-linked inputs, it can also be composed of a textual cue, mismatch of sender and recipient domains, or the existence of urgency- and credential-request words. The purpose of this layer is not to build the biggest possible feature set, but to build a defensible feature set which is not in conflict with the privacy and performance limits of the extension.

The classification layer uses one or more supervised-learning models on the prepared features. In development, various candidate models may be tested offline, but the deployed extension would preferably employ one of the selected models or a simple decision path, which is efficient enough to be used in the browser. The output of the classification layer should be more than a raw binary label. It must also offer at least a confidence-oriented interpretation or a translation rule to bring the label to levels of risk that are understandable to the user. The reason is that it must be capable of differentiating between situations which are evidently secure, those which are evidently suspicious, and those where the system perceives moderate concern but not necessarily unambiguous maliciousness.

The last layer is user feedback. This is where the most noticeable aspect of the architecture is and in real life, this is where the user will evaluate the system. The feedback layer must be designed to present the outcome in a clear way, not overwhelm the user with technical information and it must be clear what the user must do next. In case of a suspicious page or email-linked context, a warning can be visibly presented in the form of a pop-up, a blocked or caution state, or even an instant banner with a brief explanation. In case of less risk, the extension does not need to be provided with a complete interruption but with a smaller visual status cue. This layer thus is an attempt to translate technical detection into human decision support.

Such modular architecture is advantageous in a project of undergraduate interest since it helps in progressive development. Each of the layers can be tested individually by the developer before complete integration. It is possible to fine-tune feature design without having to rewrite the user interface, and it is also possible to fine-tune warning design without needing to change the

classification theory completely. Another factor why a supervised-learning-based extension is a suitable course is this modularity: controlled complexity is encouraged instead of compelling all components to adapt simultaneously.

3.2.6 Limitations of the Data Gathering Design

Even though the survey technique is appropriate when it comes to requirement collection, the technique has its own limitations that must be considered. The first one is bias in sampling. Since the questionnaire was shared online, using convenience, the sample is not a controlled random sample of all potential users. The findings can still be used in identifying the requirements, but one should not take them as estimates of how the whole population is aware about phishing or the adoption of extensions. This is a reasonable limitation in an investigation report but must be borne in mind when generalizing the results.

The second weakness is self-report bias. The survey is based on what the participants respond to their knowledge, their confidence and their behaviors in the past. Self-confidence in phishing identification might not be an accurate measure of real-life performance. And likewise, other users might report false alarms or report suspicious experiences in different ways. Nonetheless, the survey will still be useful since the project requires perceived user needs and expectations just as much as it requires objective behavioral truth. That is, despite the imperfection of self-report, it does reflect what users think they require of the extension.

The third weakness is that there is no observational or experimental validation. The survey at hand does not examine the ability of the respondents to be able to notice phishing in controlled conditions, as well as respond to the presence of the extension warning. Such activities should be in the following stage of the project. The current report thus relies on the survey to know the expectations and requirements, but it is realized that the prototype testing will be required in the future to determine whether the expectations are realized in practice.

One last weakness is related to time. The phishing scams are dynamic and the perception of browser extensions can change in the course of time. The survey thus provides a helpful yet time-limited user complaints picture. This drawback also justifies the worth of Agile development

journey where requirement interpretation can be reviewed and adjusted in the implementation process.

CHAPTER 4: CONCLUSION

4.1 Discussion of Achievement of the First Project.

4.1.1 Justification of the Adequacy of Investigation and Research for Project Achievement

The primary aim of the initial stage of the project was to identify whether the suggested system, which is a browser extension that detects phishing sites and emails based on supervised learning, is relevant, feasible, and grounds its foundation in the literature and needs of the users sufficiently. That objective has been attained based on the work done in this investigation report. The project has now transcended a very general concept of cybersecurity and has evolved into a well-organized proposal that has been backed by the recent academic findings, operational requirement analysis, a chosen approach, and a realistic technical path.

The literature review presents solid reasons that phishing is a high volume, active, and technologically developing threat. Recent studies surveyed in Chapter 2 indicate that phishing detection is still a key area of research and supervised learning is a realistic and viable method to detect websites and email-related phishing. It is also clear in the review that browser-side delivery is a direction that is valid and current and not an out-of-date or isolated concept. This is important as it implies that the project will be grounded on the latest studies as opposed to old assumptions or outdated anti-phishing frameworks.

The appropriateness of the phase of investigation is also reinforced by the examination of similar systems. The report has determined the place of the proposed system within the current field by comparing the recent browser-extension systems, real-time detection architectures, email-centric classifiers, generalization studies, and explainable-feature frameworks. This comparison step is significant since it does not allow the project to be ambiguous about what it contributes. The suggested system has now been presented as a managed-learning-based browser-extension, undertaking to integrate practical browser-side security, in addition to an authentic, lightweight detection process of websites and email-affiliated phishing indicators.

The technical research is also well developed to the stage of investigation. Chapter 2 has defined the functional and non-functional requirements of the system, rationalized the main

software and hardware environment, considered browser-extension architecture issues, and considered the effects of security, privacy and runtime constraints on the design. These results are sufficiently presented to inform the next implementation step without unnecessarily influencing the project to make decisions that will be discovered during testing. In this, the technical research has reached the proper balance: it is concrete enough to be put into practice, yet it is not so hard that it cannot be refined.

The empirical requirement analysis is perhaps the most powerful part of the first project phase. This survey gathered 44 valid responses, which is above the minimum number that the guideline specifies. More to the point, the findings are most applicable to the direction of the project. Respondents are heavy users of browsers and email, are highly exposed to suspicious emails and websites, take phishing seriously, and are highly supportive of a fast, easy-to-use, and capable-of-delivering-direct-warnings browser extension. These results make sure that the project is not founded upon the technical curiosity. It is based on a definite user requirement which has been captured by way of direct data collection.

The sufficiency of the investigation is also due to the methodology chapter. The report has identified the way that the project can progress into development in a responsible manner by making comparisons between Agile, RAD, and Kanban and then settling on Agile as the most appropriate methodology. This is not some minor administrative point. The chosen methodology aligns with the fact that the project will require iteration, model refinement, interface refinement, and improvement with the help of the backlog. Since the needs have already been analyzed in the investigation stage, the project is implemented with a more solid process base than would otherwise have been the case.

Combined, these accomplishments indicate that the initial stage of the project has yielded sufficient investigation and research to warrant the next step in implementation. There is a clear problem statement, evidence-based scope, defended technical direction, documented methodology, validated set of user expectations and a realistic development timeline. The point is that the initial component of the project could be evaluated as successful enough considering the goals of an investigation report.

4.1.2 Justification for Further Exploration and Improvement in Research and Design Areas

Despite the fact that the initial stage is sufficient and effective, it also unveils the number of areas that need to be investigated more before the project can be taken to the implementation phase. The former is dataset strategy. It has already been determined in the investigation that the quality of datasets, the proportion of the classes, and the generalization are key problems of phishing detection. The following step should then look deeper into the selected datasets, particularly regarding overlapping, duplicate patterns, stale samples, and representativeness of web and email phishing samples. It should not aim at collecting as much data as possible without any consideration but developing a strategy to create a dataset that enables defensible supervised-learning evaluation.

The second one is featuring engineering. A combination of the literature and the survey indicates that the practical indicators such as suspicious URLs, credential requests, unknown senders, and unexpected links ought to be highlighted by a useful browser extension. Nevertheless, how these cues are to be coded to be model trained is yet to be closely experimented. It is possible that some of them might be redundant, some might be weak predictors, and some might be valuable statistically, but too expensive or too intrusive to implement on the extension-side. The second step will hence require a trade-off between predictive strength and privacy, cost of running, and explanation value.

The third area is threshold and warning design. The survey indicates that users would like to see obvious and conspicuous warnings, but it does not inform the developer of the exact aggressiveness that the extension should take when marking material. Excessively conservative threshold can be bypassed by phishing attacks, whereas excessively aggressive can generate false alarms and decrease trust. This balance will have to be investigated in the implementation stage. Ideally, the chosen threshold ought to be data-driven by both classification measures and logic of usability since optimal statistical environment may not necessarily result in the most agreeable user experience.

A fourth area deals with permissions and operational trust of browser-extensions. Privacy sensitive design has been defined as a requirement in the present report, and the technical implementation remains to be explored further. This is particularly when it comes to detection of emails within webmail environments. The second step should know what can be accessed in ethical and practical terms, what processing can be done locally and how can the extension be able to communicate its behaviors in a transparent manner where the users do not feel like they are being intruded upon. This is not just a technical challenge but a challenge of trust building.

The fifth opportunity of work in the future is prototype testing in real interactions. It gathers user expectations and self-reported experience and is not yet testing user reactions to actual warnings produced by a functional extension. This should be followed by at least constrained prototype-based tests, be it scenario tests, controlled walkthroughs or pilot-like usability tests. This will assist in confirming that the styles of warnings that are preferred in the questionnaire are still effective when used by the users in a realistic browsing flow.

Another sixth area is system explainability and learning by users. The survey clarifies that most users desire explanations, besides having simple warnings. Similarly, in literature, explainable phishing detection is becoming more important. The implementation step, then, ought to look into the extent to which explanatory content is practical. Explanations that are insufficient can decrease trust whereas excessive explanation can overwhelm the user. One of the key future design issues and center of importance to the educational value of the extension is finding the right level of interpretability.

Lastly, the sustainability of work in the future should be taken into account. The threats posed by phishing evolve very fast and this makes the effectiveness of the extension also rely on the ease with which it can be updated in terms of its model, thresholds and warning logic. An effective prototype must thus not just work on the original data; it must also be designed so that re-training, tuning and re-feature modification can be easily done in the future. This once again justifies the choice of Agile during the implementation phase, as the project is probably going to be well-served by repetitive refinement, as opposed to a single fixed build.

Conclusively, the initial section of the project has managed to meet its investigation objectives but has also accurately raised the major questions that will need to be answered next. This is a good result. An excellent investigation report is one that does not state that it has been

established that all uncertainty has been eliminated. Rather, it minimizes uncertainty to a level that allows the following stage to have meaning and be manageable. This is what has been done here. The project has now proceeded with a better theoretical foundation, a better understanding of its user requirement base and a better vision of what still needs to be fine-tuned to create a useful and defensible browser-extension phishing detector.

4.1.3 Overall Reflection on the Investigation Phase

Another valuable lesson that the investigation stage teaches with respect to the design of a cybersecurity project is that effective protection tools should be approached as socio-technical systems. Not all phishing detectors are classifiers. It is an interface which lies between user trust and cyber deception. As demonstrated in the literature review, technical models are able to recognize malicious patterns in an ever more sophisticated manner, yet practical protection remains contingent upon the manner in which the findings are provided, interpreted, and responded to by the users. This is clearly confirmed by the survey. Higher intelligence was not the only demand made by the respondents, speed, clarity, explanation and confidence were also sought. This implies that the success of the project will be a matter of the development of technical performance, and the human usability in tandem rather than independently.

The other revelation during the investigation process is the fact that realism is more important than inflation. One could imagine a significantly larger anti-phishing system that includes browsing history analysis, cloud-based scanning, powerful transformer models, attachment studies, and central reporting dashboards. Yet, the exploration has revealed that a more narrowed strategy is more in line with the academic breadth as well as the user needs of the project. The most important need of the users is straightforward warnings, minimal interaction, and predictable behaviors. This reinforces the notion that the next best thing to do is not to expand the architecture, but to render the core process of detection and warning reliable and properly implemented.

The project plan that is created in this report is hence not only important as a schedule but also a statement of design discipline. It demonstrates that the project is going to be developed in

well-set steps instead of feature-growth anarchy. The timeline is an intentional sequence of tasks: initial investigation and requirement definition, dataset and feature preparation, model comparison, browser-extension interface design, integration, testing and refinement. This kind of staged strategy is particularly significant in projects relating to phishing since it helps minimize the chances that implementation choices may be taken before the problem is well-understood.

The relationship between technical feasibility and ethical responsibility has been also explained during the investigation phase. Since the project relates to phishing, trust, and a possible sensitive context (browsers or emails), accuracy is not the only key in assessing the design. The extension should also be able to work in such a manner that does not impinge on privacy and whose usage is not unnecessary. The current direction of the project has this strength. The report puts privacy, minimal permissions and transparent warning design towards the middle of the requirement analysis instead of delaying the consideration of ethics until later in the implementation process. That provides the next stage with a far more solid ethical underpinning than would be provided by considering privacy as an a posteriori.

Generally, the initial stage can thus be regarded as successful in a more profound meaning than merely getting the necessary chapters of the investigation report. It has turned a general concept into a systematic and evidence-based project with a definite direction, realistic scope, and plausible continuation course. The following stage will also require a lot of technical effort, yet it will not commence without preconceptions. It will start with the logical comprehension of the menace, the users, the viable models, the interface anticipations, and the development approach that is best adapted to handle the work.

References

Asiri, S., Xiao, Y., Alzahrani, S., Li, S., & Li, T. (2023). A survey of intelligent detection designs of HTML URL phishing attacks. IEEE Access, 11, 6421–6443. https://doi.org/10.1109/ACCESS.2023.3237798

Asiri, S., Xiao, Y., Alzahrani, S., & Li, T. (2024). PhishingRTDS: A real-time detection system for phishing attacks using a deep learning model. Computers & Security, 141, 103843. https://doi.org/10.1016/j.cose.2024.103843

Anti-Phishing Working Group. (2025). Phishing Activity Trends Report, Fourth Quarter 2024. https://docs.apwg.org/reports/apwg_trends_report_q4_2024.pdf

Federal Bureau of Investigation, Internet Crime Complaint Center. (2025). 2024 IC3 Annual Report. https://www.ic3.gov/AnnualReport/Reports/2024_IC3Report.pdf

Fares, H., Kilani, J., Fagroud, F. E., Toumi, H., Lakrami, F., Baddi, Y., & Aknin, N. (2024). Machine learning approach for email phishing detection. Procedia Computer Science, 251, 746–751. https://doi.org/10.1016/j.procs.2024.11.179

Jain, S., Jotwani, V., & Agrawal, S. (2024). Comparative analysis of agile frameworks: Evaluating their impact on software development lifecycle efficiency. International Journal of Intelligent Systems and Applications in Engineering, 12(23s), 3126–3137. https://ijisae.org/index.php/IJISAE/article/view/7627

Kavya, S., Karthik, S., Elakkiya, K., Manikandan, R., & Sudhakar, S. (2024). Staying ahead of phishers: A review of recent advances and challenges in phishing attack detection. Artificial Intelligence Review. https://doi.org/10.1007/s10462-024-11055-z

Linh, D. M., Ha, H. D., Chau, H. M., Vu, Q. S., & Tran, T.-N. (2024). Real-time phishing detection using deep learning methods by extensions. International Journal of Electrical and Computer Engineering, 14(3), 3021–3035. https://doi.org/10.11591/ijece.v14i3.pp3021-3035

Meléndez, R., Ptaszynski, M., & Masui, F. (2024). Comparative investigation of traditional machine-learning models and transformer models for phishing email detection. Electronics, 13(24), 4877. https://doi.org/10.3390/electronics13244877

Naqvi, B., Perova, K., Farooq, A., Makhdoom, I., Oyedeji, S., & Porras, J. (2023). Mitigation strategies against phishing attacks: A systematic literature review. Computers & Security, 132, 103387. https://doi.org/10.1016/j.cose.2023.103387

Omari, K. (2023). Phishing detection using gradient boosting classifier. Procedia Computer Science, 230, 120–127. https://doi.org/10.1016/j.procs.2023.12.067

Rashid, F., Doyle, B., Han, S. C., & Seneviratne, S. (2024). Phishing URL detection generalisation using unsupervised domain adaptation. Computer Networks, 245, 110398. https://doi.org/10.1016/j.comnet.2024.110398

Safi, A., & Singh, S. (2023). A systematic literature review on phishing website detection techniques. Journal of King Saud University - Computer and Information Sciences, 35(2), 590–611. https://doi.org/10.1016/j.jksuci.2023.01.004

Shafin, S. S. (2025). An explainable feature selection framework for web phishing detection with machine learning. Data Science and Management, 8, 127–136. https://doi.org/10.1016/j.dsm.2024.08.004

Sitorus, G. B. M. T., Efendi, R. A. G., Napitupulu, J. F., Pranoto, H., & Hermawan, E. S. (2025). Bridging the gap: A comparative evaluation of Agile, Waterfall, and hybrid methodologies in modern software projects. Procedia Computer Science, 269, 1259–1268. https://doi.org/10.1016/j.procs.2025.09.067

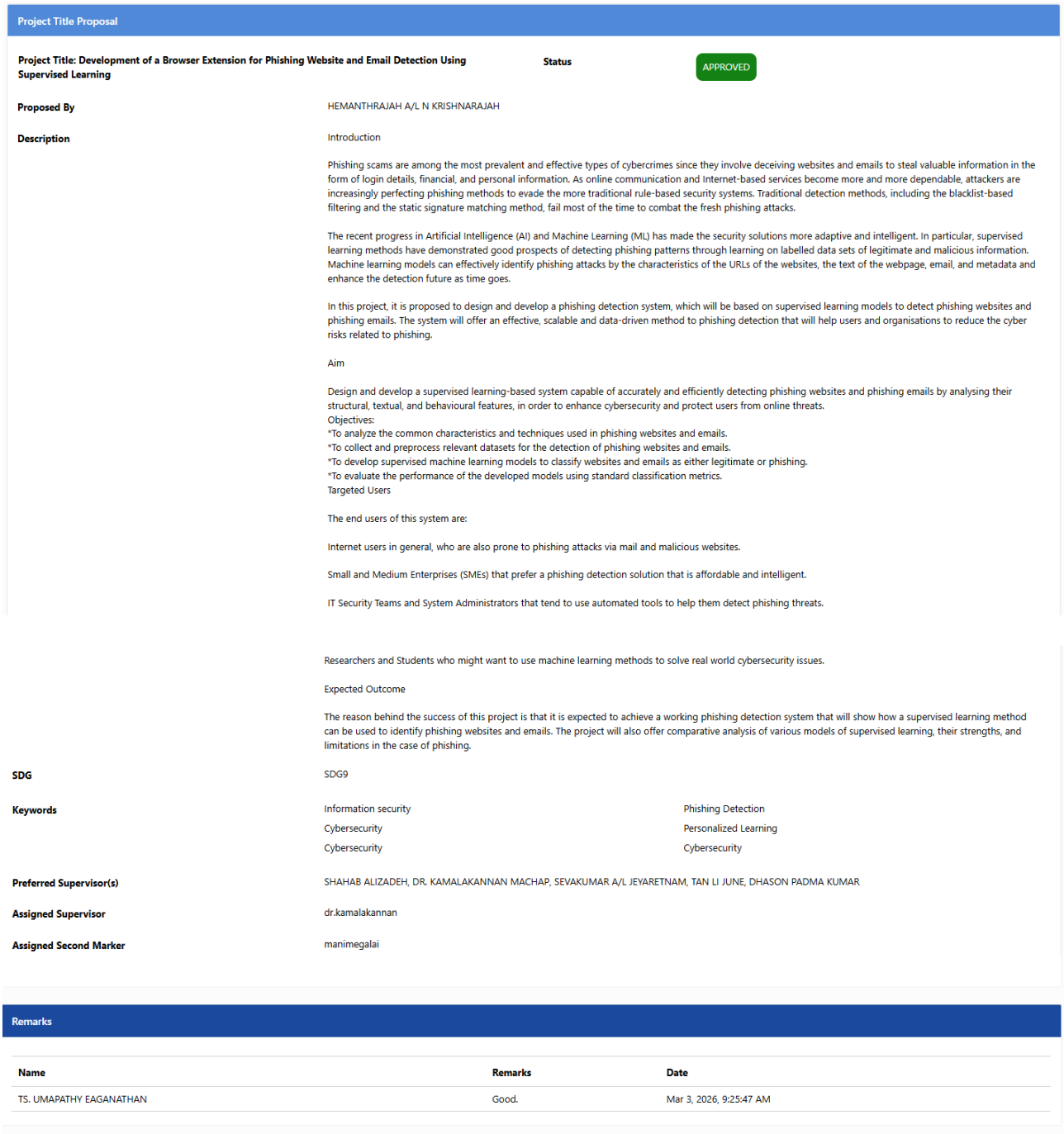
Tetteh, S. G. (2024). Empirical study of agile software development methodologies: A comparative analysis. Asian Journal of Research in Computer Science, 17(5), 30–42. https://doi.org/10.9734/ajrcos/2024/v17i5436

Wilk-Jakubowski, J. L., & Pawlik, A. (2025). Machine learning and neural networks for phishing detection: A review. Electronics, 14(18), 3744. https://doi.org/10.3390/electronics14183744

Wu, C.-Y., Kuo, C.-C., & Yang, C.-S. (2023). Phishing detection with browser extension based on machine learning. In 2023 18th Asia Joint Conference on Information Security (AsiaJCIS) (pp. 81–87). IEEE. https://doi.org/10.1109/AsiaJCIS60284.2023.00023

APPENDICES

Appendix A: PPF – Title Registration Proposal



Project Title Proposal

Project Title: Development of a Browser Extension for Phishing Website and Email Detection Using Supervised Learning

Status APPROVED

Proposed By HEMANTHRAJAH A/L N KRISHNARAJAH

Description

Introduction

Phishing scams are among the most prevalent and effective types of cybercrimes since they involve deceiving websites and emails to steal valuable information in the form of login details, financial, and personal information. As online communication and Internet-based services become more and more dependable, attackers are increasingly perfecting phishing methods to evade the more traditional rule-based security systems. Traditional detection methods, including the blacklist-based filtering and the static signature matching method, fail most of the time to combat the fresh phishing attacks.

The recent progress in Artificial Intelligence (AI) and Machine Learning (ML) has made the security solutions more adaptive and intelligent. In particular, supervised learning methods have demonstrated good prospects of detecting phishing patterns through learning on labelled data sets of legitimate and malicious information. Machine learning models can effectively identify phishing attacks by the characteristics of the URLs of the websites, the text of the webpage, email, and metadata and enhance the detection future as time goes.

In this project, it is proposed to design and develop a phishing detection system, which will be based on supervised learning models to detect phishing websites and phishing emails. The system will offer an effective, scalable and data-driven method to phishing detection that will help users and organisations to reduce the cyber risks related to phishing.

Aim

Design and develop a supervised learning-based system capable of accurately and efficiently detecting phishing websites and phishing emails by analysing their structural, textual, and behavioural features, in order to enhance cybersecurity and protect users from online threats.
Objectives:
*To analyze the common characteristics and techniques used in phishing websites and emails.
*To collect and preprocess relevant datasets for the detection of phishing websites and emails.
*To develop supervised machine learning models to classify websites and emails as either legitimate or phishing.
*To evaluate the performance of the developed models using standard classification metrics.
Targeted Users

The end users of this system are:

Internet users in general, who are also prone to phishing attacks via mail and malicious websites.

Small and Medium Enterprises (SMEs) that prefer a phishing detection solution that is affordable and intelligent.

IT Security Teams and System Administrators that tend to use automated tools to help them detect phishing threats.

Researchers and Students who might want to use machine learning methods to solve real world cybersecurity issues.

Expected Outcome

The reason behind the success of this project is that it is expected to achieve a working phishing detection system that will show how a supervised learning method can be used to identify phishing websites and emails. The project will also offer comparative analysis of various models of supervised learning, their strengths, and limitations in the case of phishing.

SDG SDG9

Keywords Information security, Cybersecurity, Cybersecurity, Phishing Detection, Personalized Learning, Cybersecurity

Preferred Supervisor(s) SHAHAB ALIZADEH, DR. KAMALAKANNAN MACHAP, SEVAKUMAR A/L JEYARETNAM, TAN LI JUNE, DHASON PADMA KUMAR

Assigned Supervisor dr.kamalakannan

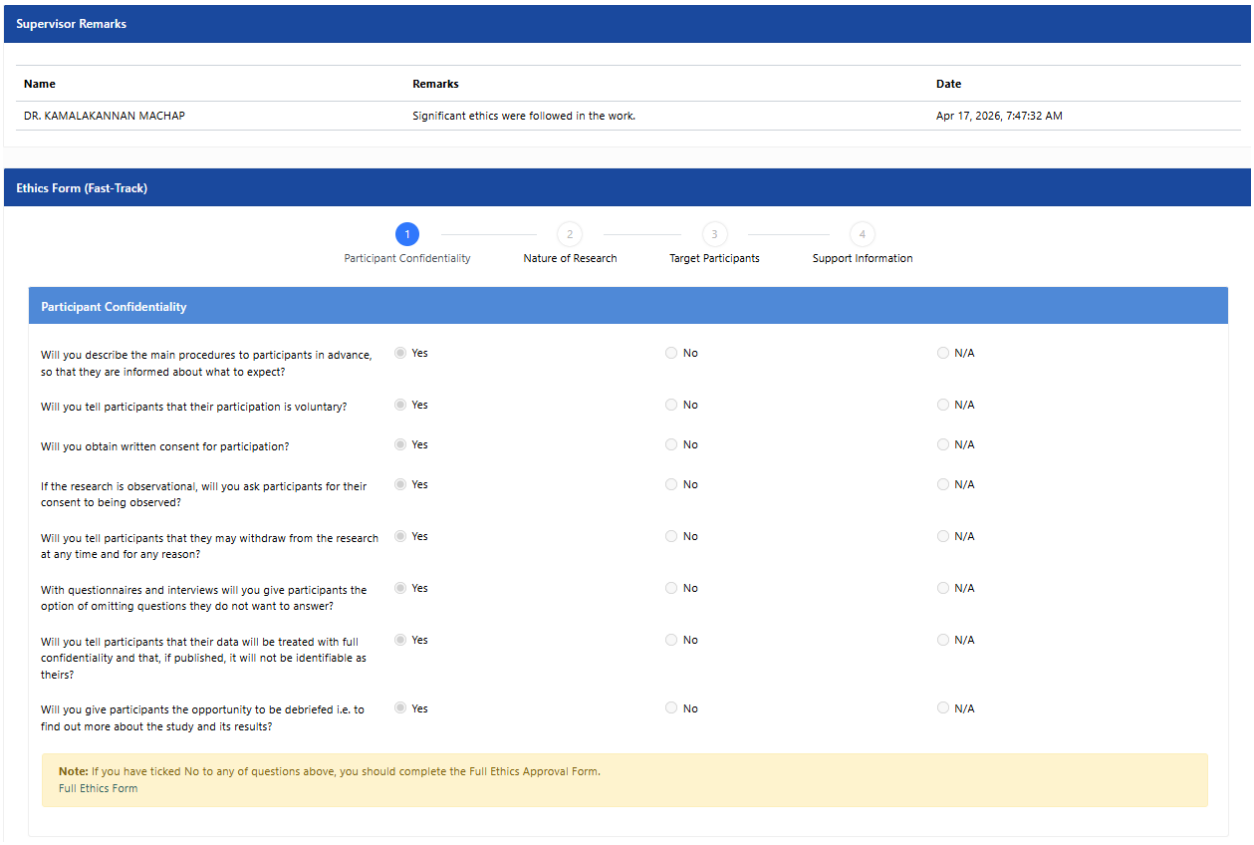
Assigned Second Marker manimegalai

Remarks

Name	Remarks	Date
TS. UMAPATHY EAGANATHAN	Good.	Mar 3, 2026, 9:25:47 AM

Figure 35: PPF

Appendix B: Ethics Form



Supervisor Remarks

Name	Remarks	Date
DR. KAMALAKANNAN MACHAP	Significant ethics were followed in the work.	Apr 17, 2026, 7:47:32 AM

Ethics Form (Fast-Track)

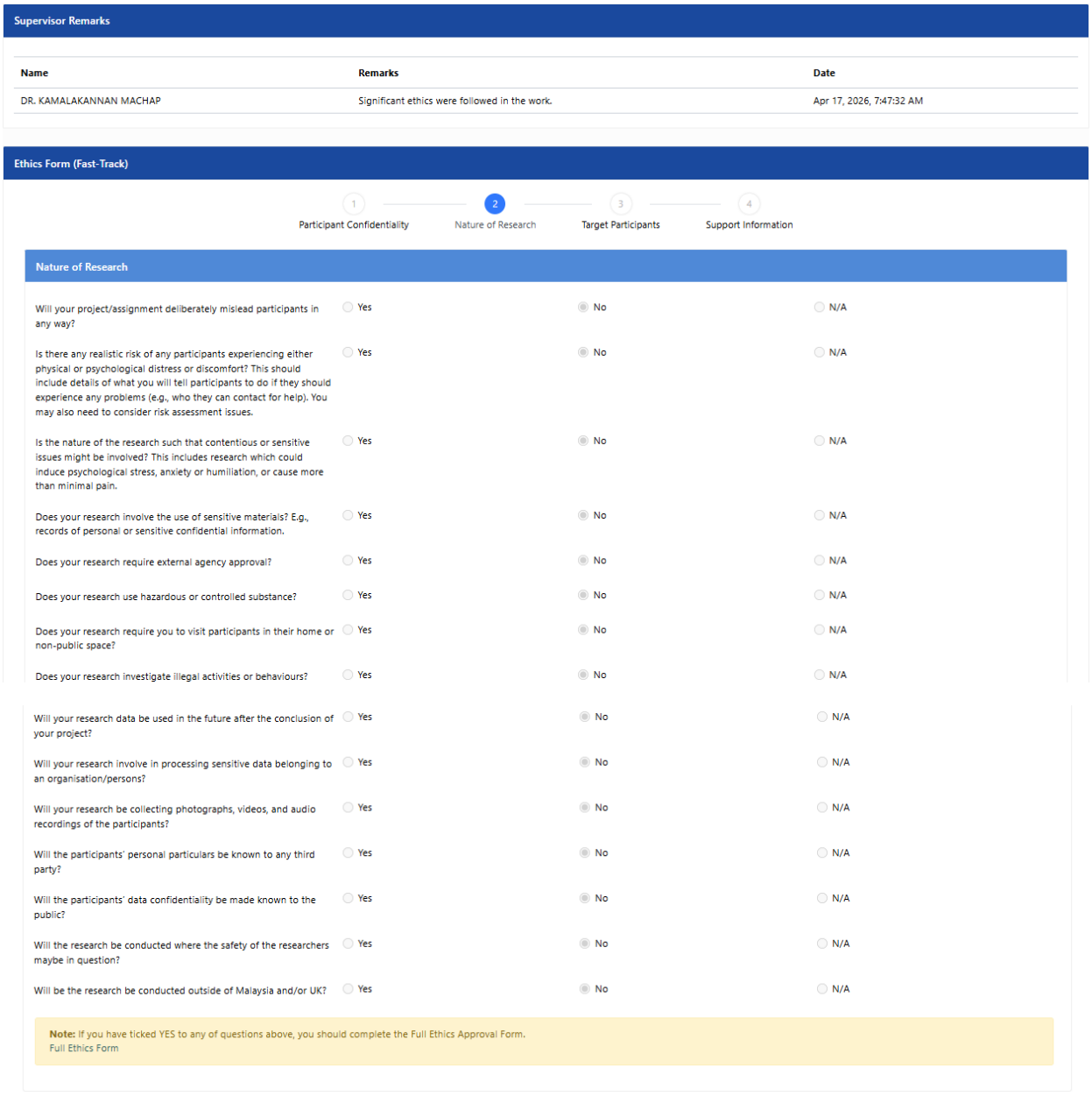
1 Participant Confidentiality — 2 Nature of Research — 3 Target Participants — 4 Support Information

Participant Confidentiality

	Yes	No	N/A
Will you describe the main procedures to participants in advance, so that they are informed about what to expect?	◉	○	○
Will you tell participants that their participation is voluntary?	◉	○	○
Will you obtain written consent for participation?	◉	○	○
If the research is observational, will you ask participants for their consent to being observed?	◉	○	○
Will you tell participants that they may withdraw from the research at any time and for any reason?	◉	○	○
With questionnaires and interviews will you give participants the option of omitting questions they do not want to answer?	◉	○	○
Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?	◉	○	○
Will you give participants the opportunity to be debriefed i.e. to find out more about the study and its results?	◉	○	○

Note: If you have ticked No to any of questions above, you should complete the Full Ethics Approval Form.
Full Ethics Form

Figure 36: Ethics Form Page 1



Supervisor Remarks

Name	Remarks	Date
DR. KAMALAKANNAN MACHAP	Significant ethics were followed in the work.	Apr 17, 2026, 7:47:32 AM

Ethics Form (Fast-Track)

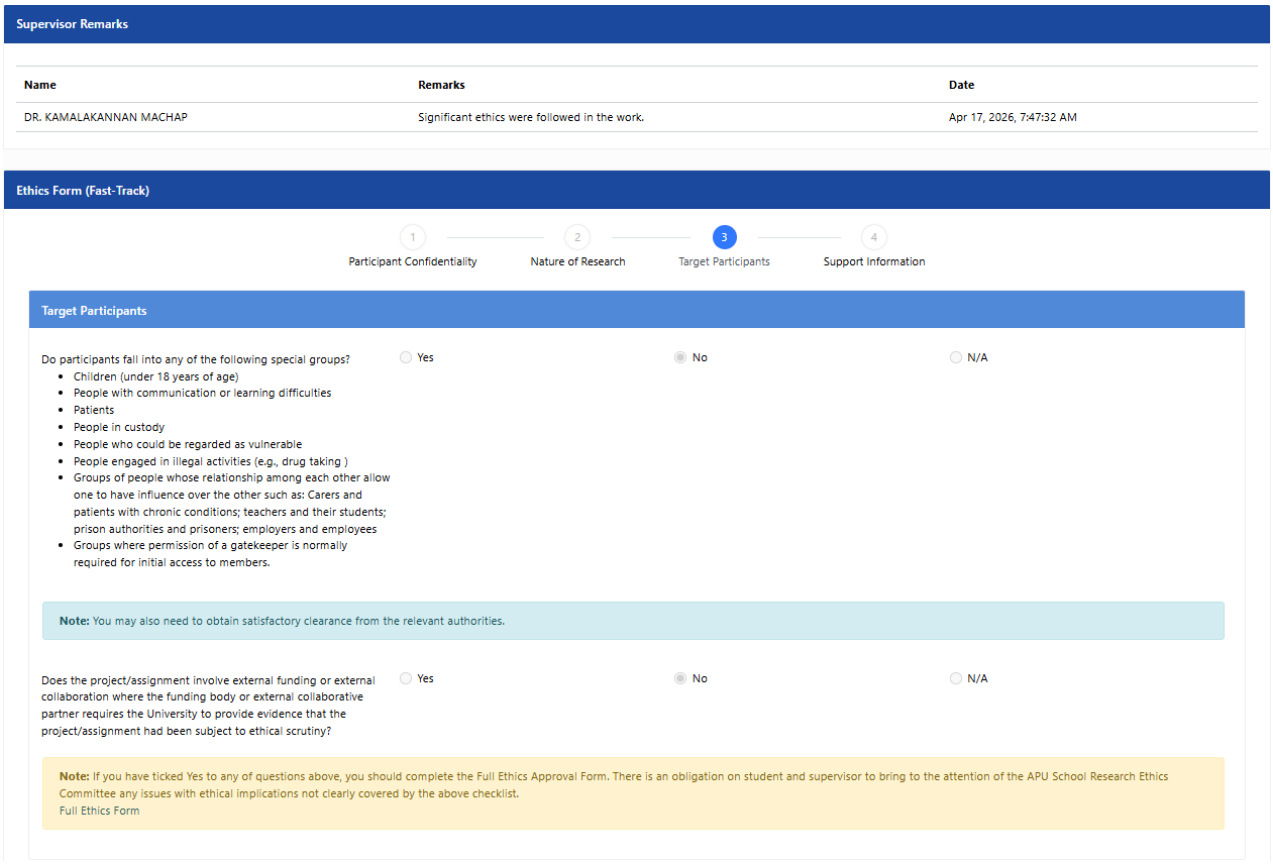
1 Participant Confidentiality — 2 Nature of Research — 3 Target Participants — 4 Support Information

Nature of Research

Will your project/assignment deliberately mislead participants in any way?	Yes	No	N/A
Is there any realistic risk of any participants experiencing either physical or psychological distress or discomfort? This should include details of what you will tell participants to do if they should experience any problems (e.g., who they can contact for help). You may also need to consider risk assessment issues.	Yes	No	N/A
Is the nature of the research such that contentious or sensitive issues might be involved? This includes research which could induce psychological stress, anxiety or humiliation, or cause more than minimal pain.	Yes	No	N/A
Does your research involve the use of sensitive materials? E.g., records of personal or sensitive confidential information.	Yes	No	N/A
Does your research require external agency approval?	Yes	No	N/A
Does your research use hazardous or controlled substance?	Yes	No	N/A
Does your research require you to visit participants in their home or non-public space?	Yes	No	N/A
Does your research investigate illegal activities or behaviours?	Yes	No	N/A
Will your research data be used in the future after the conclusion of your project?	Yes	No	N/A
Will your research involve in processing sensitive data belonging to an organisation/persons?	Yes	No	N/A
Will your research be collecting photographs, videos, and audio recordings of the participants?	Yes	No	N/A
Will the participants' personal particulars be known to any third party?	Yes	No	N/A
Will the participants' data confidentiality be made known to the public?	Yes	No	N/A
Will the research be conducted where the safety of the researchers maybe in question?	Yes	No	N/A
Will be the research be conducted outside of Malaysia and/or UK?	Yes	No	N/A

Note: If you have ticked YES to any of questions above, you should complete the Full Ethics Approval Form.
Full Ethics Form

Figure 37: Ethics Form Page 2



Supervisor Remarks

Name	Remarks	Date
DR. KAMALAKANNAN MACHAP	Significant ethics were followed in the work.	Apr 17, 2026, 7:47:32 AM

Ethics Form (Fast-Track)

1 Participant Confidentiality — 2 Nature of Research — 3 Target Participants — 4 Support Information

Target Participants

Do participants fall into any of the following special groups?
- Children (under 18 years of age)
- People with communication or learning difficulties
- Patients
- People in custody
- People who could be regarded as vulnerable
- People engaged in illegal activities (e.g., drug taking)
- Groups of people whose relationship among each other allow one to have influence over the other such as: Carers and patients with chronic conditions; teachers and their students; prison authorities and prisoners; employers and employees
- Groups where permission of a gatekeeper is normally required for initial access to members.

Yes — No — N/A

Note: You may also need to obtain satisfactory clearance from the relevant authorities.

Does the project/assignment involve external funding or external collaboration where the funding body or external collaborative partner requires the University to provide evidence that the project/assignment had been subject to ethical scrutiny?

Yes — No — N/A

Note: If you have ticked Yes to any of questions above, you should complete the Full Ethics Approval Form. There is an obligation on student and supervisor to bring to the attention of the APU School Research Ethics Committee any issues with ethical implications not clearly covered by the above checklist.
Full Ethics Form

Figure 38: Ethics Form Page 3

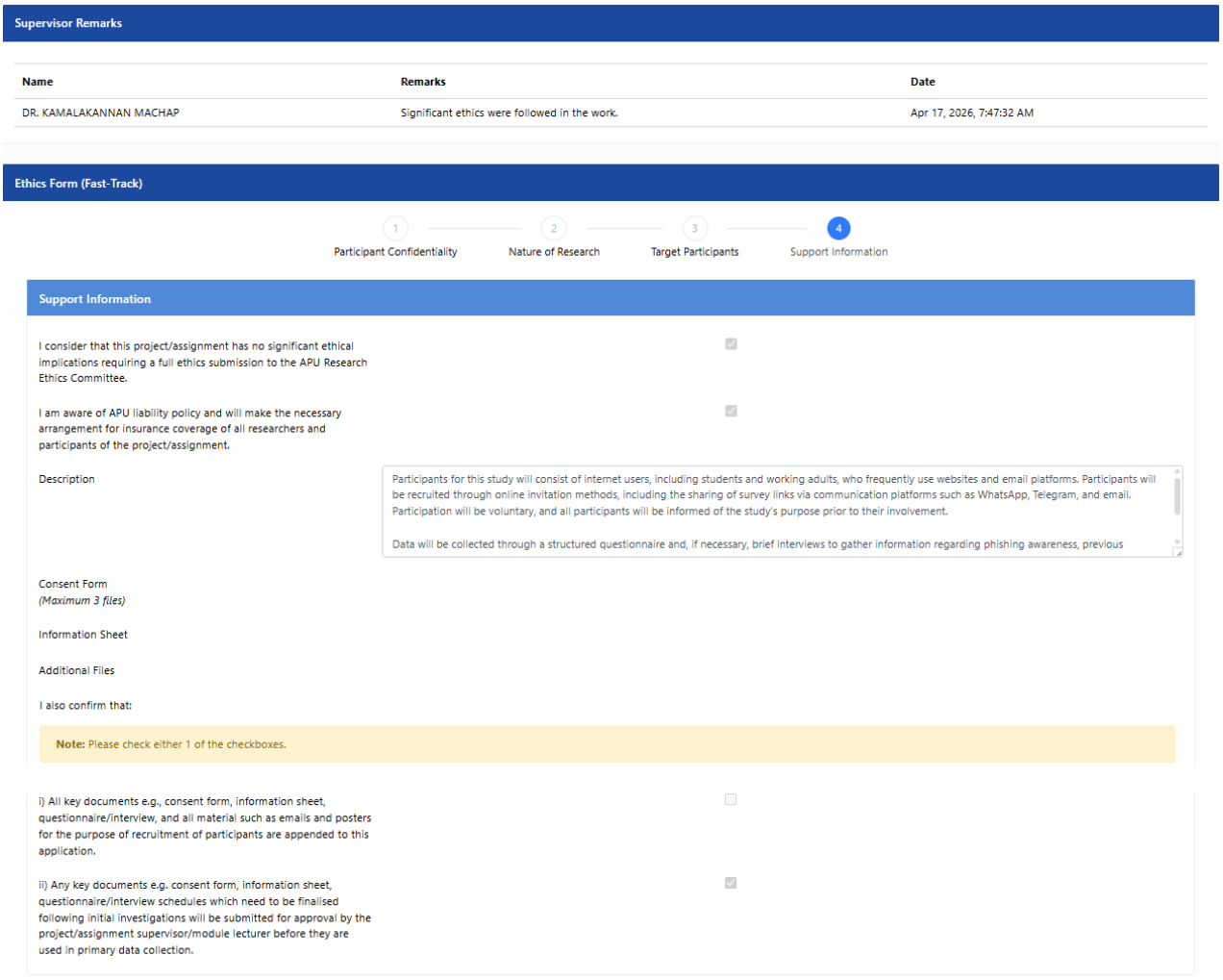
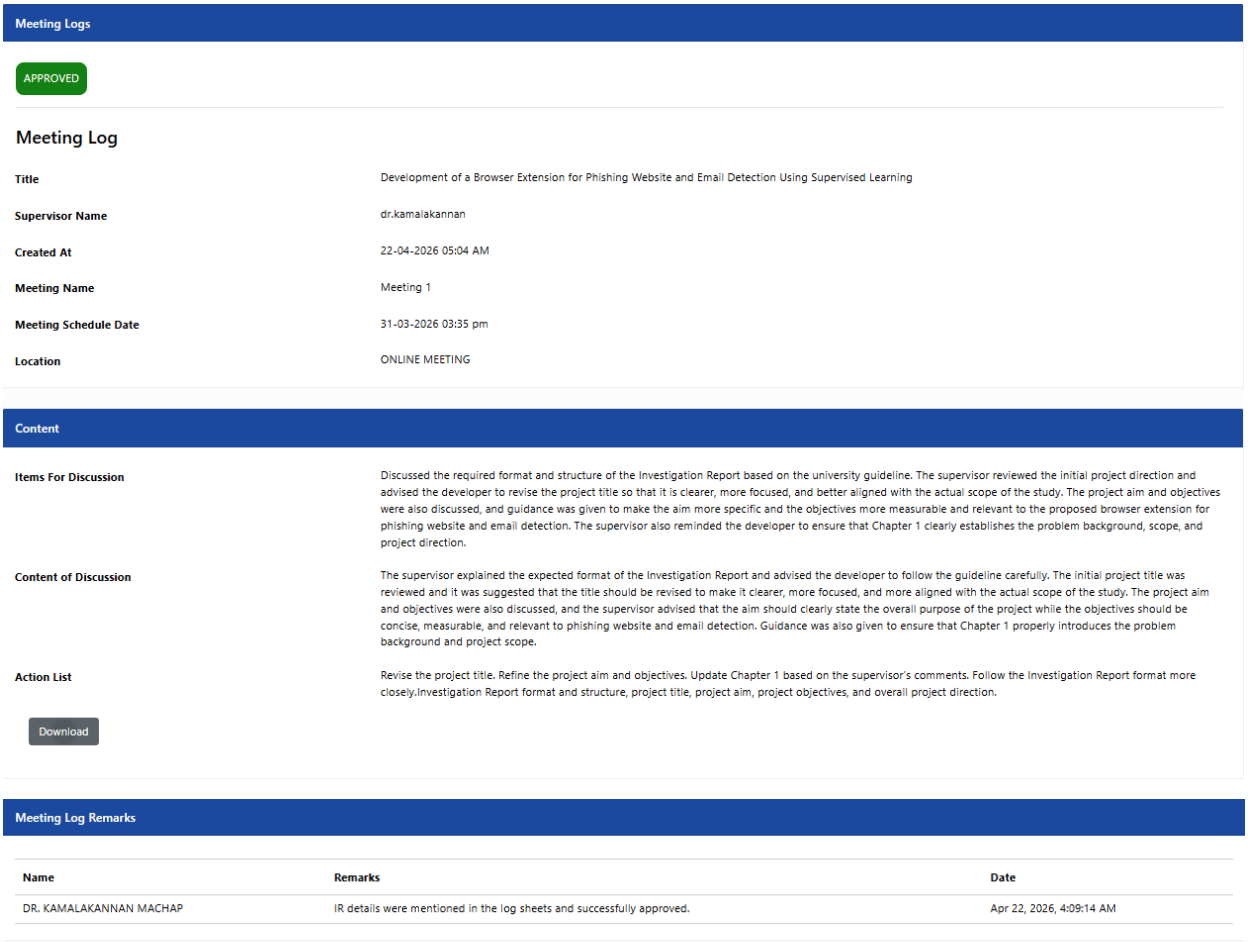


Figure 39: Ethics Form Page 4

Appendix C: Log Sheets



Meeting Logs

APPROVED

Meeting Log

Title	Development of a Browser Extension for Phishing Website and Email Detection Using Supervised Learning
Supervisor Name	dr.kamalakannan
Created At	22-04-2026 05:04 AM
Meeting Name	Meeting 1
Meeting Schedule Date	31-03-2026 03:35 pm
Location	ONLINE MEETING

Content

Items For Discussion	Discussed the required format and structure of the Investigation Report based on the university guideline. The supervisor reviewed the initial project direction and advised the developer to revise the project title so that it is clearer, more focused, and better aligned with the actual scope of the study. The project aim and objectives were also discussed, and guidance was given to make the aim more specific and the objectives more measurable and relevant to the proposed browser extension for phishing website and email detection. The supervisor also reminded the developer to ensure that Chapter 1 clearly establishes the problem background, scope, and project direction.
Content of Discussion	The supervisor explained the expected format of the Investigation Report and advised the developer to follow the guideline carefully. The initial project title was reviewed and it was suggested that the title should be revised to make it clearer, more focused, and more aligned with the actual scope of the study. The project aim and objectives were also discussed, and the supervisor advised that the aim should clearly state the overall purpose of the project while the objectives should be concise, measurable, and relevant to phishing website and email detection. Guidance was also given to ensure that Chapter 1 properly introduces the problem background and project scope.
Action List	Revise the project title. Refine the project aim and objectives. Update Chapter 1 based on the supervisor's comments. Follow the Investigation Report format more closely.Investigation Report format and structure, project title, project aim, project objectives, and overall project direction.

Download

Meeting Log Remarks

Name	Remarks	Date
DR. KAMALAKANNAN MACHAP	IR details were mentioned in the log sheets and successfully approved.	Apr 22, 2026, 4:09:14 AM

Figure 40: Meeting Log 1

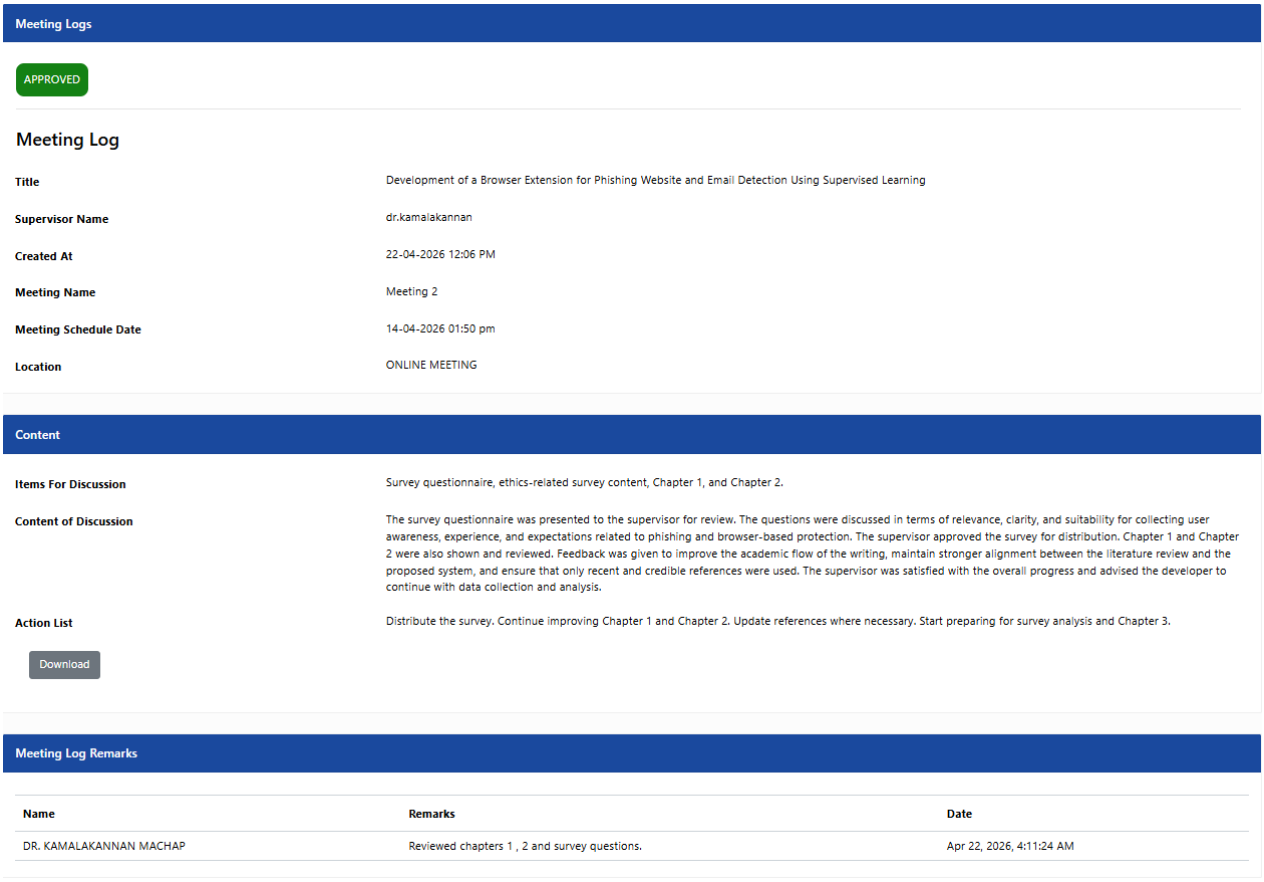


Figure 41: Meeting Log 2

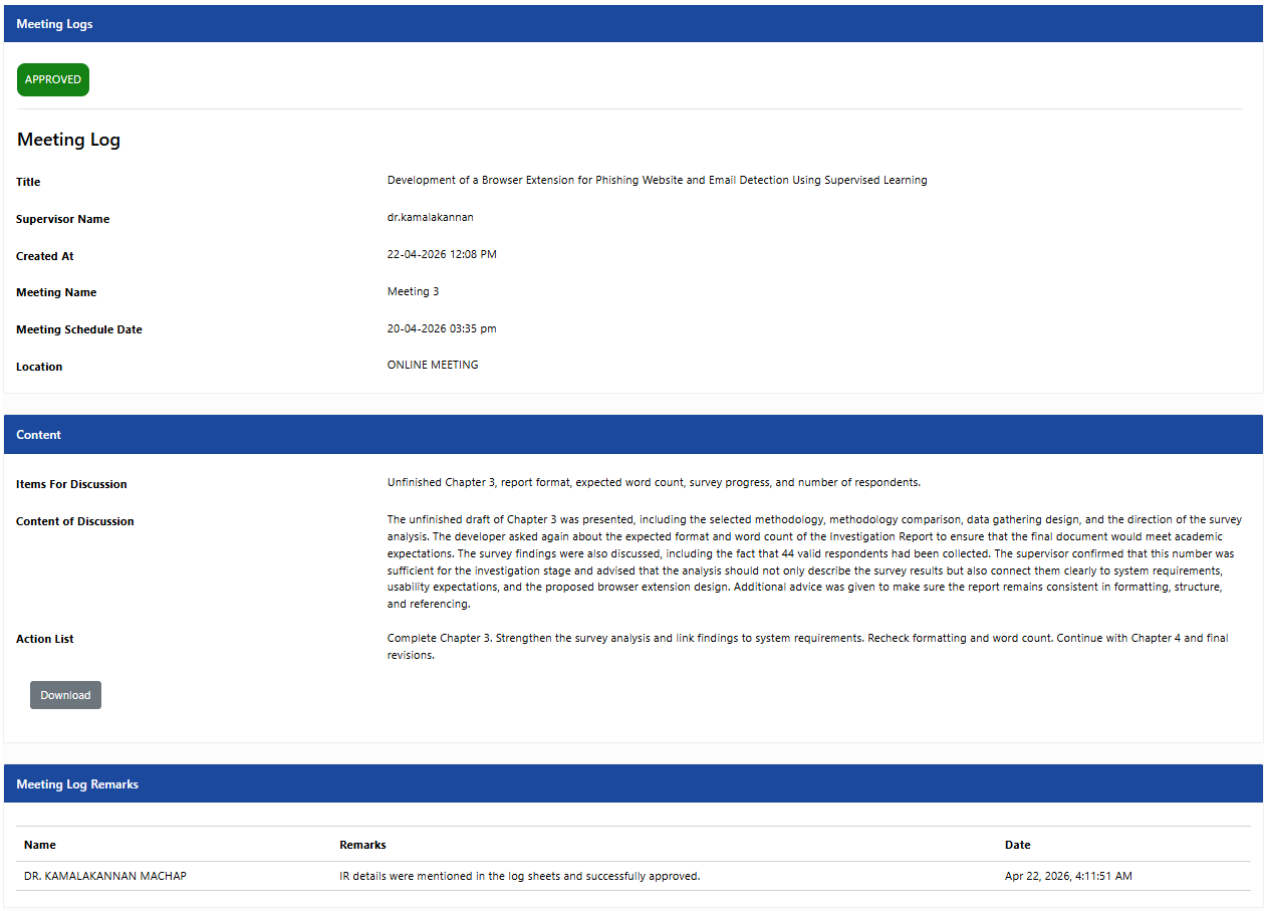


Figure 42: Meeting Log 3

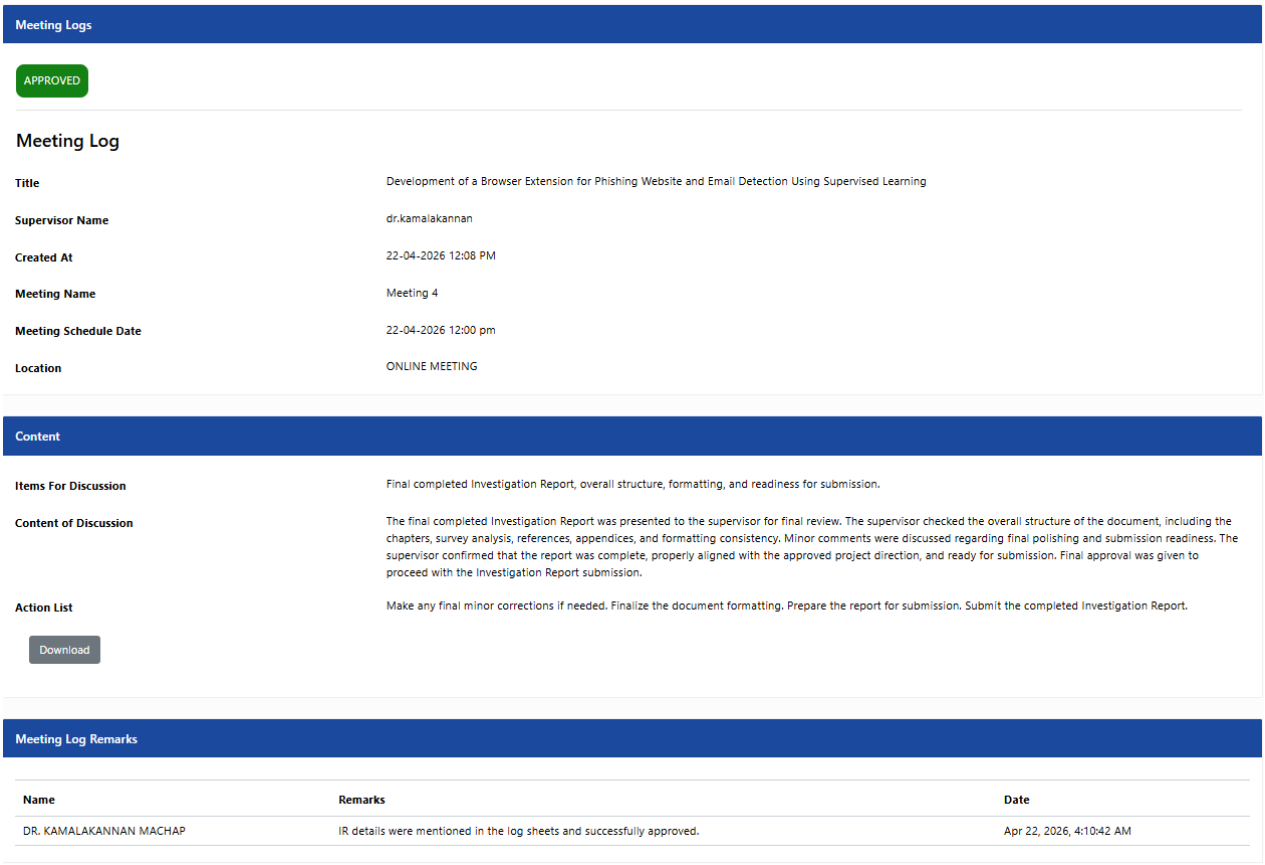


Figure 43: Meeting Log 4

Appendix D: Gantt Chart

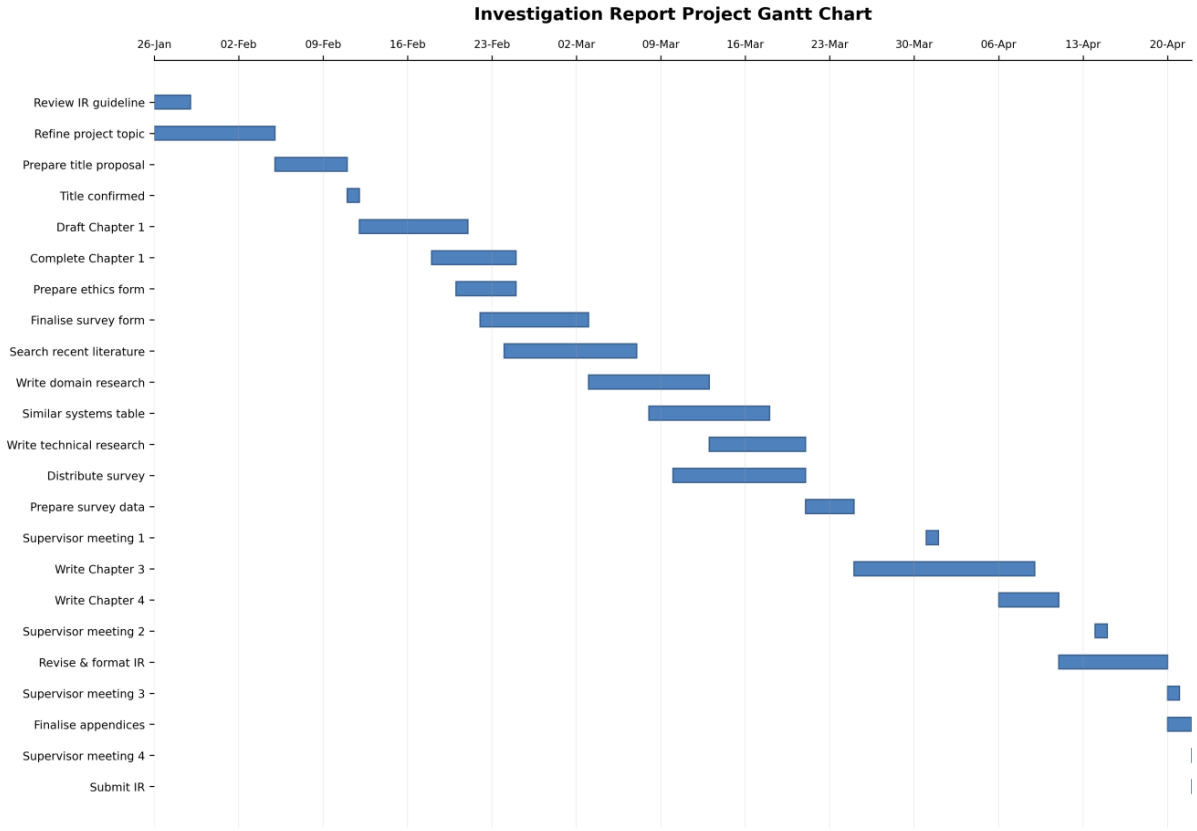


Figure 44: Project Gantt Chart Appendix

Appendix E: Respondent Demographic Profile

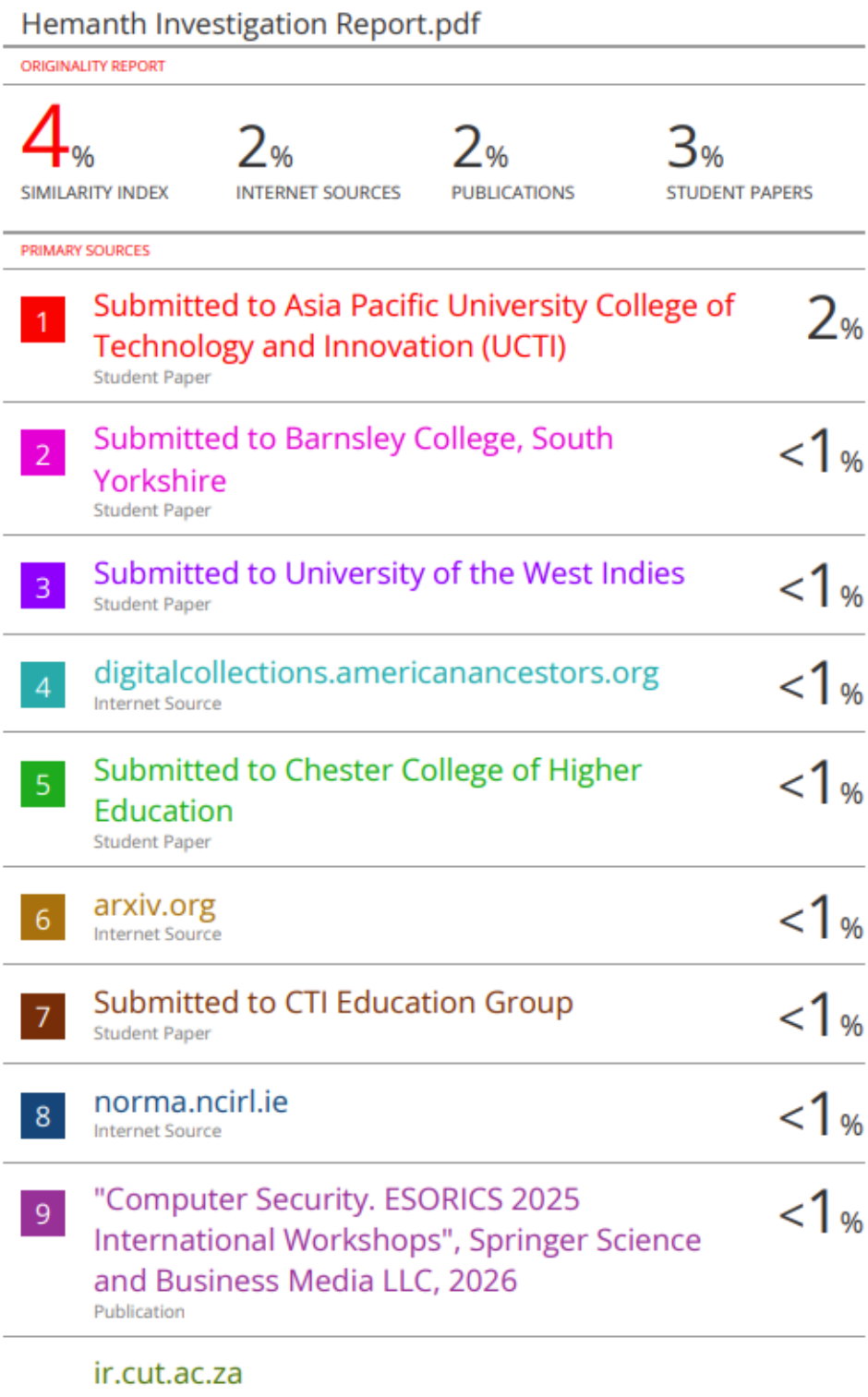
To preserve confidentiality, respondents are represented using pseudonymous identifiers rather than real names. The demographic profile below is derived directly from the valid survey responses.

Participant	Gender	Age Group	Occupation
Respondent 01	Male	18 - 24	Student
Respondent 02	Male	18 - 24	Student
Respondent 03	Male	18 - 24	Student
Respondent 04	Male	18 - 24	Employed
Respondent 05	Male	18 - 24	Student
Respondent 06	Male	18 - 24	Student
Respondent 07	Male	18 - 24	Student
Respondent 08	Male	18 - 24	Employed
Respondent 09	Male	18 - 24	Student
Respondent 10	Female	18 - 24	Student
Respondent 11	Female	18 - 24	Student
Respondent 12	Male	18 - 24	Student
Respondent 13	Female	18 - 24	Student
Respondent 14	Male	18 - 24	Student
Respondent 15	Male	18 - 24	Student
Respondent 16	Female	18 - 24	Student
Respondent 17	Female	18 - 24	Student
Respondent 18	Female	45 and above	Employed
Respondent 19	Female	18 - 24	Student
Respondent 20	Male	18 - 24	Employed
Respondent 21	Female	18 - 24	Student
Respondent 22	Female	18 - 24	Student
Respondent 23	Male	25 - 34	Unemployed
Respondent 24	Female	35 - 44	Employed
Respondent 25	Female	25 - 34	Employed
Respondent 26	Male	18 - 24	Employed
Respondent 27	Female	18 - 24	Student

Respondent 28	Male	18 - 24	Student
Respondent 29	Male	45 and above	Employed
Respondent 30	Female	45 and above	Self-employed
Respondent 31	Female	45 and above	Unemployed
Respondent 32	Female	45 and above	Employed
Respondent 33	Female	35 - 44	Employed
Respondent 34	Female	35 - 44	Employed
Respondent 35	Male	25 - 34	Employed
Respondent 36	Male	45 and above	Self-employed
Respondent 37	Male	18 - 24	Student
Respondent 38	Male	45 and above	Employed
Respondent 39	Female	35 - 44	Employed
Respondent 40	Male	18 - 24	Student
Respondent 41	Male	18 - 24	Student
Respondent 42	Male	45 and above	Self-employed
Respondent 43	Male	18 - 24	Student
Respondent 44	Female	45 and above	Unemployed

Table 9: Respondent Demographic Profile

Appendix F: Turnitin Similarity Report



Hemanth Investigation Report.pdf

ORIGINALITY REPORT

4%	2%	2%	3%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Asia Pacific University College of Technology and Innovation (UCTI) Student Paper	2%
2	Submitted to Barnsley College, South Yorkshire Student Paper	<1%
3	Submitted to University of the West Indies Student Paper	<1%
4	digitalcollections.americanancestors.org Internet Source	<1%
5	Submitted to Chester College of Higher Education Student Paper	<1%
6	arxiv.org Internet Source	<1%
7	Submitted to CTI Education Group Student Paper	<1%
8	norma.ncirl.ie Internet Source	<1%
9	"Computer Security. ESORICS 2025 International Workshops", Springer Science and Business Media LLC, 2026 Publication	<1%

ir.cut.ac.za

Figure 45: First Page Turnitin Similarity Report

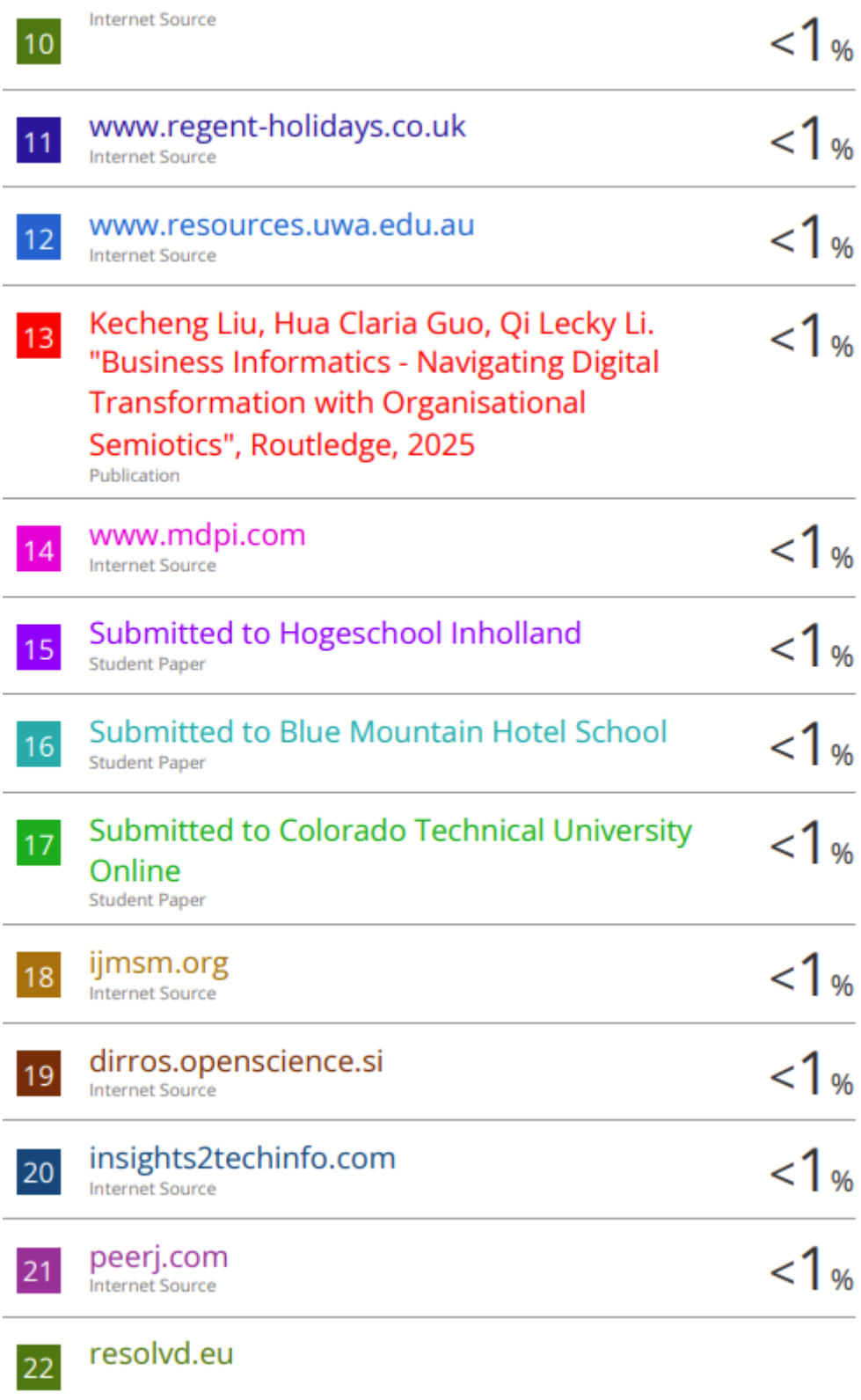


Figure 46: Second Page Turnitin Similarity Report